

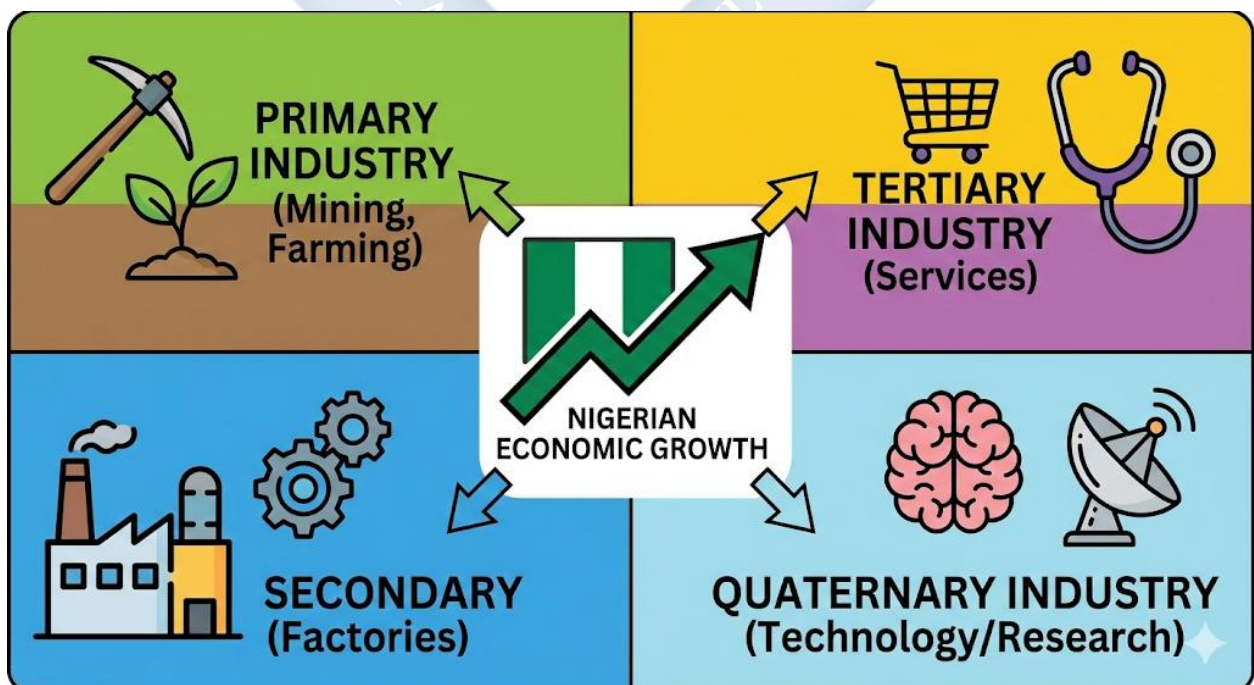
SS2 GEOGRAPHY LESSON NOTES

FIRST TERM

WEEK 1: INDUSTRY

MEANING OF INDUSTRY

- **Core Concept:** Industry is the economic engine of a nation. It encompasses all human activities involved in the **creation of value**, either by transforming raw materials into useful goods or by providing essential services.
- **Beyond the Definition:** Think of an economy as a body. Industries are the organs that keep it functioning:
 - **Primary** industries are like the digestive system, gathering raw "nutrients."
 - **Secondary** industries are the metabolic system, processing these nutrients.
 - **Tertiary** industries are the circulatory system, distributing the finished products.
 - **Quaternary** industries are the brain, processing information and driving innovation.
- **Key Idea:** The shift from primary to secondary and then to tertiary/quaternary industries is a key indicator of a country's economic development.



TYPES/CLASSIFICATION OF INDUSTRIES

a. Primary Industry: The Foundation

- **Description:** These are **extractive** or **genetic** industries. They form the base of the industrial pyramid by harvesting raw materials directly from the earth, water, and air.
- **Sub-categories:**
 - **Extractive:** Mining (e.g., crude oil in the Niger Delta, limestone in Ewekoro), quarrying, fishing.
 - **Genetic:** Agriculture (e.g., cocoa farming in Ondo, rice cultivation in Kebbi), forestry, animal husbandry.
- This sector is the historical backbone of the Nigerian economy but is prone to price fluctuations on the global market.

b. Secondary Industry: The Value Adder

- **Description:** This sector **processes and transforms** raw materials from the primary sector into finished or semi-finished goods, dramatically increasing their value and utility.
- **Sub-categories:**
 - **Heavy Industry:** Large-scale, capital-intensive, and produces large goods or raw materials for *other industries*. Examples: **Ajaokuta Steel Company, Kaduna Refinery, cement manufacturing (Dangote Cement).**
 - **Light Industry:** Smaller-scale, often consumer-oriented, producing smaller goods for the end-user. Examples: **Food processing (Dangote Flour Mills), textile manufacturing, furniture making, bottling (Coca-Cola).**
 - **Construction:** Building infrastructure like roads, bridges, and houses.

c. Tertiary Industry: The Service Enabler

- **Description:** This sector does not produce a tangible good but provides **essential services** that support the primary and secondary sectors and facilitate daily life. It is the largest and fastest-growing sector in many modern economies.
- **Examples:**

- **Trade & Commerce:** Retail shops, supermarkets, import/export.
- **Transportation & Distribution:** Logistics companies, airlines, haulage services.
- **Finance & Banking:** Banks, insurance companies, microfinance institutions.
- **Social Services:** Education, healthcare, tourism, entertainment.

d. Quaternary Industry: The Knowledge Economy

- **Description:** This is the most advanced sector, focused on **knowledge-based activities** involving information processing, research, and development. It is driven by technology and intellectual capability.
- **Examples:**
 - **Information and Communication Technology (ICT):** Software development, telecommunications (e.g., MTN, Airtel).
 - **Research & Development (R&D):** Biotechnology, pharmaceutical research.
 - **Consultancy Services:** Financial advisory, management consulting, architectural design.
 - **Nigerian Context:** The rise of tech hubs like "Yabacon Valley" in Lagos is a prime example of the quaternary sector's growth.

FACTORS INFLUENCING LOCATION OF INDUSTRIES (The "Why" Behind the "Where")

- **Concept:** Industries seek the **least-cost location**. The ideal spot minimizes total expenses and maximizes profit.

Factor	Detailed Explanation & Concrete Examples
Raw Materials	Bulky/Low Value-to-Weight: Industries locate near the source to save transport costs. E.g., a sawmill is built in a forest, a cement factory near a limestone quarry. The Ajaokuta Steel Company was sited near Itakpe iron ore deposits.
Market	Perishable/Bulky Finished Goods: Industries locate near consumers. E.g., a bakery, a bottled drink factory, newspaper printing.
Power Supply	Power-Intensive Industries: Require massive, consistent energy. E.g.,

Factor	Detailed Explanation & Concrete Examples
	Aluminum smelting (Aluminium Smelter Company of Nigeria, ALSCON) was located at Ikot Abasi due to proximity to gas sources. Unreliable power is a major disincentive in Nigeria.
Labour Supply	Labour-Intensive vs. Capital-Intensive: The textile industry in Kano/Kaduna was historically reliant on a large, available labour force. Automated car plants (capital-intensive) need fewer, but more skilled, workers.
Transportation	Footloose Industries: Modern, high-tech industries (e.g., software development) are not tied to raw materials and can locate near good transport links (e.g., airports, seaports). The Lagos-Ibadan Expressway corridor is a key transport artery attracting industries.
Government Policy	A Powerful Decider: Governments can create Growth Pole Policies . E.g., Establishing the Onne Oil & Gas Free Zone in Rivers State with tax breaks and infrastructure, or providing incentives for industries in "economically disadvantaged" areas.
Water Supply	Heavy-Usage Industries: Essential for cooling, processing, and as a raw material. E.g., Breweries (like Nigerian Breweries) and textile dyeing plants are always located near reliable water sources.
Climate	Specific Needs: The film industry ("Nollywood") is concentrated in the South due to the year-round warm, humid climate suitable for outdoor shooting.

MAJOR INDUSTRIAL REGIONS/ZONES

- **Why these axes formed?**
 - **Lagos-Ibadan Axis:** Historical port (Lagos), largest market in Nigeria, financial hub, government policy, and relatively better infrastructure.
 - **Port Harcourt–Warri Axis:** Proximity to crude oil and natural gas (Raw Material factor). Dominated by petrochemicals, refining, and related support services.

- **Kano-Kaduna Axis:** Historical centre of commerce, agricultural raw materials (cotton for textiles, groundnuts), and large labour supply.
- **Enugu-Onitsha-Aba Axis:** Proximity to coal (historically), large market (Onitsha Main Market), and a strong entrepreneurial culture.

PROBLEMS OF INDUSTRIAL DEVELOPMENT IN NIGERIA

- **Problem: Poor Infrastructure**
 - **Depth:** Unreliable power forces companies to buy and maintain expensive generators, increasing production costs by 30-40%, making Nigerian goods uncompetitive internationally.
- **Problem: Inadequate Capital**
 - **Depth:** High-interest rates and difficulty accessing loans stifle the growth of Small and Medium Enterprises (SMEs), which are the backbone of industrialisation.
- **Problem: The "Resource Curse" (Dutch Disease)**
 - **Concept:** Over-reliance on crude oil revenues has led to the neglect of other sectors like agriculture and manufacturing, making the economy vulnerable and hindering balanced industrial growth.
- **Problem: Policy Instability and Inconsistency**
 - **Example:** Frequent changes in import/export duties, tariffs, and industrial policies create an unpredictable business environment that discourages long-term investment.

SOLUTIONS TO INDUSTRIAL PROBLEMS IN NIGERIA

- **Solution: Power Sector Reform**
 - **Action:** Genuine privatisation and investment in the national grid, coupled with incentives for renewable energy sources (solar, wind) to provide stable, cheaper power.
- **Solution: Strategic Government Intervention**
 - **Action:** Beyond soft loans, the government should provide consistent **industrial policy**, invest in **specialized industrial training institutes**, and upgrade **port operations** to reduce the cost of importing raw materials and exporting finished goods.
- **Solution: Economic Diversification**

- **Action:** Deliberate policies to revive and support non-oil sectors like agriculture, solid minerals, and tourism to create a more resilient industrial base.
- **Solution: "Buy Nigerian" Campaigns**
 - **Action:** Enforcing local content laws and fostering a national culture of patronage for locally produced goods to create a ready market for industries.

EVALUATION

1. "The quaternary industry is the future of the Nigerian economy." Do you agree with this statement? Justify your answer.
2. Explain why a beverage bottling company is likely to be located in a major city like Lagos, while a cement factory is located near a limestone quarry.
3. Apart from those mentioned in the note, identify two other problems of industrial development in Nigeria and suggest one solution for each.
4. Choose one major industrial zone in Nigeria. Explain TWO factors that led to its development and ONE major challenge it currently faces.

ASSIGNMENT

1. **Case Study Analysis:** Research the **Lekki Free Trade Zone** in Lagos. Write a short report explaining:
 - The types of industries located there.
 - How at least three location factors were favorable for its establishment.
 - Its potential impact on Nigeria's industrial development.
2. **Comparative Analysis:** Identify two major industrial zones in Africa outside Nigeria (e.g., **Gauteng Province, South Africa** and **Suez Canal Economic Zone, Egypt**). Create a presentation slide comparing one of them with the **Lagos-Ibadan Axis**, highlighting similarities and differences in their development factors.

PAST WAEC/NECO QUESTIONS

WAEC 2018 Q7 (a) Define the term 'manufacturing industry'.

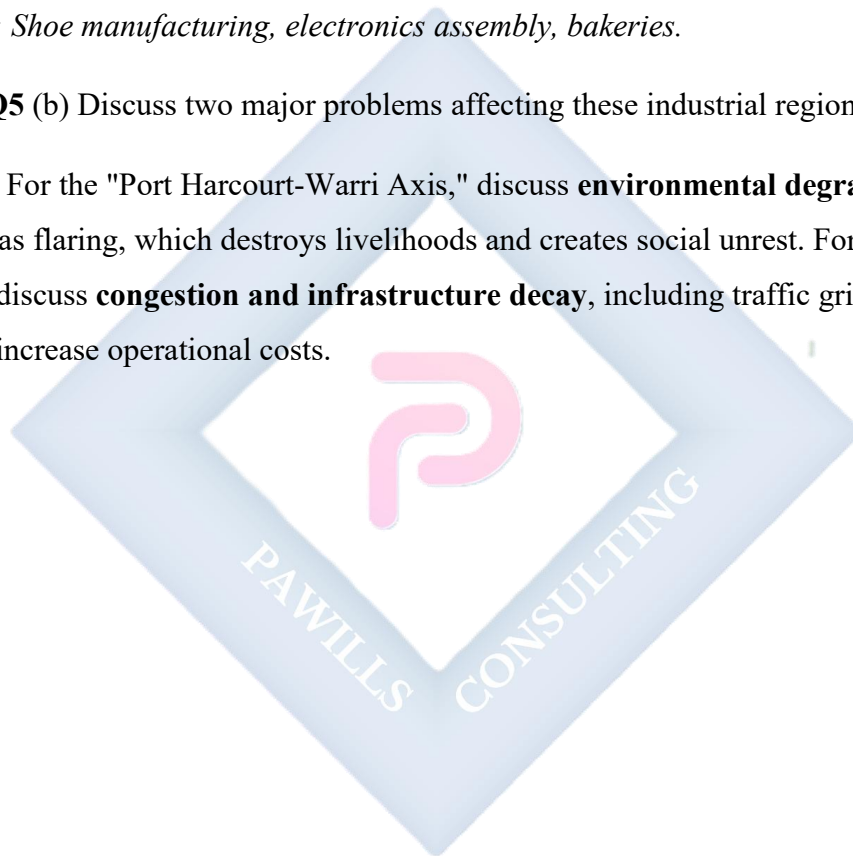
Answer Tip: Define it as a subset of the secondary industry concerned with the mechanical or chemical transformation of raw materials into new products. Give examples like car assembly, food canning, and textile production.

(b) Differentiate between heavy and light industries with examples.

***Answer Tip:** *** Heavy Industry:** Large-scale, capital-intensive, produces large goods or capital goods for other industries. *Example: Steel production, shipbuilding, oil refining.* *** Light Industry:** Smaller-scale, less capital-intensive, produces smaller consumer goods for the end-user. *Example: Shoe manufacturing, electronics assembly, bakeries.*

NECO 2019 Q5 (b) Discuss two major problems affecting these industrial regions.

***Answer Tip:** For the "Port Harcourt-Warri Axis," discuss **environmental degradation** from oil spills and gas flaring, which destroys livelihoods and creates social unrest. For the "Lagos-Ibadan Axis," discuss **congestion and infrastructure decay**, including traffic gridlock and port delays, which increase operational costs.



WEEK 2: BASIC CONCEPTS OF GIS

MEANING OF GIS

- **Core Definition:** A Geographic Information System (GIS) is a framework for gathering, managing, analyzing, and visualizing data. Its power comes from linking data to a precise location on the Earth's surface.
- **Beyond the Definition:** Think of GIS as a "smart map." A regular paper map shows you *where* things are. A GIS map shows you *where* things are and *what* they are like, allowing you to ask complex questions and discover hidden patterns.
- **Powerful Analogy: "GIS is like a geographic spreadsheet."**
 - The **spatial data** is the *location* of a cell (e.g., Column D, Row 5).
 - The **attribute data** is the *information inside* that cell (e.g., "Population: 5,000," "School Name: King's College," "Road Type: Highway"). GIS links these two things together visually on a map, allowing for dynamic analysis.



KEY CONCEPTS & TERMINOLOGY (The GIS Vocabulary) To understand GIS, you must master its core language:

1. **Spatial Data (The "Where"):** This is data that has a geographic reference.
 - **Vector Data:** Uses points, lines, and polygons (shapes) to represent features.
 - **Points:** A specific location (e.g., a school, a borehole, a hospital). *Attribute example: Hospital Name, Number of Beds.*
 - **Lines:** A linear feature (e.g., a road, a river, a pipeline). *Attribute example: Road Name, Speed Limit, Number of Lanes.*
 - **Polygons:** An enclosed area (e.g., a state boundary, a local government area, a lake, a farm). *Attribute example: State Name, Population, Crop Type.*
 - **Raster Data:** Composed of a grid of cells (pixels), where each cell has a value. Think of it like a digital photograph.
 - **Examples:** Satellite imagery, aerial photographs, Digital Elevation Models (showing height), and scanned maps.
2. **Attribute Data (The "What"):** This is the descriptive, non-spatial information about the geographic features. It is stored in a table and linked to the spatial data.
3. **Layer:** A GIS map is built by stacking different thematic datasets, each in its own layer. For example, a single map project could have separate layers for:
 - Roads (Line)
 - Buildings (Polygon)
 - Schools (Point)
 - Local Government Boundaries (Polygon)
 - Satellite Image (Raster)
 - **Concept:** You can turn layers on and off to focus on specific information, just like using tracing paper.
4. **Georeferencing:** The process of assigning real-world coordinates to a map or image that doesn't have them (like a scanned paper map), so it can be viewed and analyzed with other spatial data in the correct location.

COMPONENTS OF GIS (The Essential Parts) A functional GIS requires five key components to work together:

1. **Hardware:** The physical devices. This includes the computer running the GIS software, data storage (hard drives, servers), and input/output devices like GPS units, scanners, and printers.
2. **Software:** The programs that provide the tools to work with spatial data. Examples include:
 - **Proprietary (Paid):** ArcGIS (the industry standard), ArcGIS Pro.
 - **Open-Source (Free):** QGIS (a very powerful and popular free alternative).
3. **Data:** The core of the system. Without data, the hardware and software are useless. This includes both spatial and attribute data from various sources.
4. **People:** The most important component. GIS requires skilled individuals to manage the system, create maps, perform analysis, and interpret the results. Roles include GIS Analysts, Technicians, and Database Managers.
5. **Methods/Procedures:** The plans, rules, and workflows that ensure the GIS is used effectively and accurately. This includes standards for data collection, analysis techniques, and map creation.

FUNCTIONS OF GIS (What it Actually Does) GIS performs a sequence of operations:

1. **Data Capture/Input:** How data gets into the system. This can be done by:
 - Digitizing (tracing over maps or images on a computer).
 - Using GPS coordinates collected in the field.
 - Importing existing digital data (e.g., spreadsheets with addresses that can be "geocoded").
2. **Data Management & Storage:** Organizing the data in a structured database so it can be easily retrieved, updated, and queried.
3. **Data Manipulation & Analysis:** This is the "intelligence" of GIS. It allows you to:
 - **Query:** Ask questions of the data. E.g., "Show me all schools in Lagos Mainland LGA with more than 500 students."
 - **Spatial Analysis:** Find patterns and relationships. E.g., "Identify all areas within 1 km of a health clinic that are not served by a clean water source."

- **Overlay Analysis:** Combine different layers to create new information. E.g., Overlaying soil type, slope, and land use layers to find the most suitable land for a new housing project.
- 4. **Data Visualization & Output:** Presenting the results in an understandable way. This includes creating maps, charts, reports, and interactive web maps.

APPLICATIONS OF GIS (Real-World Impact) GIS is used in virtually every field. Here are detailed examples relevant to Nigeria:

- **Urban Planning:** The Lagos State Government uses GIS to manage land use, plan new road networks, and monitor urban sprawl.
- **Disaster Management:** During flooding, GIS can map flooded areas, identify the most vulnerable populations (by overlaying flood maps with population density), and plan the most efficient evacuation routes.
- **Public Health:** The Nigeria Centre for Disease Control (NCDC) can use GIS to track disease outbreaks like Cholera or Lassa Fever. By mapping reported cases, they can identify the source and pattern of spread to target interventions.
- **Agriculture (Precision Farming):** GIS can map soil pH and nutrient levels across a large farm. This allows farmers to apply fertilizer only where it is needed, saving money and reducing environmental pollution.
- **Business & Marketing:** A supermarket chain like Shoprite can use GIS to analyze population density, average income, and competitor locations to scientifically select the best site for a new store.
- **Security:** Security agencies can map crime incidents to identify hotspots and deploy resources more effectively.

SOURCES OF DATA FOR GIS

- **Remote Sensing:** Data collected from satellites (e.g., Landsat, Sentinel) or aircraft.
- **Global Positioning System (GPS):** Field data collection using handheld GPS devices or smartphones.

- **Existing Maps & Surveys:** Topographic maps from the Office of the Surveyor General, land use maps.
- **Census & Statistical Data:** Population data from the National Population Commission, economic data from the National Bureau of Statistics.
- **Online Repositories:** Open-source data portals from organizations like the World Bank, UN, and Nigerian government agencies.

EVALUATION (Week 2)

1. Define GIS in your own words and explain why it is more powerful than a static paper map.
2. With the aid of a diagram, differentiate between vector and raster data types.
3. Describe THREE real-world applications of GIS in solving problems in Nigeria.
4. List the five components of a GIS and explain the role of the "People" component.

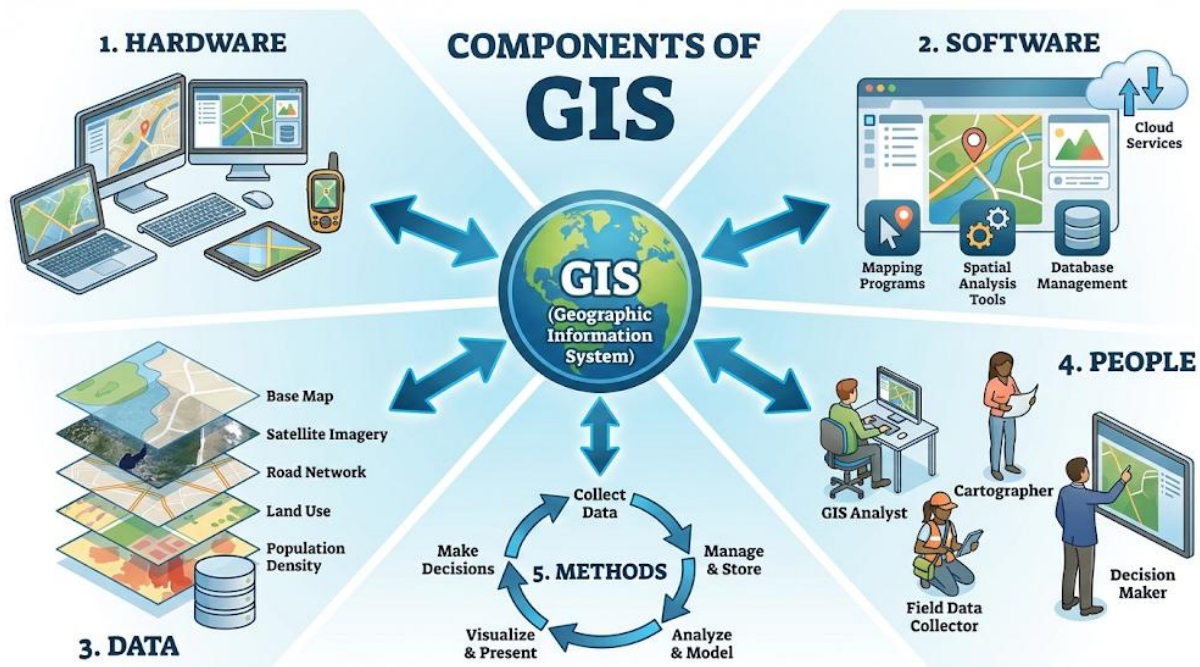
ASSIGNMENT (Week 2)

1. **Hands-On Activity:** Imagine you are a GIS analyst tasked with finding the best location for a new public library in your community.
 - List FIVE spatial data layers you would need for your analysis.
 - For each layer, state whether it would be a Point, Line, or Polygon, and one piece of attribute data it should contain.
 2. **Research:** Find and describe one specific example of how a Nigerian state or federal agency is using GIS. (Hint: Look up "Lagos State GIS," "NLNG pipeline monitoring," or "NCDC disease surveillance").
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WEEK 3: COMPONENTS OF GIS

MEANING OF GIS (Recap)

- GIS is an integrated system of hardware, software, data, people, and methods designed to capture, store, manipulate, analyze, manage, and present all types of geographically referenced information.



DEEP DIVE INTO THE FIVE COMPONENTS

1. HARDWARE: The Physical Machine

- **Explanation:** This is the tangible infrastructure that powers the GIS. The choice of hardware depends on the scale of the project—from a single laptop to a powerful network of servers.
- **Detailed Breakdown:**
 - **Central Processing Unit (Computer):** The workhorse. For complex analysis and 3D visualization, a computer with a fast processor, ample RAM (16GB+), and a good graphics card is essential.
 - **Data Storage:** Where all the spatial data lives. This can be:
 - **Local:** Hard drives or SSDs on a single computer.
 - **Network-Attached Storage (NAS):** For a team sharing data.

- **Servers & Cloud Storage:** For large organizations and web-based GIS (e.g., storing data on Amazon Web Services or Microsoft Azure).
- **Input Devices:** Tools for getting data *into* the system.
 - **GPS Receiver:** For collecting precise location coordinates in the field.
 - **Digitizing Tablet:** A specialized pad for manually tracing features from a map into digital format.
 - **Scanner:** For converting paper maps into digital images that can then be georeferenced.
- **Output Devices:** Tools for presenting the results.
 - **High-Resolution Monitor:** Crucial for clear visualization.
 - **Printer/Plotter:** For producing high-quality paper maps.

2. SOFTWARE: The Brain and the Tools

- **Explanation:** The GIS software provides the environment and the specific tools (functions) to work with spatial data. It is the interface between the user and the data.
- **Detailed Breakdown of Software Capabilities:**
 - **User Interface:** The menus, toolbars, and windows that allow the user to interact with the system.
 - **Data Management Tools:** For creating, editing, and organizing geographic datasets and databases.
 - **Spatial Analysis & Modeling Tools:** The core analytical engine. This includes tools for:
 - **Buffering:** Creating a zone of a specific distance around a feature (e.g., a 100m buffer around a river to define a no-build zone).
 - **Overlay:** Combining layers (e.g., overlaying a soil map and a land ownership map).
 - **Network Analysis:** Finding the shortest path, optimizing delivery routes, or modeling water flow through pipes.
 - **Cartographic & Visualization Tools:** Tools for designing and producing maps with legends, scale bars, and north arrows.
- **Software Examples:**

- **ArcGIS Suite (Esri):** The commercial leader. Includes ArcGIS Pro (desktop), ArcGIS Online (cloud), and ArcGIS Enterprise. Used by large corporations and government agencies.
- **QGIS:** A powerful, free, and open-source alternative. It has a vast ecosystem of plugins and is extremely popular among researchers, NGOs, and students.
- **GRASS GIS:** Another open-source option, very strong in geospatial analysis and image processing.

3. DATA: The Lifeblood of the System

- **Explanation:** Data is the fundamental resource. The famous phrase "Garbage In, Garbage Out" (GIGO) is especially true for GIS. The quality of the analysis is directly dependent on the quality of the input data.
- **Data Quality Considerations:**
 - **Accuracy:** How close the data is to the true value. Is the location of this school correct to within 10 meters?
 - **Precision:** The level of measurement detail. A coordinate with 6 decimal places is more precise than one with 2.
 - **Completeness:** Does the dataset include all the features it should? Is every road in the city represented?
 - **Consistency:** Is the data collected and formatted in a uniform way throughout the dataset?
 - **Lineage:** The history of the data—where it came from, how it was collected, and any transformations it has undergone.
- **Data Management Concept - The Geodatabase:** Modern GIS software uses a *geodatabase*, which is a structured database designed to store and manage both spatial and attribute data together efficiently. It can enforce rules and relationships between datasets.

4. PEOPLE: The Most Critical Component

- **Explanation:** Technology is useless without the human expertise to operate it, ask the right questions, and interpret the results. People design, implement, and use the system.

- **Roles in the GIS World:**

- **GIS Manager/Coordinator:** Oversees the entire GIS program, budgets, and strategy.
- **GIS Analyst:** The problem-solver. They design analytical procedures, run spatial models, and interpret the results to answer specific questions.
- **GIS Technician:** Focuses on data creation, digitizing, editing, and producing maps.
- **Database Administrator (DBA):** Manages the geodatabase, ensures its performance and security.
- **End-User:** The person who uses the final GIS products (maps, apps) to make decisions, like a town planner, a police chief, or a forest ranger.

5. METHODS/PROCEDURES: The Rulebook

- **Explanation:** These are the standardized plans, rules, and workflows that ensure the GIS is used consistently and accurately. It's the "how-to" guide for the organization.
- **What Methods Include:**
 - **Data Standards:** Defining a common coordinate system (e.g., WGS84, Minna Datum) for all data to ensure layers align correctly.
 - **Metadata Standards:** Rules for creating "data about the data" (metadata), which describes the source, accuracy, and content of a dataset.
 - **Standard Operating Procedures (SOPs):** Step-by-step instructions for common tasks, like how to update the road network or how to perform a specific analysis.
 - **Quality Assurance/Quality Control (QA/QC):** Processes for checking data and results for errors.

INTERRELATIONSHIP OF COMPONENTS

All five components are interdependent. For example:

- A **GIS Analyst (People)** uses a **Workstation (Hardware)** to run **QGIS (Software)** to analyze **Population Census Data (Data)** following a specific **SOP for Demographic Analysis (Methods)**.
- If any one component is weak (e.g., outdated software, poorly trained people, inaccurate data), the entire system fails to deliver reliable results.

EVALUATION (Week 3)

1. Create a table that summarizes the five components of GIS, providing a detailed description, specific examples, and the key function of each.
2. Explain the difference between a GIS *Technician* and a GIS *Analyst*.
3. Why is "Data" often considered the most expensive and challenging component of a GIS to maintain?
4. What is the purpose of "Methods/Procedures" in a GIS? What might happen if this component is ignored?

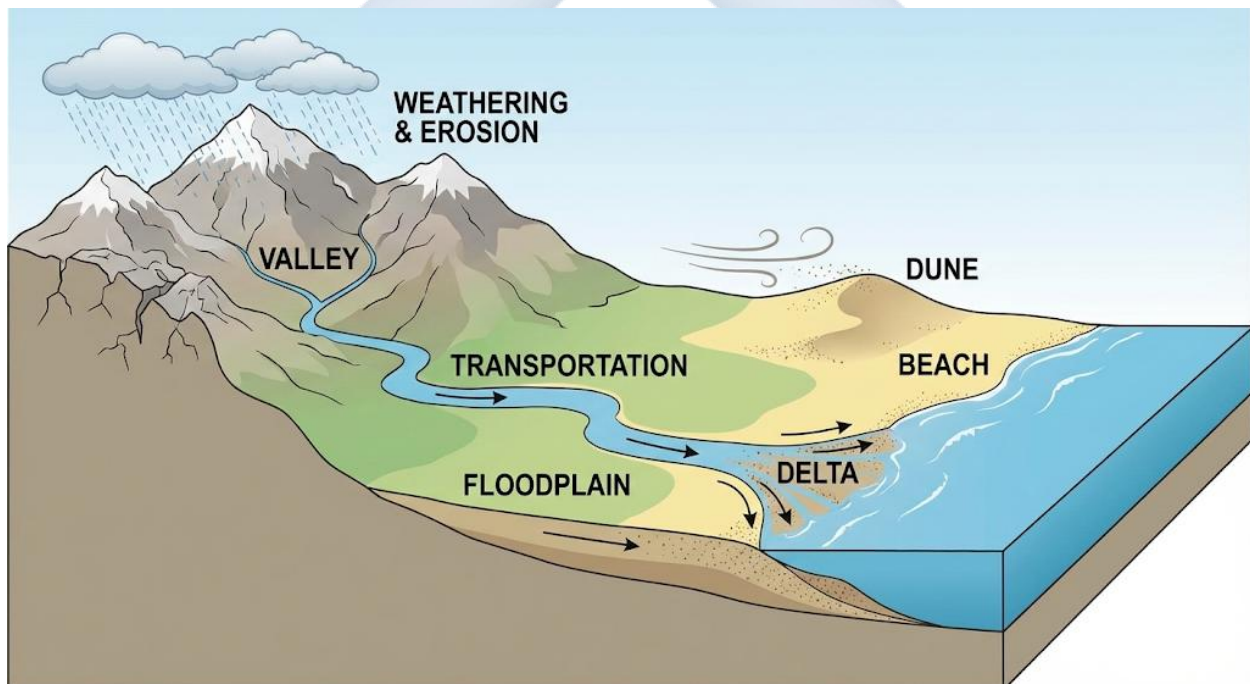
ASSIGNMENT (Week 3)

1. **Scenario Analysis:** You have been hired to set up a basic GIS unit for the Ministry of Agriculture in your state to track crop yields.
 - Propose the specific **Hardware** and **Software** you would acquire, justifying your choices (consider budget constraints).
 - Identify at least three key **Data** layers you would need to collect.
 - Suggest one important **Method/Procedure** you would implement to ensure data quality.
2. **Research:** Compare QGIS and ArcGIS. Create a list of two advantages and two disadvantages of each software platform.

WEEK 4: EARTH'S EXTERNAL PROCESSES AND LANDFORM DEVELOPMENT

CORE CONCEPT: The Great Sculptors of Our Planet

- **The Geological Battle:** The Earth's surface is a dynamic battlefield between two sets of forces:
 - **Internal Processes (Tectonics):** Build up the land, creating mountains, plateaus, and rift valleys. They are driven by the Earth's internal heat.
 - **External Processes (Denudation):** Wear down the land, sculpting the features created by tectonic forces. They are driven by solar energy and gravity.
- **The Denudation Cycle:** This is the continuous process of weathering, erosion, transportation, and deposition that levels the land over geological time.



I. EARTH'S EXTERNAL PROCESSES

1. WEATHERING: The In-Situ Breakdown

- **Concept:** The disintegration and decomposition of rocks *at or near the Earth's surface without movement*. It prepares rocks for erosion.

- **Types of Weathering:**

- **a) Mechanical/Physical Weathering:**

- **Process:** The physical breakdown of rock into smaller fragments without changing its chemical composition.
 - **Detailed Mechanisms:**
 - **Freeze-Thaw Action (Frost Wedging):** Water seeps into cracks, freezes (expanding by 9%), and exerts pressure, widening the cracks. Common in high-altitude areas like the Jos Plateau.
 - **Exfoliation (Onion Weathering):** In arid areas, rocks heat and expand by day, then cool and contract at night. This repeated stress causes the outer layers to peel off.
 - **Salt Crystallization:** In coastal areas, saltwater evaporates, leaving behind salt crystals that grow and exert pressure on rock pores.

- **b) Chemical Weathering:**

- **Process:** The decomposition of rocks through chemical reactions, altering their mineral composition.
 - **Detailed Mechanisms:**
 - **Hydrolysis:** Reaction between rock minerals and water. For example, feldspar in granite turns into kaolin clay, weakening the rock structure.
 - **Oxidation:** Reaction with oxygen. Iron-rich minerals "rust," giving rocks a reddish-brown colour (common in lateritic soils in Nigeria).
 - **Carbonation:** Rainwater (weakly acidic from CO₂) dissolves limestone and marble. This creates **karst topography** with features like:
 - **Limestone Pavements:** Flat, exposed rock with grikes (deep fissures) and clints (blocks).
 - **Sinkholes:** Collapsed surface into an underground cavity.
 - **Caves & Caverns:** Underground chambers formed by dissolved limestone.

- **Nigerian Example:** The **Somoro caves** in Kogi State are a classic example of carbonation and solution weathering.
- **c) Biological Weathering:**
 - **Process:** Breakdown caused by living organisms.
 - **Examples:**
 - **Plant Roots:** Growing into cracks and prying rocks apart.
 - **Burrowing Animals:** Earthworms and other creatures expose rock to other weathering agents.
 - **Microorganisms:** Lichens and fungi secrete acids that slowly dissolve rock.

2. EROSION, TRANSPORTATION & DEPOSITION: The Movement of Weathered Material

- **Concept:** These are a continuous sequence performed by geomorphic agents (water, wind, ice, waves).
- **The Sequence:**
 - a. **Erosion:** The *detachment and removal* of weathered material.
 - b. **Transportation:** The *movement* of eroded material.
 - c. **Deposition:** The *dropping* of transported material when the agent loses energy.

II. AGENTS OF DENUDATION AND THEIR LANDFORMS (Expanded)

1. RUNNING WATER (Fluvial Processes)

- **Context:** The most dominant and widespread agent of denudation, especially in Nigeria's humid and seasonal climates.
- **Erosional Landforms:**
 - **V-Shaped Valley:**
 - **Formation:** In a river's upper course, vertical erosion (downcutting) is dominant. The river's energy cuts downwards faster than the valley sides can be weathered, creating a characteristic V-shape.

- **Waterfalls & Gorges:**
 - **Formation:** A waterfall forms when a river flows over a band of resistant rock lying on top of softer rock. The softer rock is eroded more quickly (by hydraulic action and abrasion), undercutting the hard cap. Eventually, the caprock collapses, and the waterfall retreats upstream, leaving a steep-sided **gorge**. *Nigerian Example:* The **Gurara Waterfalls** in Niger State.
- **Interlocking Spurs:**
 - **Formation:** In the upper course, the river winds around areas of resistant rock, creating a series of 'interlocking' ridges that project from either side of the valley.
- **Depositional Landforms:**
 - **Floodplain:**
 - **Formation:** A wide, flat valley floor on either side of a river. Formed by the deposition of alluvium (silt) during floods.
 - **Levees:**
 - **Formation:** Natural, raised banks along a river channel. During floods, the heaviest material is deposited first at the river's edge, building up these embankments over time.
 - **Delta:**
 - **Formation:** When a river enters a standing body of water (sea/lake), its velocity drops, and it deposits its sediment load.
 - **Conditions Needed:**
 1. The river must carry a large sediment load.
 2. The rate of deposition must exceed the rate of removal by sea currents or tides.
 3. The sea must be relatively shallow at the river mouth.
 - *Nigerian Example:* The **Niger Delta** is one of the world's largest arcuate (bird's foot) deltas.

2. WIND (Aeolian Processes)

- **Context:** Dominant in arid and semi-arid regions (e.g., the Sahel region of Northern Nigeria).
- **Erosional Landforms:**
 - **Yardangs:**
 - **Formation:** Streamlined, ridge-like features carved from soft rock, aligned parallel to the prevailing wind direction. The wind abrades the softer rock, leaving the harder rock as ridges.
 - **Zeugen:**
 - **Formation:** Mushroom-shaped rocks with a hard cap rock protecting a softer stem from wind abrasion. The base is eroded more quickly.
 - **Deflation Hollows:**
 - **Formation:** Broad, shallow depressions formed when wind removes (deflates) loose, fine-grained sediment from the surface.
- **Depositional Landforms:**
 - **Sand Dunes:**
 - **Formation:** Mounds or ridges of sand deposited by wind. They form when wind hits an obstacle and loses energy.
 - **Types:**
 - **Barchan Dunes:** Crescent-shaped, with horns pointing downwind. Form in areas with limited sand.
 - **Seif Dunes:** Long, narrow, and ridge-like, parallel to the wind direction.

3. WAVES (Marine Processes)

- **Context:** Shape the coastline. Nigeria's coastline from Lagos to the Niger Delta is heavily influenced by wave action.
- **Erosional Landforms (at the *headland*, where rock resistance is higher):**
 - **Sequence of Formation (Cliff → Cave → Arch → Stack → Stump):**
 - i. Waves attack lines of weakness in a cliff, forming a **cave**.
 - ii. Two caves, back-to-back, may unite to form a natural **arch**.

iii. The arch collapses, leaving an isolated pillar of rock called a **stack**.

iv. The stack is further eroded into a **stump**, visible only at low tide.

- **Wave-Cut Platform:**

- **Formation:** A gently sloping, rocky surface at the base of a cliff. Formed as the cliff retreats landwards due to wave erosion at its base.

- **Depositional Landforms (in bays or where longshore drift occurs):**

- **Beaches:** Accumulations of sand and shingle deposited by waves.

- **Spit:**

- **Formation:** A long, narrow ridge of sand or shingle projecting from the mainland into the sea. Formed by longshore drift depositing material where the coastline changes direction.

- **Tombolo:**

- **Formation:** A spit that connects an island to the mainland. *Example:* The Ayre of Cara in Scotland (no major example in Nigeria, but the concept is key).

4. GLACIERS (Glacial Processes)

- **Context:** Not active in Nigeria, but crucial for understanding global landforms.

- **Erosional Landforms:**

- **U-Shaped Valley (Glacial Trough):**

- **Formation:** A glacier, being a massive body of ice, widens and deepens an existing V-shaped river valley, creating a characteristic U-shape with a flat floor and steep sides.

- **Cirque (Corrie):**

- **Formation:** A deep, armchair-shaped hollow on a mountainside where a glacier originated. Formed by freeze-thaw and plucking at the back wall.

- **Arete & Pyramidal Peak (Horn):**

- **Formation:** An **arete** is a sharp ridge formed when two cirques erode back-to-back. A **pyramidal peak** (like the Matterhorn) forms when three or more cirques erode a mountain from different sides.

- **Depositional Landforms:**

- **Moraine:** Ridges of unsorted rock and debris (till) carried and deposited by a glacier. Types include lateral, medial, and terminal moraines.

EVALUATION (Week 4 - Enhanced)

1. Describe, with the aid of a series of labelled diagrams, the formation of a stack and stump from a headland.
2. Explain TWO key differences between the processes of weathering and erosion.
3. "The Niger Delta is a gift of the River Niger to Nigeria." Using your knowledge of fluvial deposition, explain this statement by describing the conditions necessary for delta formation.
4. Compare and contrast the formation of a V-shaped valley (fluvial) and a U-shaped valley (glacial).

ASSIGNMENT (Week 4 - Applied)

1. **Fieldwork/Research Project:** "A Study of Weathering in Our Local Environment."
 - Students are to find, photograph, and identify three different examples of weathering in their community (e.g., potholes - mechanical, rusting metal gates - chemical, cracked walls by roots - biological).
 - For each example, write a short report describing the feature, the type of weathering, and the process causing it.
 2. **Diagrammatic Assignment:** Draw a fully annotated and labelled diagram of a river's course from its source to its mouth, showing the following features: V-shaped valley, interlocking spurs, waterfall, meander, floodplain, levee, and delta.
-

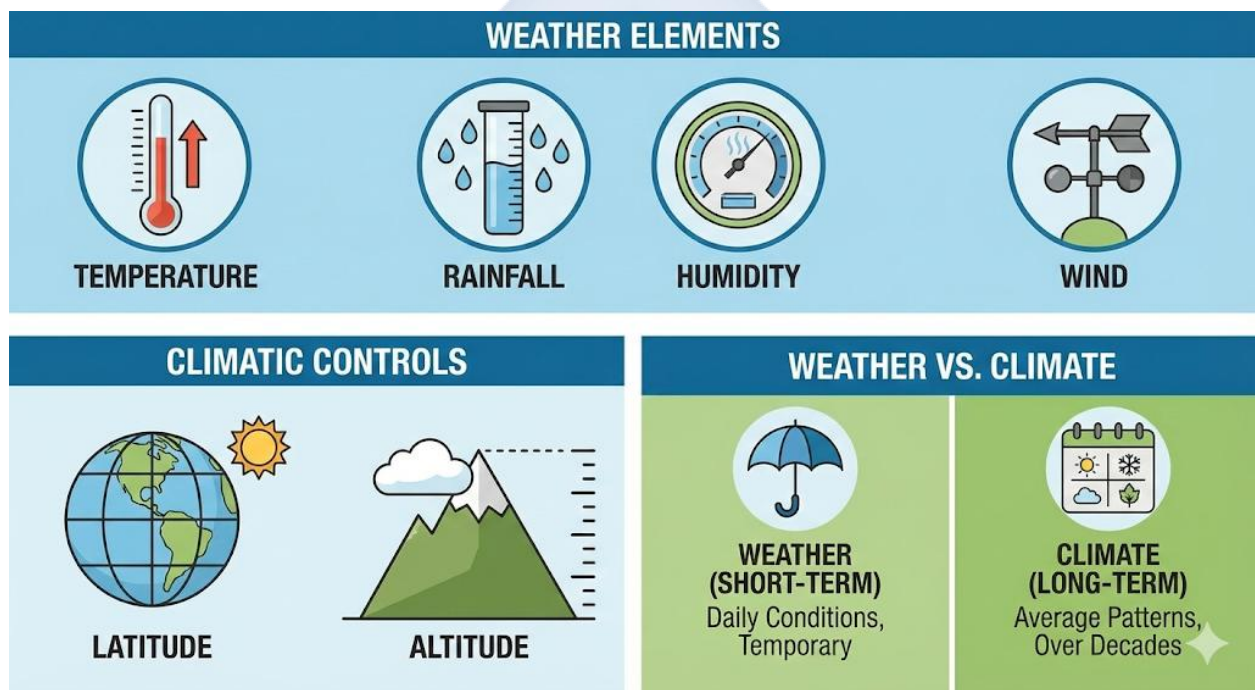
WEEK 6: CLIMATE

CORE CONCEPT: Understanding the World's Weather Patterns

- **Climate vs. Weather Revisited:**

- **Weather:** The atmospheric conditions at a specific time and place (e.g., "It is raining in Abuja today.").
- **Climate:** The long-term average of weather conditions (over 30-35 years) for a particular region (e.g., "Nigeria has a tropical savanna climate.").

- **The Global Climate System:** Climate is the result of the complex interplay between the atmosphere, hydrosphere (oceans), lithosphere (land), and cryosphere (ice).



MAJOR CLIMATIC TYPES OF THE WORLD (Using the Köppen System as a Framework)

1. TROPICAL CLIMATES (Group A)

- **Driving Force:** The Inter-Tropical Convergence Zone (ITCZ) – a belt of low pressure near the equator where the Northeast and Southeast Trade Winds converge, causing intense heating and convectional rainfall.

○ **Af - Tropical Rainforest Climate:**

- **Location & Cause:** 5°-10° N&S of the equator (Amazon Basin, Congo Basin, S.E. Asia). The ITCZ is permanently or nearly permanently overhead.
- **Characteristics:**
 - **Temperature:** High and uniform all year (26°C - 28°C). Very small annual range (~2°C).
 - **Rainfall:** Heavy (over 2000 mm) and convectional, falling almost every day in the afternoon. No dry season.
 - **Diurnal Range:** The difference between day and night temperatures is greater than the difference between seasons.
- **Vegetation:** Dense, multi-layered evergreen rainforest with incredible biodiversity.
- **Nigerian Example:** The climate of Calabar and parts of the Niger Delta.

○ **Aw - Tropical Savannah Climate:**

- **Location & Cause:** 5°-15° N&S of the equator (Surrounding the rainforest zone). Nigeria's dominant climate. Governed by the *seasonal migration of the ITCZ*.
- **Characteristics:**
 - **Temperature:** High throughout the year, but with a more pronounced peak just before the rainy season.
 - **Rainfall:** Distinct Wet & Dry Seasons.
 - **Wet Season:** When the ITCZ (and its associated rain belts) moves into the area. Heated by the overhead sun, causing convectional rainfall and thunderstorms.
 - **Dry Season:** When the ITCZ moves away, and the area comes under the influence of the dry, dusty **Continental Tropical (cT) air mass** – the Harmattan wind.
- **Vegetation:** Savanna grassland with scattered drought-resistant trees (e.g., acacia, baobab).
- **Nigerian Example:** Ibadan, Abuja, Enugu.

2. DRY CLIMATES (Group B)

- **Driving Force:** The Sub-Tropical High-Pressure Belt (STHP), where air is sinking, warming, and inhibiting cloud formation.
 - **BWh - Hot Desert Climate:**
 - **Location & Cause:** Centred on the Tropics of Cancer and Capricorn (Sahara, Arabian, Australian Deserts). Permanently under the influence of the STHP.
 - **Characteristics:**
 - **Temperature:** Very high daytime temperatures (can exceed 50°C). Nights can be very cold due to rapid radiation loss under clear skies. *Extreme diurnal range.*
 - **Rainfall:** Very low and unreliable (<250 mm per year).
 - **Vegetation:** Sparse xerophytic (drought-adapted) plants like cacti and thorny shrubs.
 - **BSh - Tropical Steppe (Semi-Arid) Climate:**
 - **Location:** Fringes of the desert (e.g., Sahel region).
 - **Characteristics:** Slightly more rainfall (250-500 mm) than the desert, but highly variable. Prone to droughts.
 - **Nigerian Example:** Maiduguri, Sokoto.

3. TEMPERATE CLIMATES (Group C)

- **Driving Force:** The battle between tropical and polar air masses, influenced by the Westerly winds.
 - **Cs - Mediterranean Climate:**
 - **Location & Cause:** Between 30°-40° N&S, on western margins (California, Central Chile, Cape Town, S. Europe). Seasonal shift in pressure belts.
 - **Characteristics:**
 - **Summer:** Hot and dry (influenced by the STHP).

- **Winter:** Mild and wet (influenced by the migrating Polar Front and associated cyclones).
 - **Vegetation:** Sclerophyllous (hard-leaved), evergreen shrubs and trees (e.g., olive trees, cork oak) adapted to summer drought.
- **Cfa - Humid Subtropical Climate:**
 - **Location:** Eastern sides of continents (S.E. USA, S.E. China, S.E. South America).
 - **Characteristics:** Hot, humid summers and mild winters. Rainfall throughout the year, often with a summer maximum from convectional storms.

4. CONTINENTAL & POLAR CLIMATES (Groups D & E)

- **Dfb - Humid Continental Climate:** Found in the interiors of large continents (e.g., N. USA, Russia). Warm summers, very cold winters. Four distinct seasons.
- **ET - Tundra Climate:** Found in the Arctic/Antarctic regions. Very short, cool summers; long, severely cold winters. Permanently frozen subsoil (permafrost). Vegetation: mosses, lichens, low shrubs.

FACTORS AFFECTING CLIMATE (Expanded)

1. **Latitude:** Determines the angle of the sun and the amount of solar energy received. Low latitudes (near equator) are hot, high latitudes (poles) are cold.
2. **Altitude:** Temperature decreases with height (**Environmental Lapse Rate** ~6.5°C per 1000m). This is why the Obudu Plateau is cooler than Calabar.
3. **Distance from the Sea (Continentality):** Land heats and cools faster than water. Coastal areas have a **maritime climate** (small temperature range). Inland areas have a **continental climate** (large temperature range).
4. **Ocean Currents:**
 - **Warm Currents** (e.g., Gulf Stream): Warm and moisten adjacent coasts, raising winter temperatures.
 - **Cold Currents** (e.g., Benguela, Canary): Cool and dry adjacent coasts, often causing coastal fog and desert conditions (e.g., Namib Desert).

5. **Prevailing Winds:** Determine the characteristics of the air mass affecting a region (e.g., the moist South-West monsoon vs. the dry Harmattan).
6. **Aspect:** In mountainous regions, the slope facing the sun (south-facing in the N. Hemisphere) is warmer than the shaded slope.

CLIMATIC REGIONS OF NIGERIA (A Synthesis)

- **Concept:** Nigeria is a perfect case study of a **latitudinal climate gradient**, from the humid south to the arid north, driven primarily by the seasonal migration of the ITCZ.

Climatic Zone	Köppen	Location	Rainfall & Seasons	Vegetation	Example
Tropical Rainforest	Af	Coastal & S.E.	>2000mm. Very short or no dry season.	Evergreen Rainforest	Calabar, Bonny
Tropical Rainforest	Am	Coastal Belt	1500-2000mm. Slightly longer dry season.	Moist Evergreen Forest	Lagos, Port Harcourt
Guinea Savannah	Aw	Middle Belt	1000-1500mm. Distinct wet & dry seasons.	Tall Grass, Scattered Trees	Ibadan, Abuja
Sudan Savannah	Aw	Northern Zone	600-1000mm. Longer, more severe dry season.	Short Grass, Shorter Trees	Kano, Sokoto
Sahel Savannah	BSh	Extreme North	250-600mm. Very long dry season, unreliable rain.	Semi-Desert Scrub, Thorny Bushes	Maiduguri, Nguru

EVALUATION (Week 6 - Analytical)

1. Draw a labelled diagram to explain the seasonal rainfall pattern in the Tropical Savannah (Aw) climate of Nigeria, making reference to the movement of the ITCZ and the Harmattan wind.
2. Compare and contrast the Tropical Rainforest (Af) climate and the Hot Desert (BWh) climate under the following headings: (a) Temperature pattern, (b) Rainfall amount and seasonality, (c) Natural vegetation.
3. Explain how **altitude** has modified the climate of the Jos Plateau compared to the surrounding lowlands.
4. Using specific examples, explain how **ocean currents** can influence the climate of a coastal region.

ASSIGNMENT (Week 6 - Applied)

1. Climograph Analysis:

- Students are provided with climographs (temperature and rainfall graphs) for two cities: one in the Mediterranean (e.g., Rome) and one in the Tropical Savannah (e.g., Kano).
- **Task:** Write a report analyzing the two graphs, identifying the climate type for each, and explaining the differences in their temperature and rainfall patterns with reference to the climatic factors learned.

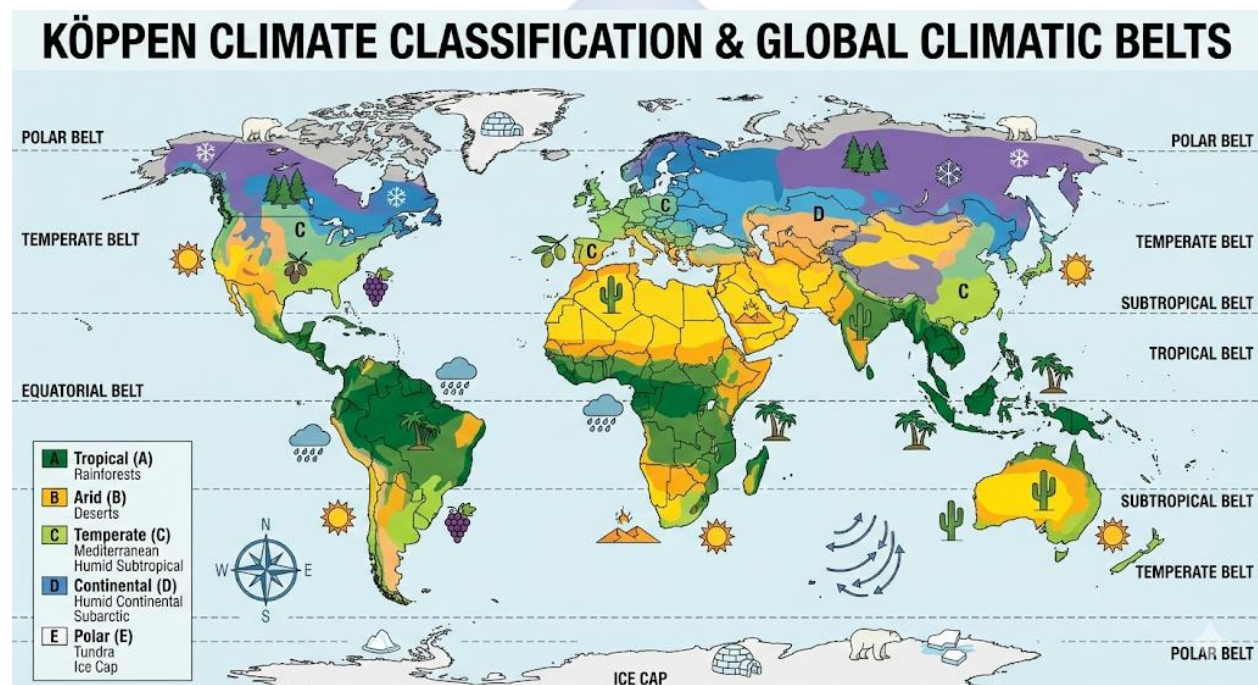
2. Research Project: "The Impact of Climate on Life in Nigeria."

- Choose ONE climatic zone in Nigeria (e.g., Sahel, Rainforest).
 - Research and write a report on how this climate influences:
 - Agricultural practices.
 - Types of houses built.
 - Cultural festivals or economic activities.
 - Environmental challenges (e.g., desertification, flooding).
-

WEEK 8: CLIMATIC CLASSIFICATION

CORE CONCEPT: Making Sense of Global Weather Patterns

- **Why Classify Climate?** The Earth has an incredible diversity of weather patterns. Climatic classification is the geographer's tool for organizing this complexity into a manageable and understandable system. It helps us identify patterns, predict conditions, understand ecosystems, and plan for the future.
- **The Goal:** To group regions with similar climatic characteristics, allowing for comparison, analysis, and prediction on a global scale.



DEFINITION OF CLIMATIC CLASSIFICATION

- **Core Definition:** The systematic categorization of the world's climates based on long-term atmospheric conditions (typically over 30-35 years), using specific, measurable criteria such as temperature, precipitation, and their annual cycles.
- **Key Idea:** It moves beyond describing single weather events to defining the *statistical average* and *extremes* of atmospheric conditions that characterize a region.

BASES FOR CLIMATIC CLASSIFICATION (The "What" We Measure)

- **1. Temperature:**
 - **Measures:** Mean annual temperature, average temperature of the warmest and coldest months, and annual temperature range.
 - **Significance:** Determines the potential for evaporation, the types of crops that can grow, and the need for heating or cooling.
- **2. Precipitation:**
 - **Measures:** Total annual precipitation, its seasonal distribution (e.g., winter-dry, summer-dry, uniform), and its reliability.
 - **Significance:** The single most critical factor for determining life zones (biomes). It dictates water availability for agriculture, industry, and domestic use.
- **3. Vegetation (as an Indicator):**
 - **Concept:** Natural vegetation is a visible, integrated expression of the long-term climate. The presence of rainforests, deserts, or grasslands directly reflects temperature and rainfall patterns. This is a key principle in the Köppen system.
- **4. Air Masses and Pressure Belts:**
 - **Concept:** This is a *causal* basis. Classifying climates based on the dominant air masses (e.g., maritime tropical, continental polar) and pressure systems (e.g., ITCZ, Sub-Tropical High) that control them. This explains *why* a climate is the way it is.
- **5. Evapotranspiration:**
 - **Concept:** Used in more advanced systems like Thornthwaite's. It measures the total water loss from the soil (evaporation) and plants (transpiration). A high evapotranspiration rate in a hot region means even moderate rainfall can be insufficient, leading to arid conditions.

MAJOR CLIMATIC CLASSIFICATION SYSTEMS (The "How" We Classify)

1. Greek/Classical Classification (The Ancient View)

- **Basis:** Purely **latitude** and the angle of the sun.
- **Zones:**

- **Torrid Zone (Tropical):** Between the Tropic of Cancer and Capricorn. Believed to be perpetually hot.
- **Temperate Zones:** Between the tropics and the polar circles. Believed to have moderate, habitable climates.
- **Frigid Zones (Polar):** Within the polar circles. Believed to be perpetually cold and frozen.
- **Critical Analysis & Limitations:**
 - **Oversimplification:** It ignores the profound effects of altitude, ocean currents, and continentality. For example, Nairobi (on the equator in Kenya) has a temperate climate due to its high altitude, and the UK (at 55°N) is much warmer than it should be due to the Gulf Stream.
 - **Conclusion:** While historically important, it is geographically inaccurate and not used in modern science.

2. Köppen-Geiger Climatic Classification (The Modern Standard)

- **Basis:** Temperature and Precipitation, with natural vegetation as a key indicator.
- **Structure:** A logical, letter-based code system.
 - **First Letter:** Main climate group (5 major groups).
 - **Second Letter:** Precipitation pattern.
 - **Third Letter:** Temperature pattern.
- **Detailed Breakdown of the System:**

1st Letter (Main Group)	2nd Letter (Precipitation)	3rd Letter (Temperature)
A - Tropical ($T_{cold} \geq 18^{\circ}\text{C}$)	f - Fully humid (no dry season)	h - Hot arid (low-latitude)
B - Arid (Evaporation > Precipitation)	m - Monsoonal (short dry season)	k - Cold arid (mid-latitude)
C - Temperate ($-3^{\circ}\text{C} < T_{cold} < 18^{\circ}\text{C}$)	s - Dry summer (Mediterranean)	a - Hot summer ($T_{hot} \geq 22^{\circ}\text{C}$)
D - Continental ($T_{cold} \leq -3^{\circ}\text{C}$)	w - Dry winter (e.g.,	b - Warm summer ($T_{hot} <$

1st Letter (Main Group)	2nd Letter (Precipitation)	3rd Letter (Temperature)
	Savanna)	22°C)
E - Polar (Thot < 10°C)		c - Cool summer (< 4 months > 10°C)

- **Illustrative Examples:**

- **Af:** *Tropical Rainforest*. Hot all year, heavy rain all year (e.g., Amazon, Congo).
- **Aw:** *Tropical Savannah*. Hot all year, with a distinct winter dry season (e.g., Most of Nigeria, Brazilian Highlands).
- **BWh:** *Hot Desert*. Low rainfall, hot all year (e.g., Sahara, Arabian Desert).
- **Csa:** *Mediterranean*. Hot, dry summers; mild, wet winters (e.g., Southern Italy, California).
- **Dfb:** *Humid Continental*. Warm summers, severe, snowy winters (e.g., Ontario, Russia).
- **ET:** *Tundra*. Very short, cool summers; long, severe winters (e.g., Alaska, Northern Siberia).
- **Strengths:** Quantitative, widely used, excellent for mapping and general comparison.
- **Weaknesses:** Can be too broad, doesn't account for climate causes like wind patterns.

3. Thornthwaite's Classification (The Water-Balance Approach)

- **Basis: Moisture Efficiency**, using a complex calculation of Precipitation (P) and Potential Evapotranspiration (PE).
- **Core Concept:** It's not just how much rain falls, but how much is available for plants after accounting for loss to the atmosphere.
 - **Moisture Index (Im):** If $P > PE$, the climate is humid. If $PE > P$, the climate is arid.
- **Significance:** This system is highly valued in **agriculture, hydrology, and water resource management** because it directly addresses water surplus and deficit.
- **Comparison with Köppen:** While Köppen is better for general geography and mapping, Thornthwaite is more practical for understanding agricultural potential and drought risk.

NIGERIA IN THE GLOBAL CLASSIFICATION (A Case Study)

- **Application of Köppen:**
 - **South (Coastal): Af and Am** (Tropical Rainforest). E.g., Lagos, Port Harcourt, Calabar.
 - **Central & Most of the Country: Aw** (Tropical Savannah). E.g., Ibadan, Abuja, Enugu, Kano.
 - **Extreme North-East: BSh** (Hot Semi-Arid/Steppe). E.g., Maiduguri.
- **The Causal Factor (The "Why"):** This latitudinal pattern is a direct result of Nigeria's position astride the migrating **Inter-Tropical Convergence Zone (ITCZ)**, which brings moist maritime air (rain) when it is overhead and dry, dusty continental air (Harmattan) when it moves away.

EVALUATION (Week 8 - Analytical)

1. Explain TWO major limitations of the Greek system of climatic classification that make it unsuitable for modern use.
2. Decode the following Köppen climate codes and name one country where each is found: **Aw, Csa, BWh, Dfb**.
3. Why is the Thornthwaite classification system considered more useful than Köppen's for agricultural planning?
4. Using the Köppen system, classify the climate of your school's location and justify your answer with reference to local temperature and rainfall patterns.

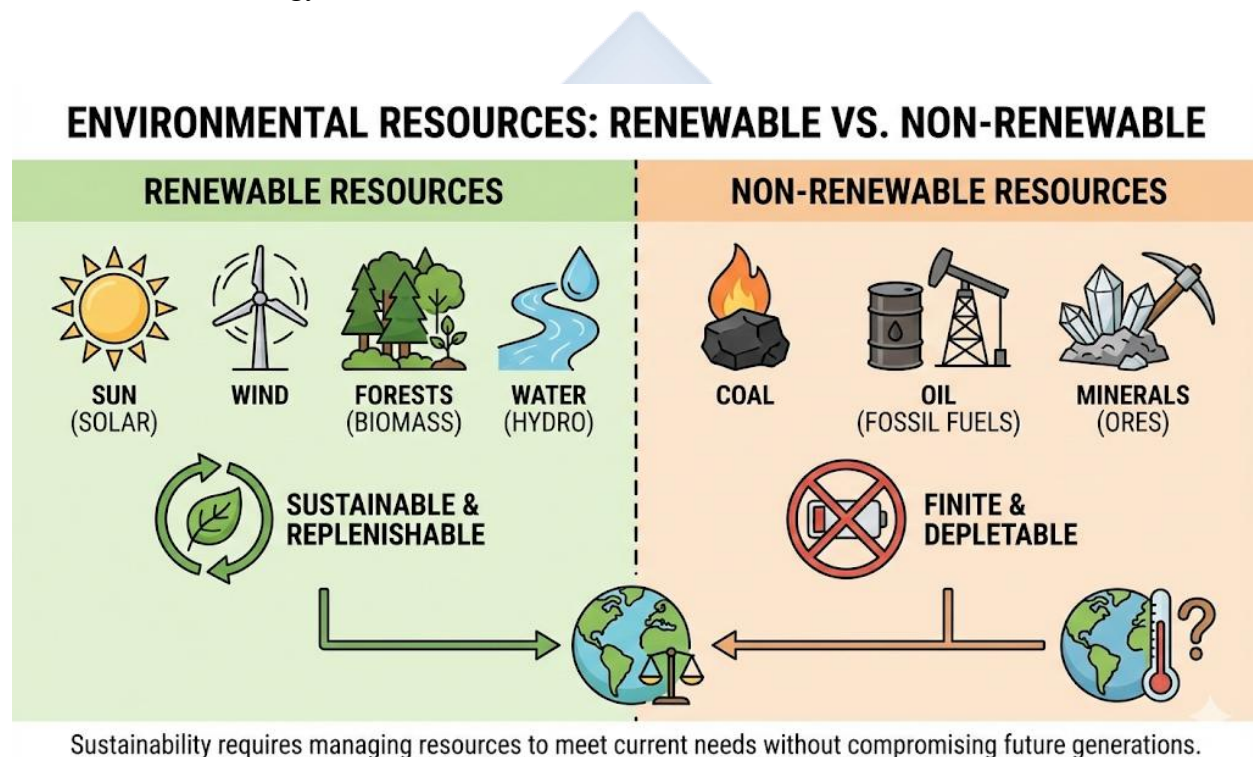
ASSIGNMENT (Week 8 - Applied)

1. **Map Activity:** Draw a world map and colour-code it to show the distribution of the major Köppen climate groups (A, B, C, D, E). Annotate the map with labels for at least two specific regions for each group.
2. **Research and Debate:** "Is the Köppen system still the best tool for classifying climate in the 21st century?"
 - Students are to research a modern alternative (e.g., the Bergeron classification based on air masses, or a modified Köppen system).
 - Prepare a two-minute argument for or against the motion.

WEEK 9: ENVIRONMENTAL RESOURCES

CORE CONCEPT: The Foundation of Human Civilization

- **Definition:** Environmental resources are all the materials and energy sources found in the natural environment that are useful to humans in fulfilling their needs and wants.
- **Key Idea:** The value of a resource is not fixed; it is defined by **human technology, culture, and economic systems**. For example, uranium was merely a rock until the 20th century; now it is a critical energy resource.



CRITICAL DISTINCTIONS: Resource vs. Reserve

- **Resource:** The total amount of a material that exists in the earth, whether discovered or not. (e.g., All the gold in the Earth's crust).
- **Reserve:** The portion of a resource that has been **discovered, can be extracted with current technology, and is economically profitable to extract**. (e.g., The gold in a specific, operational mine).

- **Concept:** Reserves can shrink or grow based on new technology, market prices, and new discoveries.

CLASSIFICATION OF ENVIRONMENTAL RESOURCES (A Multi-layered View)

A. Based on Renewability (The Most Critical Classification)

- **1. Renewable Resources:**

- **Definition:** Resources that can be replenished naturally over relatively short periods (on a human timescale).
- **The Critical Caveat:** They are only renewable if used at a rate slower than their rate of regeneration. Overuse can destroy them.
- **Examples:**
 - **Perpetual/Inexhaustible:** Solar energy, wind, tides, flowing water.
 - **Potentially Renewable:** Forests, freshwater, soil, fish stocks, clean air. These require careful management.

- **2. Non-Renewable Resources:**

- **Definition:** Resources that exist in a fixed quantity or are consumed much faster than nature can replenish them. They are, for all practical purposes, finite.
- **Sub-categories:**
 - **Fossil Fuels:** Formed over millions of years from organic matter (e.g., Crude oil, natural gas, coal).
 - **Minerals:** Metallic (e.g., iron ore, gold, tin) and non-metallic (e.g., limestone, clay, gemstones).

B. Based on Origin

- **Biotic Resources:** Obtained from the biosphere (living and organic material).
 - *Examples:* Forests, animals, crops, fossil fuels (derived from ancient life).
- **Abiotic Resources:** Non-living, inorganic materials.
 - *Examples:* Minerals, rocks, air, water, sunlight.

TYPES AND IMPORTANCE OF ENVIRONMENTAL RESOURCES (With Nigerian Context)

Resource Type	Sub-types & Nigerian Examples	Importance & Specific Uses
Mineral	Energy: Crude Oil (Niger Delta), Natural Gas, Coal (Enugu). Metallic: Iron Ore (Kogi), Tin & Columbite (Jos). Non-Metallic: Limestone (Ewekoro), Kaolin, Bitumen (Ondo).	Economic Backbone: Oil provides ~90% of foreign exchange. Industrial Raw Materials: Limestone for cement, iron ore for steel. Employment: Mining and processing industries.
Water	Surface: Rivers (Niger, Benue), Lakes (Kainji). Groundwater: Aquifers, Boreholes.	Life-Sustaining: Drinking, sanitation. Agriculture: Irrigation for dry-season farming. Energy: Hydroelectric power (Kainji, Jebba dams). Transportation: Inland waterways.
Forest	High Forest: Mahogany, Iroko (Cross River, Ondo). Savanna Woodland: Shea, Baobab.	Biodiversity: Habitat for unique flora and fauna. Economic: Timber, fuelwood, medicinal plants. Environmental Services: Carbon sequestration, soil conservation, climate regulation.
Soil/Land	Agricultural Land: Fertile soils of the Middle Belt. Clay/Sand: For construction and industry.	Food Security: Basis for agriculture. Settlement & Infrastructure: Land for housing, roads, industries.
Atmospheric	Sunlight, Wind, Rainfall.	Energy: Solar panels, wind turbines. Agriculture: Rain-fed agriculture. Climate: Determines living conditions and ecosystems.

PROBLEMS OF RESOURCE DEVELOPMENT IN NIGERIA (A Deeper Look)

1. The "Resource Curse" (Dutch Disease):

- **Concept:** An over-reliance on a single non-renewable resource (oil) leads to the neglect of other sectors (like agriculture), currency appreciation that hurts other exports, corruption, and economic instability.

2. Environmental Degradation:

- **Case Study - Niger Delta:** Oil spills poison farmland and aquatic ecosystems, gas flaring causes air pollution and acid rain, leading to loss of livelihoods (fishing, farming) and health problems.
- **Case Study - Northern Nigeria:** Deforestation for fuelwood and overgrazing accelerate desertification and soil erosion.

3. Policy Inconsistency and Corruption:

Lack of transparent management of resource revenues, leading to mismanagement and funds not benefiting the general population.

4. Technological Limitations:

Inadequate technology for deep mining, efficient processing, and exploring new reserves limits the optimal utilization of resources.

5. Overdependence on Crude Oil:

Makes the national economy highly vulnerable to global oil price shocks.

RESOURCE CONSERVATION AND SUSTAINABLE DEVELOPMENT

• Conservation vs. Preservation:

- **Conservation:** The *sustainable use and management* of natural resources to prevent exploitation, destruction, or neglect. (E.g., Selective logging, regulated fishing quotas).
- **Preservation:** Protecting nature from *any* human use to maintain it in its pristine state. (E.g., Designating a strict nature reserve).

• Strategies for Nigeria:

- a. **Diversification of the Economy:** Reducing dependence on oil by investing in agriculture, manufacturing, and tourism.
- b. **Afforestation and Reforestation:** Massive tree-planting campaigns and sustainable forestry practices to combat deforestation.

- c. **Investment in Renewable Energy:** Harnessing Nigeria's vast solar, wind, and hydro potential to provide clean energy and reduce gas flaring.
- d. **Strengthening Regulations and Enforcement:** Empowering agencies like NESREA (National Environmental Standards and Regulations Enforcement Agency) to hold polluters accountable.
- e. **Promoting Recycling and Waste Management:** Developing a circular economy to reduce pressure on virgin resources.
- f. **Public Awareness and Education:** Fostering a culture of environmental stewardship from a young age.

EVALUATION (Week 9 - Critical Thinking)

1. Distinguish between a *resource* and a *reserve*, using crude oil in Nigeria as an example.
2. Explain why forests and fisheries are classified as "potentially renewable" rather than "perpetual" resources. What human action can make them non-renewable?
3. Analyze the statement: "Nigeria is rich in environmental resources, but this wealth has become a curse." Do you agree? Justify your answer.
4. Propose THREE specific, actionable strategies the government could implement to promote the conservation of soil resources in the Sudan Savanna region of Nigeria.

ASSIGNMENT (Week 9 - Solutions-Focused)

1. **Project: "A Sustainable Development Plan for My State":**
 - Students will choose their state of origin or residence.
 - They will identify ONE key environmental resource in that state (e.g., Limestone in Ogun, Crude Oil in Bayelsa, Agricultural Land in Benue).
 - They will then create a brief report outlining:
 - The current economic importance of the resource.
 - Two major environmental or social problems linked to its exploitation.
 - A detailed proposal for how to manage this resource more sustainably for future generations.

2. **Debate Preparation:** "This house believes that the discovery of oil in the Niger Delta has done more harm than good to the people of the region."

- Students will be divided into proposers and opposers to research and prepare arguments for a formal debate.



SS2 GEOGRAPHY LESSON NOTES

SECOND TERM

1. Renewable and Non-Renewable Resources
2. Environmental Problems
3. Environmental Conservation
4. Agriculture
5. Midterm Examination
6. Transport and Communication in Nigeria
7. Midterm break
8. Manufacturing Industries in Nigeria
9. Commercial Activities in Nigeria

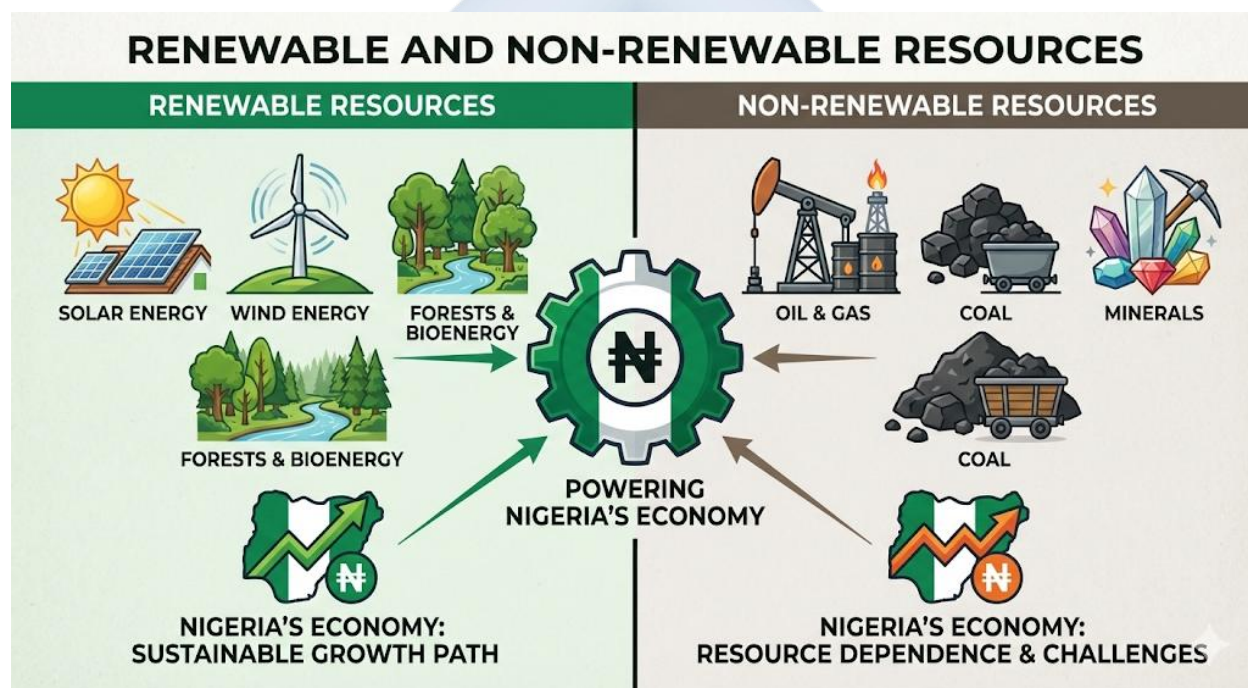


WEEK 1 - RENEWABLE AND NON-RENEWABLE RESOURCES

Lesson Duration:

Learning Objectives: By the end of this lesson, students should be able to:

1. Define environmental resources.
2. Differentiate between renewable and non-renewable resources with clear examples.
3. Analyze the importance and limitations of each resource type.
4. Explain the concept of sustainability in resource management.



1. INTRODUCTION: OUR DEPENDENCE ON NATURE

Imagine a day without sunlight, water, petrol for vehicles, or wood for furniture. Our daily lives and the entire global economy are built upon materials provided by our environment. These naturally occurring, useful materials are called **Environmental Resources**. They are the foundation for our survival, economic development, and overall well-being. To manage these

resources wisely, we must understand their nature, particularly their ability to replenish themselves.

Key Concept: Resources are not just "things"; they are valuable assets that require careful management for present and future generations.

2. MEANING AND CLASSIFICATION OF RESOURCES

Environmental Resources are all the biotic (living) and abiotic (non-living) components of the environment that are accessible and useful to humans for satisfying their needs and wants. This includes water, air, soil, minerals, forests, wildlife, and even sunlight.

Classification Based on Renewability: The most critical way to classify resources is by their rate of replenishment.

1. **Renewable Resources**
 2. **Non-Renewable Resources**
-

3. RENEWABLE RESOURCES

Definition: Natural resources that can be replenished or regenerated naturally within a relatively short period relative to a human lifespan. They are often called "flow" resources.

Key Insight: While these resources are replenishable, they are **not inexhaustible**. If used faster than they can be replaced (e.g., overfishing, deforestation), they can be depleted or degraded.

Examples and Detailed Explanations:

- **Solar Energy:** Energy from the sun. It is the most abundant energy source. Used via solar panels (photovoltaics) for electricity and solar thermal systems for heating.
- **Wind Energy:** Kinetic energy from moving air. Harnessed by wind turbines that convert it into electricity. Effectiveness is location-dependent.
- **Hydropower:** Energy from flowing water. Dams capture the energy of falling water to spin turbines and generate electricity.

- **Forests/Timber:** Trees can be replanted and grown. Sustainable forestry involves selective logging and replanting to maintain the forest cover.
- **Biomass:** Organic material from plants and animals (e.g., wood, agricultural waste, biogas). It can be burned for heat or converted to biofuels.
- **Geothermal Energy:** Heat derived from the Earth's subsurface. Used for heating buildings or generating electricity in geologically active areas.

Characteristics:

- **Sustainable:** Can be used indefinitely with proper management.
- **Low Pollution:** Generally have a lower carbon footprint and cause less air and water pollution than fossil fuels.
- **Often Intermittent:** Some, like solar and wind, are not constant and require energy storage solutions.

4. NON-RENEWABLE RESOURCES

Definition: Natural resources that exist in a fixed quantity or that are consumed much faster than nature can create them. Their formation often takes millions of years.

Key Insight: These are **stock** resources. Once used, they are essentially gone for all practical purposes, making them finite and exhaustible.

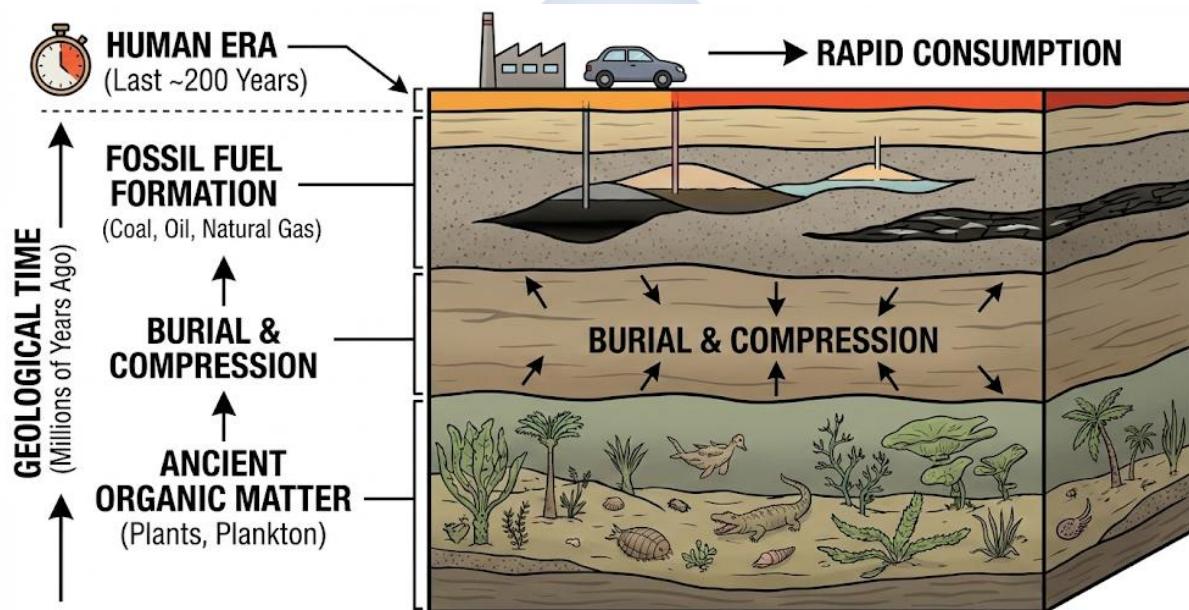
Examples and Detailed Explanations:

- **Fossil Fuels (Crude Oil, Natural Gas, Coal):** Formed from the decomposed remains of ancient plants and organisms over millions of years under intense heat and pressure. They are the primary source of the world's energy but are major contributors to greenhouse gas emissions.
- **Minerals (Metallic):** Such as iron ore (for steel), copper (for wiring), gold, and tin. Mined from the earth's crust.
- **Minerals (Non-Metallic):** Such as limestone (for cement), sand, and gravel.

- **Nuclear Fuels:** Such as Uranium. While the energy produced is immense, the mined uranium itself is finite, and radioactive waste poses long-term storage challenges.

Characteristics:

- **Finite:** Limited in supply and will eventually run out.
- **High Environmental Impact:** Extraction (mining, drilling) and consumption (burning) often cause significant pollution, habitat destruction, and greenhouse gas emissions.
- **High Energy Density:** Provide a large amount of energy per unit (e.g., a small amount of petrol can power a car for miles).



FINITE NATURE: Formed over millions of years, consumed in a geological blink.

5. IN-DEPTH DIFFERENCES: RENEWABLE VS. NON-RENEWABLE

Feature	Renewable Resources	Non-Renewable Resources
Replenishment	Naturally replenished over short periods.	Finite; cannot be replenished within a human timescale.
Availability	Virtually unlimited if managed sustainably.	Limited and depleting with use.
Environmental	Generally low; "clean" or	High (air/water pollution, greenhouse

Feature	Renewable Resources	Non-Renewable Resources
Impact	"green" energy.	gases, habitat loss).
Sustainability	Sustainable with proper management.	Inherently unsustainable in the long term.
Cost Structure	High initial investment, low operating cost.	Costs can fluctuate with extraction difficulty and market volatility.
Examples	Sunlight, wind, forests, water.	Crude oil, coal, natural gas, gold.

6. IMPORTANCE AND LIMITATIONS

A. Importance of Renewable Resources:

- **Energy Security:** Reduces dependence on imported fossil fuels.
- **Climate Change Mitigation:** Key to reducing greenhouse gas emissions.
- **Public Health:** Leads to cleaner air and water.
- **Economic Diversification:** Creates jobs in new technology sectors.

B. Limitations of Renewable Resources:

- **Intermittency:** Solar and wind power are weather-dependent.
- **High Initial Capital:** Cost of technology (e.g., solar panels, wind turbines) can be prohibitive.
- **Land Use:** Large-scale solar or wind farms require significant space.
- **Geographic Limitations:** Not all regions are suitable for all types of renewable energy.

C. Importance of Non-Renewable Resources:

- **Current Energy Backbone:** Still supply the majority of the world's energy.
- **Economic Development:** Crucial for industrialization, manufacturing, and infrastructure.
- **Revenue Generation:** A primary source of export earnings for countries like Nigeria.
- **Versatile Products:** Basis for plastics, fertilizers, chemicals, and pharmaceuticals.

D. Limitations of Non-Renewable Resources:

- **Resource Depletion:** They will inevitably run out.
 - **Environmental Degradation:** From oil spills and mine tailings to air pollution and acid rain.
 - **Price Volatility:** Prone to geopolitical and market shocks.
 - **Social Conflicts:** Can lead to displacement of communities and resource-based conflicts.
-

7. EVALUATION & ASSESSMENT

Short Answer:

3. Define "environmental resources."
4. List three examples of renewable resources and three of non-renewable resources.
5. State two key characteristics of non-renewable resources.

Critical Thinking: 4. "Solar energy is a renewable resource, but it is not without its challenges." Discuss two limitations of solar energy. 5. Why is the concept of "sustainability" more easily applied to renewable resources than to non-renewable ones?

8. ASSIGNMENT

1. **Research and Explain:** Choose one renewable and one non-renewable resource found in Nigeria. Write a paragraph on each, describing how it is extracted/used and one major environmental concern associated with its exploitation.
 2. **Compare and Contrast:** Create a detailed table (beyond the one in class) highlighting five differences between renewable and non-renewable resources.
 3. **Problem-Solving:** Propose two practical ways a local community in Nigeria could transition from relying on a non-renewable resource (like firewood/charcoal) to a renewable one.
-

WEEK 2 - ENVIRONMENTAL PROBLEMS

Learning Objectives: By the end of this lesson, students should be able to:

1. Identify and describe major environmental problems in Nigeria.
2. Analyze the causes and effects of these problems.
3. Propose viable solutions and mitigation strategies.



1. INTRODUCTION: THE PRESSURES ON OUR PLANET

An **Environmental Problem** is any disturbance or undesirable change in the natural environment that is harmful to ecosystems and human life. These problems are often triggered or exacerbated by human activities interacting with natural processes. Nigeria, with its rapid population growth, industrialization, and heavy reliance on natural resources, faces severe environmental challenges that threaten its sustainable development.

2. MEANING OF ENVIRONMENTAL PROBLEMS

Environmental problems refer to the negative alterations in the physical, chemical, and biological characteristics of the environment. These changes degrade the quality of air, water, and land, leading to a decline in the environment's capacity to support life and economic activities.

3. MAJOR ENVIRONMENTAL PROBLEMS IN NIGERIA: A DEEPER DIVE

1. DEFORESTATION

- **Meaning:** The permanent removal of forest cover for other land uses.
- **Causes in Nigeria:**
 - **Agricultural Expansion:** Slash-and-burn farming and clearing for large plantations.
 - **Logging:** Both legal and illegal for timber and fuelwood (charcoal production is a major driver).
 - **Urbanization & Infrastructure:** Building houses, roads, and industries.
- **Detailed Effects:**
 - **Loss of Biodiversity:** Destruction of habitats for unique species like chimpanzees and forest elephants.
 - **Soil Erosion:** Tree roots no longer hold the soil together.
 - **Climate Change:** Forests are "carbon sinks." Deforestation releases stored carbon, contributing to global warming.
 - **Disruption of Water Cycles:** Reduced transpiration leads to reduced and erratic rainfall.

2. POLLUTION

- **Meaning:** The introduction of contaminants into the environment.
- **Types and Nigerian Context:**

- **Air Pollution:** From vehicle emissions, gas flaring in the Niger Delta, generator fumes, and industrial smoke. Leads to respiratory diseases (asthma, bronchitis) and acid rain.
- **Water Pollution:** Caused by oil spills in the Niger Delta, discharge of untreated industrial effluents, and domestic sewage. Renders water unsafe for drinking and kills aquatic life.
- **Land/Soil Pollution:** Resulting from improper disposal of solid waste (plastics, e-waste), oil spills, and excessive use of agricultural pesticides and fertilizers.
- **Effects:** Health issues, destruction of fisheries, loss of agricultural productivity, and visual blight.

3. SOIL EROSION

- **Meaning:** The washing or blowing away of the topmost, fertile layer of soil.
- **Causes:**
 - **Deforestation** (removes protective cover).
 - **Bad Farming Practices:** Monocropping, overgrazing, and ploughing along slopes.
- **Effects:**
 - **Gullies:** Severe, large-scale gully erosion in states like Anambra and Imo destroys homes, roads, and farmlands.
 - **Loss of Fertile Land:** Reduces agricultural yield.
 - **Siltation:** Eroded soil fills up rivers and reservoirs, reducing their capacity and causing flooding.

4. DESERTIFICATION

- **Meaning:** The process by which fertile land becomes desert, typically as a result of drought, deforestation, or inappropriate agriculture. Affects Northern states like Sokoto, Kano, and Borno.
- **Causes:**
 - **Climate Change:** Leading to reduced and erratic rainfall.

- **Human Activities:** Overgrazing, over-cultivation, and deforestation for firewood.
- **Effects:**
 - **Food Insecurity:** Loss of arable land leads to crop failure and famine.
 - **Migration:** Forces farmers and herders to migrate southwards, often leading to conflicts with sedentary communities.
 - **Loss of Biodiversity:** Desert conditions cannot support diverse life.

5. FLOODING

- **Meaning:** An overflow of water that submerges usually dry land.
- **Causes:**
 - **Natural:** Heavy rainfall and river overflow.
 - **Human-Induced: Blocked drainage systems** due to poor waste management, **unplanned urban development** on floodplains, and **deforestation** which reduces water absorption.
- **Effects:** Displacement of people, destruction of property and infrastructure, outbreak of waterborne diseases (cholera, typhoid), and loss of lives.

4. CAUSES OF ENVIRONMENTAL DEGRADATION

- **Human (Anthropogenic) Causes:**
 - **Population Growth:** Increased demand for food, water, and space.
 - **Poverty:** Leads to over-exploitation of resources for immediate survival (e.g., cutting trees for charcoal).
 - **Poor Governance & Weak Enforcement:** Lack of enforcement of environmental laws and regulations.
 - **Natural Causes:**
 - Climate change, droughts, and natural wildfires.
-

5. EFFECTS OF ENVIRONMENTAL PROBLEMS

- **Economic:** Loss of income from agriculture and fisheries, high cost of rebuilding after floods.
 - **Social:** Health crises, displacement, and conflicts over scarce resources (e.g., farmer-herder conflicts).
 - **Environmental:** Irreversible loss of species and ecosystems.
-

6. SOLUTIONS AND MITIGATION STRATEGIES

1. **Afforestation/Reforestation:** Massive tree planting campaigns.
 2. **Sustainable Agriculture:** Promoting crop rotation, terracing, and organic farming.
 3. **Strict Enforcement of Laws:** Empowering agencies like NESREA.
 4. **Public Enlightenment:** Educating citizens on waste management and conservation.
 5. **Investment in Infrastructure:** Building proper drainage systems and sanitary landfills.
 6. **Promotion of Alternative Energy:** Solar, wind, and biogas to reduce reliance on fuelwood.
-

7. EVALUATION & ASSIGNMENT

Evaluation (Past Questions):

1. **(WAEC 2018):** Identify and explain three environmental problems common in Nigeria and their effects.
2. **(NECO 2020):** State four causes of deforestation and suggest three control measures.

Assignment:

1. **Case Study:** Write a one-page report on the environmental challenges in the Niger Delta, focusing on the causes and consequences of oil pollution.
2. **Action Plan:** Design a poster educating your community on the dangers of improper waste disposal and how to practice recycling.

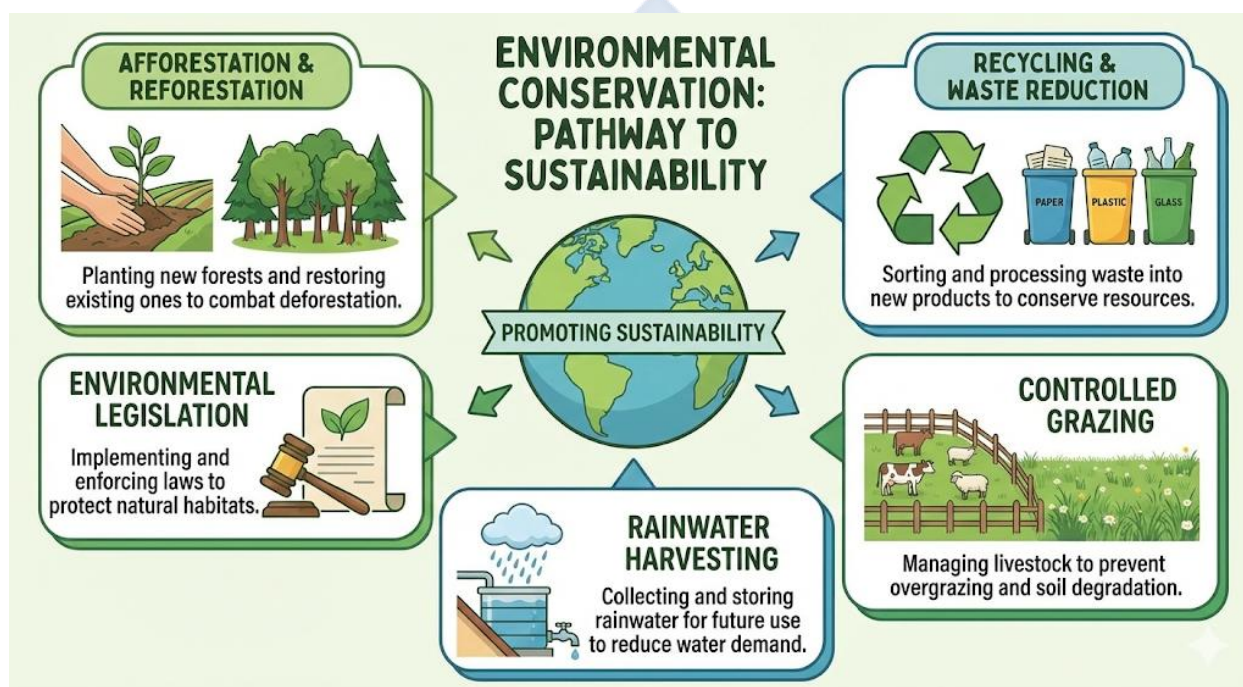
3. **Problem-Solving:** Propose three specific strategies the government could implement to reduce the impact of flooding in a major city like Lagos.
-



WEEK 3 - ENVIRONMENTAL CONSERVATION

Learning Objectives: By the end of this lesson, students should be able to:

1. Define environmental conservation and distinguish it from preservation.
2. Explain the importance of conservation for ecological balance and human survival.
3. Identify and describe various methods of environmental conservation.
4. Analyze the roles of key organizations in promoting conservation in Nigeria and globally.



1. INTRODUCTION: FROM EXPLOITATION TO STEWARDSHIP

For decades, humanity's relationship with the environment has been largely one-sided: take, use, and discard. The resulting environmental problems we face today—deforestation, pollution, species loss—are a direct consequence of this approach. **Environmental Conservation** is the necessary paradigm shift. It is the ethical and practical philosophy of using the Earth's resources responsibly so that they remain available for future generations. It is not about locking away resources, but about managing them wisely for long-term sustainability.

Key Concept: Conservation is the bridge between our current needs and our future survival.

2. MEANING OF ENVIRONMENTAL CONSERVATION

Environmental Conservation is the sustainable use, management, and protection of vital natural resources and ecosystems. It aims to prevent habitat destruction, species extinction, and resource depletion.

- **Conservation vs. Preservation:** It's important to distinguish these two related ideas.
 - **Conservation** emphasizes the *sustainable use* of resources. For example, a conservationist supports logging that follows a plan to replant trees and maintain the forest's health.
 - **Preservation** emphasizes *protecting nature from use*. A preservationist would argue for leaving a forest completely untouched, with no logging allowed.

The Core Aims of Conservation are to:

- **Maintain Ecological Balance:** Ensure that natural cycles (water, carbon, nitrogen) continue to function.
 - **Protect Biodiversity:** Safeguard the variety of plant and animal life, which is crucial for ecosystem resilience.
 - **Ensure Resource Availability:** Guarantee that resources like water, timber, and fertile soil are available for the long term.
 - **Mitigate Climate Change:** Protect and expand forests and wetlands that absorb carbon dioxide.
-

3. THE CRITICAL IMPORTANCE OF ENVIRONMENTAL CONSERVATION

1. **Safeguarding Ecosystem Services:** Nature provides "free services" we often take for granted. Forests purify our air and regulate rainfall. Wetlands filter our water and control floods. Bees and other insects pollinate our crops. Conservation protects these essential services.

2. Economic Stability and Growth:

- **Agriculture:** Conservation through soil management and water protection ensures long-term agricultural productivity.
 - **Tourism:** Well-conserved natural areas like Yankari Game Reserve and Obudu Cattle Ranch are significant sources of revenue through ecotourism.
 - **Fisheries:** Protecting mangrove forests and regulating fishing ensures fish stocks do not collapse, securing livelihoods and food sources.
3. **Climate Regulation:** Forests act as "carbon sinks," absorbing CO₂ from the atmosphere. Conserving them is a direct strategy in the fight against global warming.
4. **Public Health:** Clean air, clean water, and stable climates reduce the prevalence of respiratory illnesses, waterborne diseases, and malnutrition.
5. **Cultural and Aesthetic Value:** Natural landscapes are often central to cultural identities and provide recreational and spiritual benefits that contribute to human well-being.

4. METHODS AND STRATEGIES OF ENVIRONMENTAL CONSERVATION

1. Afforestation and Reforestation:

- **Afforestation:** Planting trees in areas that have not been forested in recent history (e.g., planting trees in arid regions to combat desertification).
- **Reforestation:** Replanting trees in areas that have been recently deforested.
- **Nigerian Example:** The "Great Green Wall" initiative is a large-scale afforestation project aimed at restoring land and combating desertification across the Sahel region, including Northern Nigeria.

2. Sustainable Agriculture:

- **Crop Rotation:** Planting different crops sequentially to improve soil health and reduce pest buildup.
- **Organic Farming:** Using natural fertilizers (compost, manure) and pest control instead of synthetic chemicals.

- **Agroforestry:** Integrating trees and shrubs into farmland, which improves soil fertility and provides additional income.

3. **Waste Management and Recycling:**

- Moving from a "take-make-dispose" model to a **circular economy** where waste is minimized, and materials are reused and recycled.
- This reduces pollution and the demand for virgin raw materials, thereby conserving resources and energy.

4. **Water Conservation:**

- Techniques include rainwater harvesting, fixing leaky pipes, using drip irrigation in agriculture, and protecting watersheds from pollution.

5. **Environmental Legislation and Policy:**

- Laws and regulations are essential to control pollution, manage waste, and protect endangered species and critical habitats. Enforcement is key.

6. **Public Awareness and Education:**

- This is the foundation of all conservation efforts. When people understand the value of their environment, they are more likely to protect it. This can be done through school curricula, media campaigns, and community workshops.

5. KEY ORGANIZATIONS IN ENVIRONMENTAL CONSERVATION

A. National Organizations:

1. NESREA (National Environmental Standards and Regulations Enforcement Agency):

- **Role:** Nigeria's primary environmental watchdog. It is responsible for enforcing all environmental laws, regulations, and standards. This includes monitoring industrial pollution, regulating waste management, and ensuring environmental impact assessments are conducted for new projects.

2. NCF (Nigerian Conservation Foundation):

- **Role:** Nigeria's foremost non-governmental organization dedicated to nature conservation. It focuses on biodiversity conservation, policy advocacy, and environmental education. It manages protected areas like the Lekki Conservation Centre.

3. **Forestry Research Institute of Nigeria (FRIN):**

- **Role:** Conducts research on sustainable forest management, agroforestry, and biodiversity conservation. It provides the scientific backbone for forest conservation policies.

B. International Organizations:

1. **UNEP (United Nations Environment Programme):**

- **Role:** The leading global environmental authority. It sets the global environmental agenda, promotes sustainable development, and provides technical assistance to countries.

2. **WWF (World Wide Fund for Nature):**

- **Role:** A global NGO working to halt the degradation of the planet's natural environment. It works on issues ranging from protecting endangered species (like elephants and gorillas) to combating climate change.

3. **IUCN (International Union for Conservation of Nature):**

- **Role:** A membership union of governments and civil society organizations. It is best known for compiling the **Red List of Threatened Species**, a critical indicator of the health of the world's biodiversity.

6. EVALUATION & ASSESSMENT

Short Answer:

1. What is the difference between conservation and preservation?
2. List three methods of environmental conservation.

3. What does the acronym NESREA stand for and what is its primary role?

Critical Thinking:

4. "Public awareness is the most important method of environmental conservation." Do you agree with this statement? Justify your answer.
 5. How does the practice of agroforestry contribute to both environmental conservation and economic development for a local farmer?
-

7. ASSIGNMENT

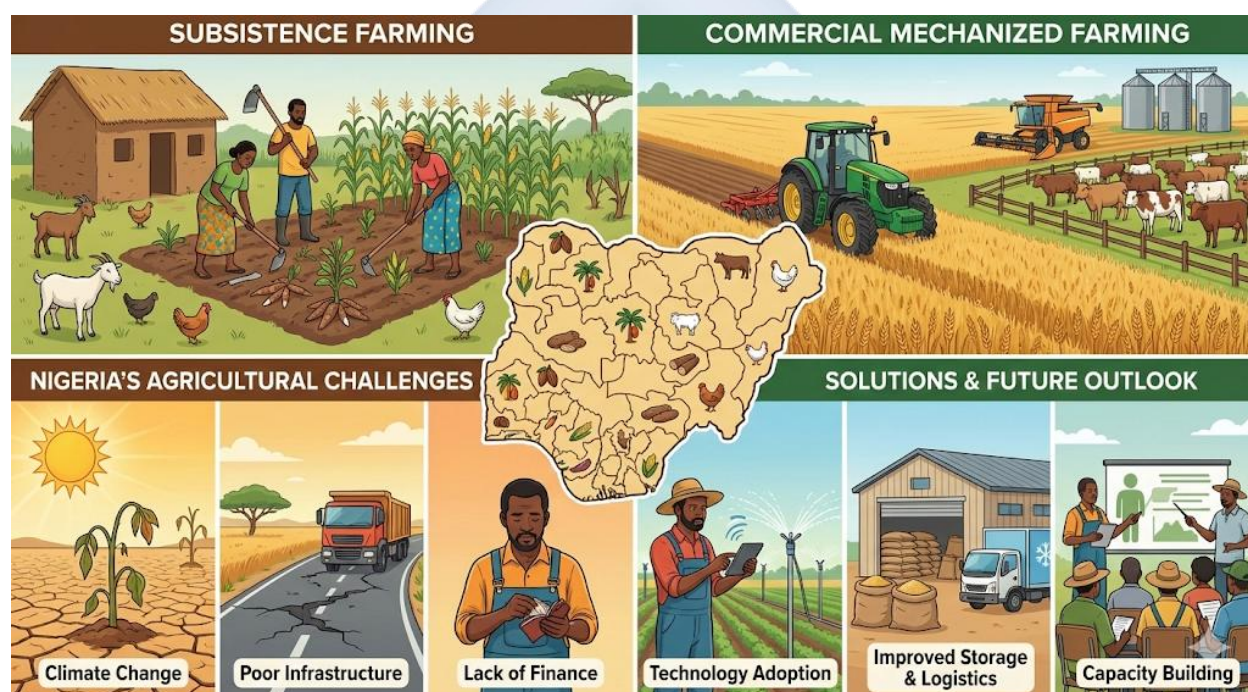
1. **Local Case Study:** Research a specific environmental conservation project in your state or a nearby state (e.g., a community forest reserve, a beach clean-up initiative, a watershed protection program). Write a one-page report describing the project, its goals, and its impact so far.
 2. **Role Play:** You are an environmental officer from NESREA. Prepare a short speech you would give to a factory owner who is dumping untreated waste into a nearby river. Your speech should explain why this is harmful and what the factory must do to comply with environmental laws.
 3. **Analyze and Propose:** Identify one major environmental problem in your local community. Propose a detailed conservation plan to address it, including:
 - The specific conservation methods you would use.
 - Which organization(s) (local, national, or international) you would partner with.
 - How you would involve the community.
-

WEEK 4 - AGRICULTURE

Lesson Duration: 60 Minutes

Learning Objectives: By the end of this lesson, students should be able to:

1. Define agriculture and appreciate its broad scope beyond mere farming.
2. Differentiate between subsistence and commercial agriculture with clear examples.
3. Describe the major agricultural systems practiced in Nigeria.
4. Analyze the critical importance of agriculture to the Nigerian economy and society.
5. Evaluate the challenges facing the agricultural sector and propose sustainable solutions.



1. INTRODUCTION: THE BEDROCK OF CIVILIZATION

Agriculture is more than just farming; it is the foundation upon which human civilizations were built. It encompasses the entire complex process of producing food, feed, fiber, and other goods through the deliberate cultivation of plants and rearing of livestock. From the food on our plates to the raw materials in our industries, agriculture remains a vital force shaping Nigeria's economy, employment, and food security.

2. MEANING AND SCOPE OF AGRICULTURE

Agriculture is the science, art, and business of cultivating soil, producing crops, and raising livestock for economic purposes. It is a multifaceted sector that includes:

- **Crop Production (Agronomy):** Growing crops like grains, vegetables, and fruits.
 - **Livestock Farming (Animal Husbandry):** Rearing animals for meat, milk, eggs, and hides.
 - **Economics and Marketing:** The business of selling and distributing agricultural products.
 - **Agricultural Engineering:** Designing tools, machinery, and infrastructure for farming.
-

3. TYPES OF AGRICULTURE: SUBSISTENCE VS. COMMERCIAL

This fundamental classification is based on the **scale of production** and the **primary purpose**.

A. Subsistence Agriculture

- **Purpose:** Production is primarily for **family consumption**. Little or no surplus is left for sale.
- **Scale:** Small-scale, often on fragmented plots of land (1-3 hectares).
- **Technology:** Relies on simple, traditional tools like hoes, cutlasses, and family labor.
- **Capital:** Very low capital investment.
- **Yield:** Low yield per hectare due to reliance on natural soil fertility and rainfall.
- **Example:** A family in a rural village growing yams, cassava, and vegetables on a small plot to feed themselves.

B. Commercial Agriculture

- **Purpose:** Production is primarily for **sale in the market** to generate profit.
- **Scale:** Large-scale operations, often covering hundreds of hectares.

- **Technology:** Highly mechanized, using tractors, combine harvesters, irrigation systems, and fertilizers.
- **Capital:** Requires substantial capital investment for land, machinery, and inputs.
- **Yield:** High yield per hectare due to the application of scientific methods.
- **Example:** A large plantation in Ogun State growing cassava for starch production or a poultry farm in Oyo State with tens of thousands of birds for egg and meat supply.

AGRICULTURAL SYSTEMS: SUBSISTENCE VS. COMMERCIAL

	 SUBSISTENCE AGRICULTURE	COMMERCIAL AGRICULTURE 
PURPOSE	 For personal and family consumption, survival.	 For profit, sale in local and international markets.
SCALE	 Small-scale, limited land area, often single plots.	 Large-scale, extensive land holdings, industrial operations.
TECHNOLOGY	 Traditional tools, simple methods, human and animal power.	 Mechanized, advanced machinery, modern techniques, chemical inputs.
CAPITAL	 Low financial investment, minimal capital inputs.	 High financial investment, significant capital for machinery, land, and inputs.
LABOR	 Family-based, labor-intensive, often seasonal.	 Hired, specialized labor, mechanized, often year-round.

4. AGRICULTURAL SYSTEMS IN NIGERIA

An agricultural system is a way of practicing agriculture, defined by its methods, inputs, and outputs.

1. Shifting Cultivation (A form of Subsistence Agriculture):

- **Process:** A farmer clears a piece of forest land by slashing and burning, cultivates it for 2-3 years until soil fertility declines, and then abandons it to clear a new plot. The old plot is left to regenerate (fallow period).
- **Where:** Common in tropical forest zones with low population density.

- **Challenge:** Increasing population pressure shortens the essential fallow period, leading to soil degradation.
2. **Plantation Agriculture (A form of Commercial Agriculture):**
- **Process:** Large-scale farming specializing in a single cash crop (monoculture) for export or large-scale local processing.
 - **Crops:** Cocoa, oil palm, rubber, tea, sugarcane.
 - **Where:** Established in parts of Ondo (cocoa), Edo (rubber), and Cross River (oil palm).
3. **Pastoral Nomadism (A form of Subsistence Agriculture):**
- **Process:** Herders (like the Fulani) move their cattle, sheep, and goats from place to place in search of fresh pasture and water. There is no permanent settlement.
 - **Where:** Practiced in the arid and semi-arid regions of Northern Nigeria.
 - **Challenge:** Often leads to conflicts with sedentary farmers over land and water resources.
4. **Ley Farming (A more Advanced System):**
- **Process:** A system that integrates crop cultivation with livestock rearing. A piece of land is used for cropping for some years and then left to pasture for animals. The animals manure the land, restoring its fertility for the next cropping cycle.
 - **Benefit:** A more sustainable and productive system.
-

5. THE MULTIFACETED IMPORTANCE OF AGRICULTURE TO NIGERIA

1. **Food Security:** Provides the staple foods (yam, cassava, rice, maize) that feed the nation's population.
2. **Employment:** It is the largest employer of labour, engaging over 35% of the workforce, especially in rural areas.
3. **Raw Material Provision:** Serves as the primary source of raw materials for agro-allied industries (e.g., cotton for textiles, hides for leather, oil palm for soap).
4. **Revenue Generation:**

- **Foreign Exchange:** Exports of cash crops like cocoa, sesame, and cashew nuts earn vital foreign currency.
 - **Government Revenue:** Through taxes and levies on agricultural products and exports.
5. **Industrial Development:** A thriving agricultural sector stimulates the growth of related industries such as food processing, fertilizer production, and packaging.
 6. **Poverty Alleviation:** Provides a primary source of income for millions of rural households.
-

6. CRITICAL CHALLENGES FACING AGRICULTURE IN NIGERIA

1. **Primitive Farming Technology:** Over-reliance on hoes and cutlasses limits productivity and farm size.
 2. **Inadequate Infrastructure:**
 - **Poor Road Networks:** Makes it difficult and expensive to transport produce from rural farms to urban markets, leading to high post-harvest losses.
 - **Unreliable Energy Supply:** Hinders the operation of processing and storage facilities.
 3. **Financing and Credit Inaccessibility:** Small-scale farmers lack collateral to access loans for improved seeds, fertilizers, and machinery.
 4. **Pests and Diseases:** Attacks by locusts, quelea birds, and diseases like Cassava Mosaic Virus destroy crops and reduce yields.
 5. **Land Tenure System:** Communal land ownership makes it difficult for individuals to acquire large tracts of land for commercial farming.
 6. **Climate Change:** Manifested through unpredictable rainfall patterns, prolonged droughts, and devastating floods, which disrupt farming cycles.
 7. **Over-reliance on Oil:** The "Oil Boom" of the 1970s led to government neglect of the agricultural sector, a phenomenon known as the "Dutch Disease."
-

7. EVALUATION & ASSESSMENT

Short Answer:

1. What is the primary goal of commercial agriculture?
2. Name two crops commonly grown under the plantation system in Nigeria.
3. List two factors that limit agricultural development in Nigeria.

Critical Thinking:

4. "The land tenure system in Nigeria is a major obstacle to commercial agriculture."
Explain this statement.
 5. How does the problem of poor transportation infrastructure specifically affect a tomato farmer in a rural village?
-

8. ASSIGNMENT

1. **Comparative Analysis:** Create a detailed poster comparing Subsistence and Commercial Agriculture. Use headings, images (or drawings), and bullet points to highlight their differences in technology, labor, capital, and output.
 2. **Research and Report:** Choose one major cash crop in Nigeria (e.g., Cocoa, Oil Palm, or Groundnut). Research and write a one-page report on:
 - Its major producing states.
 - Its importance to the Nigerian economy.
 - Two major challenges facing its farmers.
 3. **Problem-Solving Proposal:** You are an advisor to the Minister of Agriculture. Propose a concrete plan to address **one** of the following challenges:
 - The farmer-herder conflict.
 - Post-harvest losses of a specific crop. Your plan should include specific actions and suggest which government agencies would be involved.
-

WEEK 6 - TRANSPORT AND COMMUNICATION IN NIGERIA

Learning Objectives: By the end of this lesson, students should be able to:

1. Define transport and communication and understand their interrelationship.
2. Identify and describe the different modes of transport and communication in Nigeria.
3. Analyze the profound importance of transport and communication to national development.
4. Critically examine the challenges facing these sectors and propose innovative solutions.



1. INTRODUCTION: THE NERVOUS SYSTEM OF A NATION

Imagine trying to sell your farm produce without a road to the market, or a business trying to operate without phones or the internet. **Transport** and **Communication** are the circulatory and nervous systems of a country. They connect people, places, goods, and ideas, enabling every other sector of the economy to function. Together, they shrink distances, create a sense of national unity, and are indispensable for modern life.

2. MEANING AND INTERRELATIONSHIP

- **Transport:** The physical movement of people, goods, and services from one geographical location to another.
- **Communication:** The transmission of information, ideas, and emotions from a source to a receiver.

The Link: Efficient transport relies on communication for coordination (e.g., tracking shipments), and effective communication often depends on transport for the delivery of physical media (e.g., letters, devices). They are synergistic forces for development.

3. MODES OF TRANSPORT IN NIGERIA

1. Road Transport:

- **Description:** The most dominant and widely used mode. Includes motorcycles (okada), tricycles (keke napep), cars, buses, and trucks.
- **Advantages:** Highly flexible, provides door-to-door service, and is relatively cheap for short distances.
- **Disadvantages:** Prone to traffic congestion, high accident rates, and severely affected by poor road conditions.
- **Key Infrastructure:** Lagos-Ibadan Expressway, East-West Road, Second Niger Bridge.

2. Rail Transport:

- **Description:** Involves the movement of passengers and freight on trains along a fixed track.
- **Advantages:** Highly efficient for moving heavy and bulky goods over long distances; more fuel-efficient and environmentally friendly than road transport.
- **Disadvantages:** The network is still underdeveloped and outdated in many parts, leading to slow speeds and limited coverage.
- **Key Infrastructure:** The modernized Lagos-Ibadan Standard Gauge Line, the Abuja-Kaduna Rail Line, and the historic Narrow Gauge lines.

3. Air Transport:

- **Description:** The fastest mode of transport, using aircraft.
- **Advantages:** Essential for long-distance travel, perishable goods (e.g., flowers, seafood), and emergency services.
- **Disadvantages:** The most expensive mode; limited to areas with airports.
- **Key Infrastructure:** Murtala Mohammed International Airport (Lagos), Nnamdi Azikiwe International Airport (Abuja), Port Harcourt International Airport.

4. Water Transport:

- **Description:** Includes inland waterways (rivers, creeks) and ocean/sea transport.
- **Advantages:** Cheapest for moving very heavy and non-perishable goods (e.g., crude oil, containers) over long distances.
- **Disadvantages:** Slow and limited to navigable waterways and port locations.
- **Key Infrastructure:** Apapa and Tin-Can Island Ports (Lagos), Onne Port (Port Harcourt), River Niger and Benue.

4. *MODES OF COMMUNICATION IN NIGERIA*

1. Traditional Media:

- **Examples:** Town criers, drums, festivals.
- **Role:** Still relevant for local, community-based communication.

2. Print Media:

- **Examples:** Newspapers (The Punch, The Guardian), magazines, pamphlets.
- **Role:** Provides in-depth analysis and has a long-lasting record.

3. Broadcast Media:

- **Examples:** Radio (e.g., FRCN, private stations) and Television (e.g., NTA, Channels TV).

- **Role:** The most powerful tool for mass communication and public enlightenment, especially radio in rural areas.

4. Telecommunication:

- **Examples:** Mobile networks (MTN, Airtel, Glo), landline phones, and the internet.
- **Role:** Revolutionized personal and business communication through voice, SMS, and mobile data services. Enables mobile banking and e-commerce.

5. New/Digital Media:

- **Examples:** Social media (WhatsApp, Twitter, Facebook), blogs, online news platforms.
- **Role:** Allows for instant, interactive, and user-generated communication.

5. THE IMPORTANCE OF TRANSPORT AND COMMUNICATION

1. **Economic Integration:** Links producers to consumers, and farms to markets, facilitating trade and commerce.
2. **Promotion of National Unity:** Enables the movement and interaction of people from different ethnic groups, fostering a shared national identity.
3. **Employment Generation:** Creates jobs in driving, logistics, telecom services, media, and courier services.
4. **Diffusion of Ideas and Innovation:** Communication spreads new technologies, agricultural practices, and health information quickly.
5. **Support for Tourism and Social Interaction:** Makes tourist sites accessible and helps families and friends stay connected.
6. **National Security:** Essential for the rapid deployment of security forces and dissemination of critical information.

6. CHALLENGES AND THE PATH FORWARD

A. Challenges of Transport:

- **Poor Infrastructure:** Dilapidated roads, outdated rail lines, and shallow waterways.
- **Safety and Security:** Road accidents, kidnapping on highways, and piracy in the Gulf of Guinea.
- **Funding:** Inadequate public funding for maintenance and new projects.
- **Urban Congestion:** Traffic gridlock in major cities like Lagos and Abuja.

B. Challenges of Communication:

- **Digital Divide:** Poor network coverage and internet access in rural areas.
- **Cost:** High cost of data and mobile devices for the average citizen.
- **Misinformation:** The rapid spread of fake news and hate speech through digital platforms.
- **Power Supply:** Erratic electricity disrupts communication networks and access.

Solutions:

- **Public-Private Partnerships (PPPs):** To attract investment for infrastructure development (e.g., the Lekki Deep Sea Port).
- **Strict Regulation and Maintenance:** Enforcing axle-load controls for trucks and consistent road maintenance.
- **Investment in Rural Infrastructure:** Expanding broadband internet and grid electricity to rural communities.
- **Media Literacy Education:** Teaching citizens how to critically evaluate information online.

7. EVALUATION & ASSIGNMENT

Evaluation (Past Questions):

1. **(WAEC 2017):** List and explain three problems of transportation in Nigeria.
2. **(NECO 2019):** Discuss two challenges of air transport and two of rail transport in Nigeria.

Assignment:

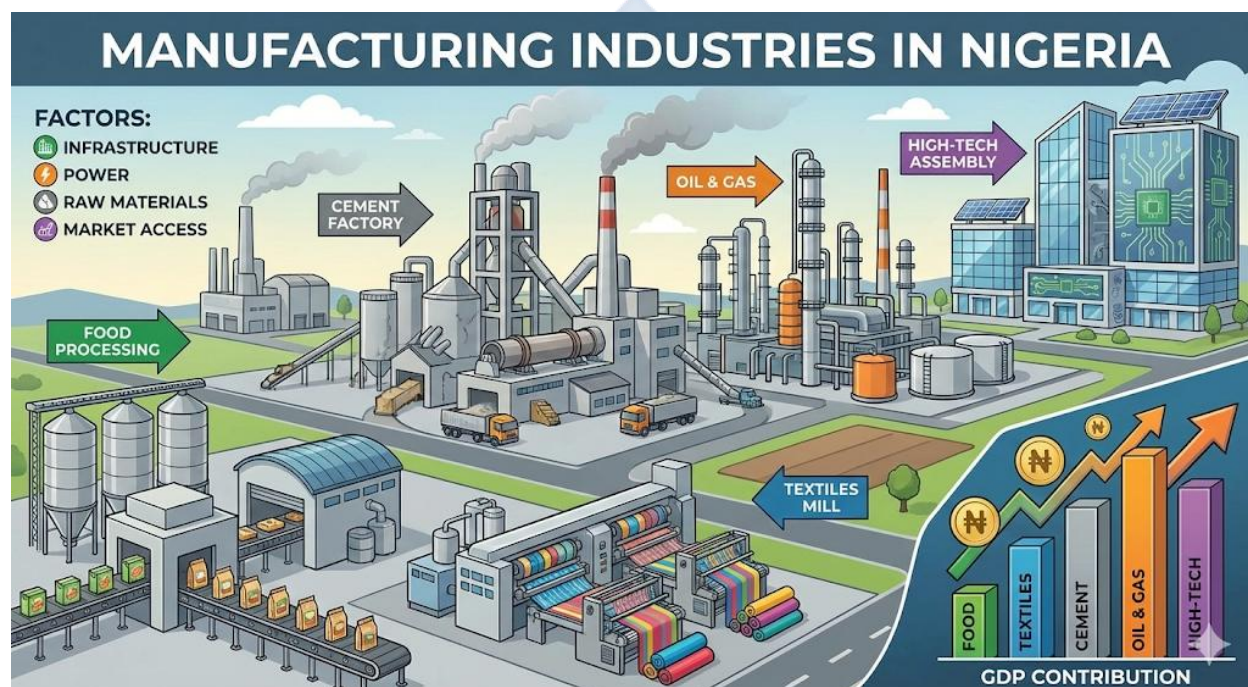
1. **Digital Presentation:** Create a short presentation comparing two modes of transport (e.g., Road vs. Rail) for moving goods from Kano to Lagos. Analyze them based on cost, speed, safety, and suitability for different types of goods.
 2. **Local Infrastructure Audit:** Write a brief report on the state of transport and communication in your local government area. Identify one major problem in each sector and suggest a practical solution.
 3. **Debate Preparation:** Prepare a one-page argument for a debate on the topic: "The invention of the mobile phone has had a more significant impact on Nigerian society than the construction of the Lagos-Ibadan Expressway."
-



WEEK 8 - MANUFACTURING INDUSTRIES IN NIGERIA

Learning Objectives: By the end of this lesson, students should be able to:

4. Define and classify manufacturing industries with relevant examples.
5. Identify major manufacturing industries and their locations in Nigeria.
6. Analyze the key factors influencing the location of industries.
7. Evaluate the importance of manufacturing to Nigeria's economy and the challenges it faces.



1. INTRODUCTION: FROM RAW MATERIALS TO FINISHED GOODS

The primary sector (like farming and mining) extracts raw materials, but it is the **manufacturing sector** that transforms these materials into valuable products we use every day. By adding value through processing, assembling, and refining, manufacturing industries are a powerful engine for economic growth, job creation, and technological advancement. A strong manufacturing base is a hallmark of a developed and diversified economy.

2. MEANING AND CLASSIFICATION OF MANUFACTURING INDUSTRIES

A **Manufacturing Industry** is an establishment that is involved in the mechanical or chemical transformation of raw materials or components into new products. The key word is **transformation**—changing the form, utility, and value of a material.

Classification of Industries:

Industries can be classified based on the raw materials they use, the products they make, and their scale:

1. Based on Raw Material Source:

- **Agro-Based Industries:** Use agricultural products as raw materials.
 - *Examples:* Tomato paste factory, fruit juice processing, textile (cotton), sugar refining, leather tanning.
- **Mineral-Based Industries:** Use mineral ores as raw materials.
 - *Examples:* Cement production (from limestone), Aluminium smelting (from bauxite), Steel rolling mills (from iron ore).
- **Chemical-Based Industries:** Involve complex chemical processes.
 - *Examples:* Petroleum refineries, petrochemical plants (producing plastics, fertilizers), pharmaceutical companies, paint factories.

2. Based on Size/Scale of Operation:

- **Heavy Industries:** Involve large-scale plants, massive machinery, and produce large, bulky products. They are capital-intensive.
 - *Examples:* Vehicle assembly plants, shipbuilding, steel and iron production.
- **Light Industries:** Produce relatively small, lightweight consumer goods. They often require less capital and can be located in urban areas.
 - *Examples:* Footwear, garment making, furniture, food processing, electronics assembly.

3. Based on Ownership:

- **Large-Scale:** Owned by large corporations (e.g., Dangote Group, Nigerian Breweries).
- **Small-Scale:** Often family-owned, with a small workforce and capital (e.g., local bakeries, soap making).

3. MAJOR INDUSTRIES AND INDUSTRIAL ZONES IN NIGERIA

Nigeria's industries are not evenly distributed; they are concentrated in zones with favourable location factors.

Industrial Zone	Key Industries	Specific Examples & Locations
Lagos State	A diverse mix; the nation's industrial hub.	Food & Beverage: Nigerian Breweries (Lagos). Textiles: United Nigerian Textiles. Vehicles: Vehicle assembly plants in Ikeja. Chemicals: PZ Cussons.
South-East (Aba, Nnewi, Onitsha)	Light manufacturing, commerce, and entrepreneurship.	Aba: Garments, footwear (Aba-made shoes). Nnewi: Automotive parts and plastics manufacturing.
South-South (P/Harcourt, Warri)	Dominated by oil and gas-related industries.	Port Harcourt: Oil refineries (Port Harcourt Refinery), petrochemicals (Eleme Petrochemicals). Warri: Refining and gas processing.
North-Central & South-West (Cement Belt)	Mineral-based, specifically cement.	Obajana (Kogi): Largest cement plant in Africa (Dangote). Ewekoro & Sagamu (Ogun): Lafarge Cement. Gboko (Benue): Dangote Cement.
Kano & Kaduna	Traditional agro-based industries.	Kano: Tanneries, textile mills, food processing. Kaduna: Refinery, textile mill, automobile assembly (Peugeot).

4. FACTORS INFLUENCING INDUSTRIAL LOCATION

The decision on where to site a factory is strategic and depends on a combination of factors:

1. Availability of Raw Materials (Material-Oriented Industries):

- Industries that use heavy, bulky, or perishable raw materials are located near the source to minimize transport costs.
- *Example:* Cement factories are located near limestone quarries (Obajana, Ewekoro). Fruit canning plants are built near fruit-growing areas.

2. Market (Market-Oriented Industries):

- Industries that produce perishable goods (like bread) or bulky products (like bottles) are located close to their consumers.
- *Example:* Beverage bottling plants are found in major cities like Lagos and Ibadan.

3. Adequate Power Supply:

- A constant and reliable source of energy is non-negotiable for most industries. Power outages disrupt production and increase costs, forcing many companies to rely on expensive diesel generators.

4. Efficient Transport Network:

- Proximity to roads, railways, ports, and airports is crucial for bringing in raw materials and distributing finished goods. This is a key reason for the dominance of Lagos.

5. Labour Supply:

- Access to a pool of skilled, semi-skilled, and unskilled labour at a reasonable cost is essential.

6. Government Policies:

- Policies can attract or deter industries. **Tax incentives, grants, and the development of Industrial Parks or Free Trade Zones (e.g., Calabar FTZ)** can encourage industrial location. Conversely, unpredictable policies can discourage investment.

7. Capital Availability:

- Access to loans, venture capital, and other financial resources is necessary to start and run an industry.

5. IMPORTANCE AND CHALLENGES OF THE MANUFACTURING SECTOR

A. Importance:

- **Economic Diversification:** Reduces over-reliance on crude oil.
- **Value Addition:** Increases the worth of raw materials exported (e.g., turning crude oil into petrol and plastics is more profitable than exporting crude).
- **Employment Generation:** Creates jobs in factories and related services.
- **Foreign Exchange Earnings/Savings:** Exporting finished goods earns forex; producing goods locally saves forex spent on imports.
- **Stimulates Other Sectors:** Boosts agriculture (providing a market), transport, and banking.

B. Critical Challenges:

- **Inadequate Infrastructure:** Erratic power supply is the single biggest challenge. Poor road networks also increase operational costs.
 - **Unstable Input of Raw Materials:** Agro-based industries suffer from seasonal and inconsistent supply of farm produce.
 - **Financing Problems:** High-interest rates and difficulty accessing loans hinder growth.
 - **Stiff Competition from Imports:** Cheaper, often subsidized, foreign goods flood the market, making it hard for local industries to compete.
 - **Unfavourable Government Policies:** Policy inconsistency and multiple taxation frustrate investors.
-

6. EVALUATION & ASSIGNMENT

Evaluation (Past Questions):

1. **(WAEC 2017):** (a) Define manufacturing industry. (b) Identify and explain four factors affecting the location of manufacturing industries in Nigeria.
2. **(WAEC 2015):** With examples, distinguish between light and heavy industries.

Assignment:

1. **Industry Profile:** Choose one major manufacturing company in Nigeria (e.g., Dangote Cement, Nigerian Breweries, Innoson Vehicles). Research and write a profile on its location, primary raw materials, finished products, and explain how at least three location factors influenced where it was sited.
 2. **Policy Memo:** You are an economic advisor. Write a one-page memo to the Minister of Industry titled "Three Key Policies to Revive Nigerian Manufacturing." Propose specific, actionable policies to address the sector's biggest challenges.
 3. **Map Work:** On an outline map of Nigeria, locate and label the following:
 - Two cement-producing towns.
 - Two cities known for textile manufacturing.
 - Two oil refinery locations.
 - One vehicle assembly plant.
-

WEEK 9 – COMMERCIAL ACTIVITIES IN NIGERIA

Learning Objectives: By the end of this lesson, students should be able to:

1. Define commerce and distinguish between trade and aids to trade.
2. Identify and explain different types of commercial activities.
3. Analyze the factors responsible for the growth of major commercial centres in Nigeria.
4. Evaluate the role of commerce in national development and the challenges it faces.



1. INTRODUCTION: THE LIFE-BLOOD OF THE ECONOMY

If production (agriculture and manufacturing) is the heart of an economy, then **commerce** is its circulatory system. Commerce ensures that goods and services reach the final consumer efficiently. It encompasses all the activities involved in buying, selling, and distributing goods, as well as all the supporting services that make trade possible. Without commerce, a farmer's yams would rot in the village, and a factory's products would sit in a warehouse.

2. MEANING OF COMMERCE AND COMMERCIAL ACTIVITIES

Commerce is the study of the sum total of all activities involving the exchange of goods and services from the producer to the final consumer. It is divided into two main branches:

1. **Trade:** The actual *buying and selling* of goods and services. This is the core of commerce.
 - **Home Trade:** Trade within a country (e.g., selling Lagos-made clothes in Kano).
 - *Wholesale Trade:* Buying in bulk from manufacturers and selling in smaller quantities to retailers.
 - *Retail Trade:* Selling in small units directly to the final consumer (e.g., supermarkets, corner shops).
 - **Foreign Trade:** Trade between different countries.
 - *Export:* Selling goods to other countries (e.g., Nigeria selling cocoa to the Netherlands).
 - *Import:* Buying goods from other countries (e.g., Nigeria buying cars from Japan).
2. **Aids to Trade (Commercial Activities):** These are the services that facilitate and support trade, making it smoother, faster, and less risky. They are the "glue" that holds the commercial system together.

3. AIDS TO TRADE (COMMERCIAL ACTIVITIES)

1. **Banking:** Provides the financial lifeblood for commerce. Banks offer:
 - **Loans and Credit:** For businesses to start or expand.
 - **Safe Keeping of Money:** Through savings and current accounts.
 - **Facilitating Payments:** Through transfers, cheques, and online transactions.
2. **Insurance:** Protects businesses from potential financial losses due to risks like fire, theft, accidents, or damage to goods in transit. It provides a safety net that encourages trade.

3. **Transportation:** Physically moves goods from where they are produced to where they are needed. Without transport, trade over any distance would be impossible.
4. **Warehousing:** Provides storage for goods until they are needed. This helps to stabilize prices by ensuring goods are available all year round, not just during harvest or production season.
5. **Communication:** Essential for exchanging information between buyers and sellers. This includes advertising to inform customers, as well as phones and the internet for placing orders and negotiating deals.
6. **Tourism:** While often considered part of the service sector, tourism is a major commercial activity that involves the buying and selling of travel, accommodation, and recreational experiences.

4. MAJOR COMMERCIAL CENTRES IN NIGERIA

The growth of these centres is not accidental; it is due to a combination of favourable factors.

Commercial Centre	Primary Factors for Growth	Key Commercial Features
Lagos	Historical Port, Political Capital (formerly), Population, Infrastructure.	Headquarters of most banks and corporations. Home to Apapa and Tin-Can ports. Largest consumer market.
Kano	Historical Trans-Saharan Trade Route, Agricultural Hub of the North.	Major market for agricultural produce (grains, hides). Kurmi Market is a historic international trade centre.
Onitsha	Strategic Location on the River Niger Bridge, Entrepreneurial Spirit.	Onitsha Main Market is one of the largest open markets in West Africa. A hub for wholesale trade.
Aba	Industrial Base (Garments, Footwear), Strategic SE	A centre for small-scale manufacturing and commerce. Famous for "Aba-made"

Commercial Centre	Primary Factors for Growth	Key Commercial Features
	Location.	goods.
Ibadan	Historical, Large Population, Administrative Centre.	Major market for agricultural produce from the surrounding region. Hosts the popular Bodija Market.
Port Harcourt	Oil and Gas Industry, Port Facility.	Centre for commerce related to the oil sector.

5. THE ROLE OF COMMERCE IN NATIONAL DEVELOPMENT

1. **Creates Employment:** Millions are employed as retailers, bankers, transporters, advertisers, and insurance agents.
2. **Improves Standard of Living:** Makes a wide variety of goods and services available to consumers, increasing choice and quality of life.
3. **Stimulates Production:** By creating a ready market, commerce encourages farmers and manufacturers to produce more.
4. **Generates Government Revenue:** Through taxes like Value Added Tax (VAT), company income tax, and customs duties on imports/exports.
5. **Promotes International Relations:** Foreign trade fosters economic and diplomatic ties between Nigeria and other nations.
6. **Development of Urban Areas:** Commercial centres grow into cities, attracting infrastructure and other development.

6. CHALLENGES FACING COMMERCE IN NIGERIA

1. **Inadequate Infrastructure:** Poor road networks, erratic power supply, and inefficient ports increase the cost and time of doing business.
2. **Insecurity:** Armed robbery, kidnapping on highways, and piracy disrupt the free flow of goods and people.

3. **Multiple Taxation:** Businesses are often burdened by numerous taxes and levies from different levels of government.
 4. **Unstable Economic Policies:** Frequent changes in fiscal and monetary policies (e.g., foreign exchange regulations) create uncertainty for traders and investors.
 5. **Counterfeiting and Piracy:** The sale of fake and substandard goods hurts legitimate businesses and poses risks to consumers.
 6. **Financial Exclusion:** Many small traders, especially in rural areas, lack access to formal banking and credit facilities.
-

7. EVALUATION & ASSIGNMENT

Evaluation (Past Questions):

1. **(NECO 2019):** (a) What is commerce? (b) List and explain any three auxiliary services that aid commerce.
2. **(WAEC 2017):** Highlight four problems affecting commercial activities in Nigeria.

Assignment:

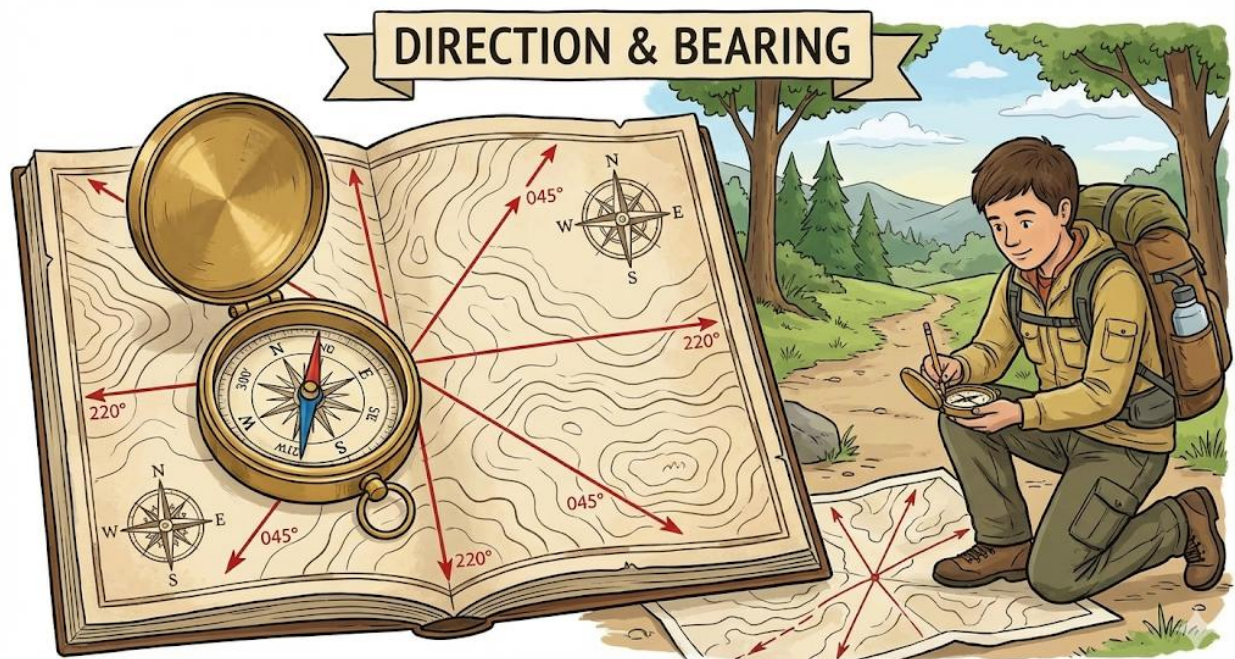
1. **Commercial Chain Analysis:** Choose a common product (e.g., a bottle of soft drink or a loaf of bread). Trace its commercial journey from the factory to your hand. Identify and describe the role of at least four different commercial activities (e.g., Transport, Wholesaling, Retailing, Advertising) in this chain.
2. **Problem-Solving Proposal:** Write a formal letter to the Chairman of your Local Government Area. Identify one major challenge facing commerce in your local market (e.g., poor sanitation, lack of security, bad access roads). Propose two concrete solutions to address this problem.
3. **Debate Preparation:** Prepare talking points for a debate on the topic: "The rise of e-commerce (online shopping) will eventually make physical markets like Onitsha Main Market irrelevant." Argue for or against the motion.

SS2 GEOGRAPHY LESSON NOTES

THIRD TERM

WEEK 1 - DIRECTION AND BEARING

Introduction to Direction and Bearing Direction and bearing are fundamental concepts in geography and navigation, allowing us to describe the precise location and orientation of places and objects on the Earth's surface. They are essential for map reading, surveying, navigation (on land, sea, and air), and describing spatial relationships.



1. Meaning of Direction and Bearing

- **Direction:** This is the general line or path along which something is moving, pointing, or facing. It is a qualitative measure, often given in relative terms like "left," "right," "towards the sea," or using cardinal points. Direction answers the question, "Which way?"
- **Bearing:** This is a precise, quantitative method of expressing direction. It is the angle measured in degrees ($^{\circ}$) from a fixed reference direction (usually True North), moving

clockwise. Bearings provide an exact numerical value for direction, eliminating ambiguity. They answer the question, "At what exact angle from North?"

2. Types of Direction To systematically describe direction, we use a compass rose.

- **Cardinal Points (Primary Directions):** The four principal points.
 - North (N) - 0° or 360°
 - East (E) - 90°
 - South (S) - 180°
 - West (W) - 270°
- **Intermediate Directions (Secondary Directions):** Points halfway between the cardinal points.
 - North-East (NE) - 45°
 - South-East (SE) - 135°
 - South-West (SW) - 225°
 - North-West (NW) - 315°
- **Tertiary Directions:** For even greater precision, directions can be further subdivided (e.g., North-North-East (NNE) at 22.5° , East-North-East (ENE) at 67.5°).

3. Compass or Angular Bearing (0° – 360°) This is the most precise system. A bearing is always a three-figure value (e.g., 045° , 180° , 005°).

- **Measurement:** Start from True North (0°), and measure the angle clockwise to the line pointing to your destination.
- **Key Bearings:**
 - North: 000° or 360°
 - East: 090°
 - South: 180°
 - West: 270°
- **Example:** A bearing of 045° is exactly North-East. A bearing of 130° is South-East.

4. The "Norths": True, Magnetic, and Grid This is a critical concept for accurate navigation, as a compass does not point to the geographic North Pole.

- **True North (Geographic North):** The direction along the Earth's surface towards the geographic North Pole, where all lines of longitude converge. It is fixed. This is the North used in cartography.
- **Magnetic North:** The direction a magnetic compass needle points towards, which aligns with the Earth's magnetic field. The Magnetic North Pole is not stationary and moves over time due to changes in the Earth's molten outer core.
- **Grid North:** The direction northwards along the grid lines (vertical lines) on a map. Because map sheets are flat but the Earth is curved, grid lines and true north lines do not always align perfectly.
- **Magnetic Declination (or Variation):** The angle between True North and Magnetic North at any given location. This value is crucial and is usually found in the map's margin. To convert a **Magnetic Bearing** (from a compass) to a **True Bearing** (for a map):
 - $\text{True Bearing} = \text{Magnetic Bearing} + \text{Magnetic Declination}$
 - (Note: If declination is West, you subtract; if East, you add. Always check the map's specific instruction).

5. Back Bearings A back bearing is the bearing from your destination back to your starting point. It is the exact opposite direction of the forward bearing.

- **Calculation:**
 - If the forward bearing is **less than 180°**, add 180°.
 - If the forward bearing is **more than 180°**, subtract 180°.
- **Example 1:** Forward Bearing from A to B is 070°. Back Bearing from B to A is $070^\circ + 180^\circ = 250^\circ$.
- **Example 2:** Forward Bearing from C to D is 300°. Back Bearing from D to C is $300^\circ - 180^\circ = 120^\circ$.

6. Use of Compass and Protractor in Map Work

- **Compass:** A field instrument containing a magnetized needle that aligns with the Earth's magnetic field, pointing to Magnetic North. Used to find directions on the ground and take bearings of visible landmarks.
- **Protractor:** A semi-circular or circular tool marked in degrees (0° to 180° or 0° to 360°). Used on maps to measure the angle (bearing) between North and a line connecting two points.

Practical Application on Topographic Maps

- **Finding Bearing from A to B on a Map:**
 - a. Draw a straight line connecting point A and point B.
 - b. Draw the North-South grid line through point A.
 - c. Place the center of the protractor on point A, aligning its base with the North-South line and the $0^\circ/180^\circ$ mark facing North.
 - d. Read the degree value where the line A-B crosses the protractor's scale. This is the grid bearing from A to B.

EVALUATION (WAEC/NECO Past Questions) *(Answers should now be more detailed based on the expanded notes)*

1. (WAEC June 2018, Q3a) Define: (i) Direction (ii) Bearing, highlighting the key difference (qualitative vs. quantitative).
2. (NECO June 2020, Q4b) Using a diagram, differentiate between True North and Magnetic North, explaining the concept of magnetic declination.
3. (WAEC June 2016, Q2c) Explain the step-by-step process of using a protractor on a map and a compass in the field to determine the true bearing between two locations.
4. (NECO 2017 Q5) Explain the difference between forward and back bearing with a calculation example.

ASSIGNMENT

1. On a map, the bearing of a hill from a school is 110° . What is the back bearing of the school from the hill? (**Answer: 290°**)

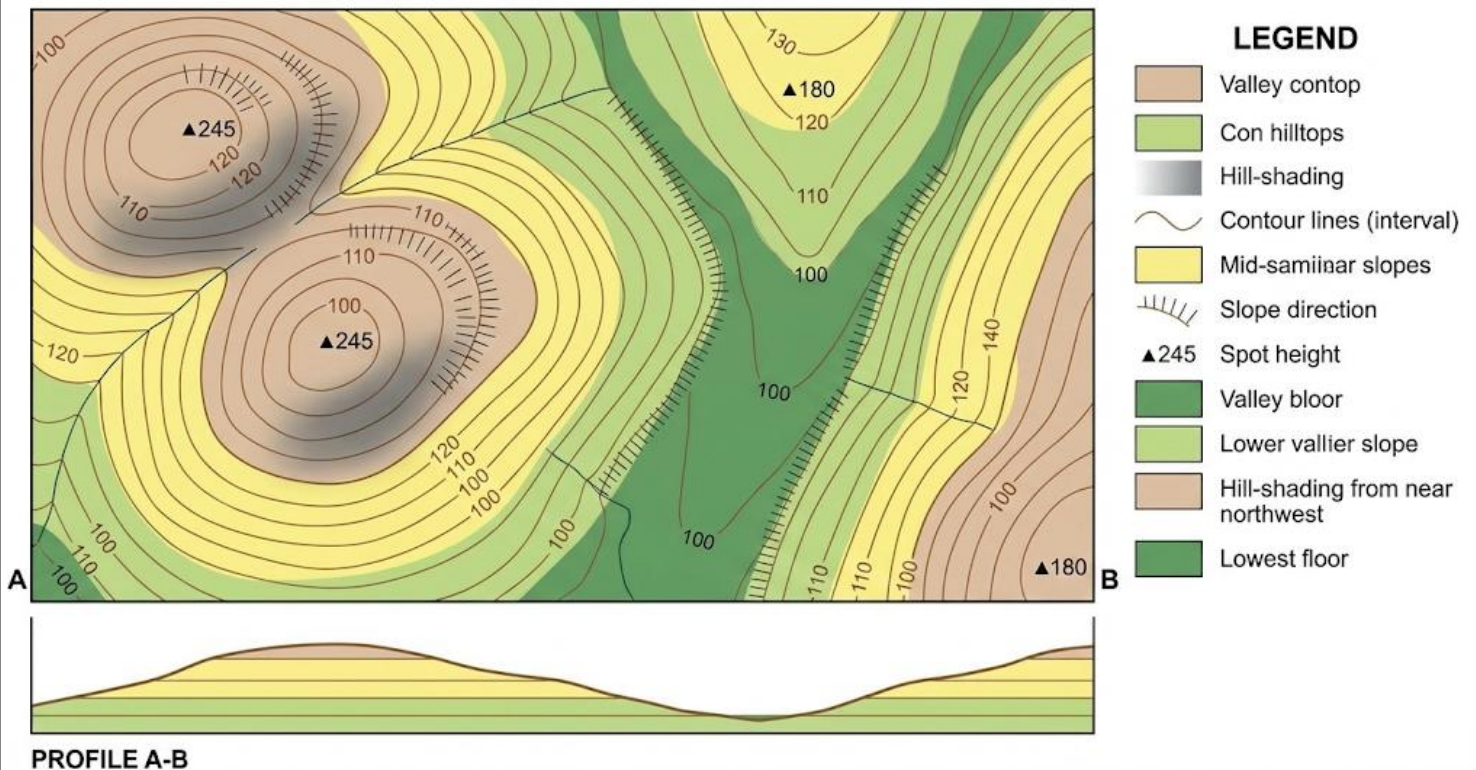
2. A student walks on a bearing of 045° . What type of direction is this? (**Answer: B. North-East**)
 3. (WAEC 2014) Draw a fully labelled 16-point compass rose.
 4. Differentiate between: a. True North and Magnetic North. b. Grid North and Magnetic North.
 5. Using a topographic map, determine the bearing of a specific feature (e.g., a dam) from a nearby settlement and classify the direction as cardinal or intermediate.
-



WEEK 2 - REPRESENTATION OF RELIEF FORMS

Introduction to Relief Relief describes the shape and configuration of the Earth's surface—its hills, valleys, plains, and mountains. It is defined by the variations in elevation (height) and slope (steepness) across a landscape. Understanding relief is vital as it influences drainage patterns,

REPRESENTATION OF RELIEF FORMS: MAP READING DIAGRAM



climate, soil formation, land use, settlement location, and transportation routes.

1. Methods of Representing Relief on Maps Since maps are two-dimensional, cartographers use various techniques to depict three-dimensional terrain.

- **a. Contour Lines:** The most important and accurate method.
 - **Definition:** Imaginary lines on a map joining points of equal height above a specified datum (usually sea level).
 - **Interpretation:**
 - **Close Contours:** Indicate a **steep slope** (e.g., cliffs, escarpments).

- **Wide Contours:** Indicate a **gentle slope** (e.g., plains).
 - **Uniform Spacing:** Indicate a **constant slope**.
 - **Concentric Circles:** Indicate a **hill** or **mountain** (height increases towards the center).
 - **V-Shaped Patterns:** Indicate a **valley**. The "V" points upstream (uphill).
 - **U-Shaped Patterns:** Indicate a **spur** (a projecting ridge of land). The "U" points downhill.
- **Contour Interval:** The vertical distance between one contour line and the next. A small interval shows detailed relief; a large interval shows generalized relief.
- **b. Spot Heights:** A precise point on the map where the exact elevation is written (e.g., 450m). They provide specific, accurate data but not a continuous picture of the terrain.
 - **c. Trigonometrical Stations (Triangulation Pillars):** Represented by a small triangle (Δ) and often a spot height. These are fixed survey points, usually at the highest locations, providing the most accurate elevation data for an area.
 - **d. Layer Tinting (Colouring):** Using different colours to represent different height bands (e.g., dark green for lowlands, yellow for foothills, brown for high mountains). It is visually intuitive but less precise than contours.
 - **e. Hachures:** Short, parallel lines drawn in the direction of the steepest slope. The thickness and density of the lines indicate steepness: thicker, closer lines for steeper slopes. This method gives a shaded relief effect but does not show exact elevations.
 - **f. Hill Shading:** A technique that simulates the effect of sunlight and shadows on the terrain, creating a realistic 3D appearance. It is often used alongside contours.

2. Interpretation of Relief Features Using Contours By analyzing contour patterns, we can identify specific landforms:

- **Hill/Mountain:** Concentric circles of contours with increasing height towards the center.
- **Ridge:** An elongated area of high land. Contours form a pattern of U's or V's pointing away from the higher ground (downhill).

- **Valley:** An elongated lowland between higher areas. Contours form a V-shape that points towards the higher ground (uphill). A river is often present in the valley.
- **Spur:** A smaller ridge projecting from the side of a hill or mountain. Contours form a U-shape pointing away from the hill.
- **Escarpment (Scarp):** A long, steep slope. Shown by very closely spaced contours on the steep face and wider spacing on the gentle back slope.
- **Plateau:** An extensive area of level highland. Shown by wide, flat areas of contours at a high elevation, often with steep edges (closely spaced contours).

3. Drawing and Interpreting Cross-Sections A cross-section is a side-profile view of the land along a straight line on a map (e.g., line A-B).

- **Steps to Draw a Cross-Section:**
 - a. Place a straight edge of paper along the line A-B on the map.
 - b. Mark on the paper every point where a contour line crosses the line A-B, and note the height of each contour.
 - c. On graph paper, choose a **vertical scale** (e.g., 1 cm = 100 m) and a **horizontal scale** (usually the same as the map scale, e.g., 1:50,000).
 - d. Plot the height points from your paper strip onto the graph paper.
 - e. Join the points with a smooth, freehand curve to represent the ground surface.
- **Vertical Exaggeration (VE):** Because the vertical dimension (height) is so small compared to the horizontal dimension (distance), we often exaggerate the vertical scale to make the relief features visible.
 - **Formula:** Vertical Exaggeration (VE) = Vertical Scale / Horizontal Scale
 - **Example:** If Horizontal Scale is 1 cm = 1000 m (1:100,000) and Vertical Scale is 1 cm = 100 m, then $VE = (100 \text{ m/cm}) / (1000 \text{ m/cm}) = 1/10$. This means the vertical scale is 10 times more exaggerated than the horizontal scale.
- **Intervisibility:** Determining whether two points on a map are visible to each other without any land obstruction.

- **Method:** Draw a cross-section between the two points. Then, draw a straight line of sight from the top of one point to the top of the other. If this line does not touch the ground profile in between, the points are intervisible. If it does, the view is blocked.
-

EVALUATION

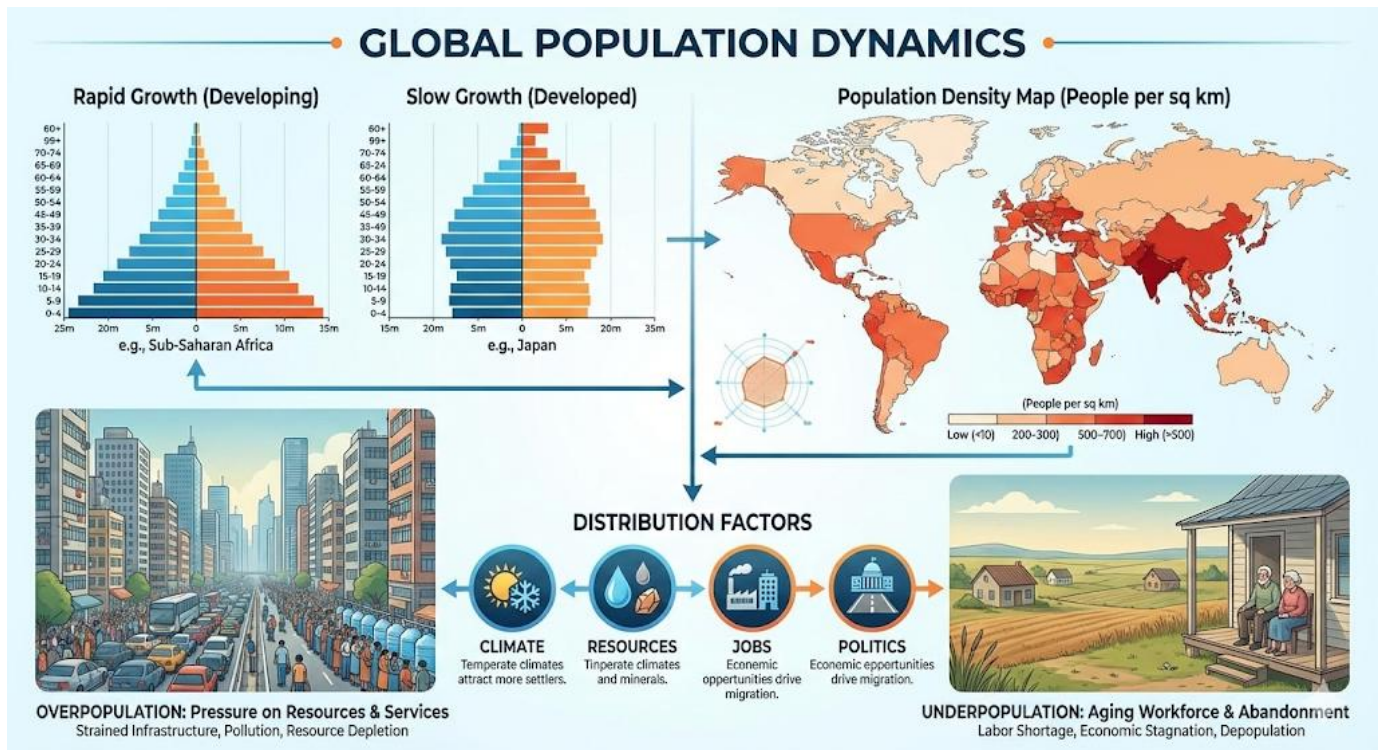
1. (WAEC 2022 Q3a) With the aid of diagrams, describe three methods of representing relief (e.g., contours, spot heights, layer tinting).
2. (WAEC 2020 Q2b) Explain contour lines and describe two landforms identifiable from their contour patterns (e.g., valley and hill).
3. (NECO 2021 Q4) Describe how to determine intervisibility between two points using a cross-section.
4. (WAEC 2018 Q2c) Explain, with a calculation, how vertical exaggeration is determined.

ASSIGNMENT

1. Define relief and list five methods of representation.
 2. Draw a simple contour map of a conical hill and explain its characteristics.
 3. Calculate the VE for a cross-section with a horizontal scale of 1:50,000 and a vertical scale of 1 cm = 50 m. (**Answer: VE = 1**)
 4. Differentiate between a spur and a valley using simple contour diagrams.
 5. Explain the procedure for determining intervisibility.
-

WEEK 3 - POPULATION

Introduction to Population Studies Population geography studies the spatial aspects of human populations—their size, composition, distribution, growth, and movement. Understanding these dynamics is crucial for national planning, resource allocation, and economic development.



1. Meaning of Population and Related Terms

- **Population:** The total number of people inhabiting a specific geographical area (e.g., a village, country, or the world) at a given time.
- **Census:** An official, systematic count of a country's population, usually conducted every 10 years. It gathers data on age, sex, occupation, literacy, etc., providing a detailed demographic snapshot.
- **Population Density:** The average number of people per unit area (usually per square kilometre).
 - **Formula:** Population Density = Total Population / Total Land Area.

- **Population Distribution:** The pattern of where people live within an area. It can be **dense** (clustered), **sparse** (scattered), or **random**.
- **Population Growth:** The change in population size over time, influenced by **birth rate**, **death rate**, and **migration**.

2. Sources of Population Data

- **National Population Census:** The primary source, providing comprehensive data.
- **Vital Registration System:** Continuous recording of births, deaths, marriages, and divorces.
- **Sample Surveys:** e.g., Demographic and Health Surveys (DHS), which collect specific data from a representative sample of the population.
- **Administrative Records:** Data from schools, hospitals, tax offices, and voter registration.

3. Factors Affecting Population Distribution and Density

- **Physical Factors:**
 - **Climate:** Temperate climates with reliable rainfall attract dense populations (e.g., Europe). Extreme climates (hot deserts, cold tundras) are sparsely populated.
 - **Relief:** Flat, low-lying plains are easier to build on and farm than steep, mountainous regions.
 - **Soil:** Fertile soils (e.g., river valleys) support agriculture and dense populations.
 - **Water Supply:** People cluster near reliable sources of fresh water (rivers, lakes).
- **Economic Factors:**
 - **Employment Opportunities:** Industrial areas, mining towns, and commercial hubs attract people.
 - **Infrastructure:** Good transport, roads, and communication networks encourage settlement.
- **Social and Political Factors:**
 - **Government Policy:** Policies can encourage settlement in certain areas.
 - **Conflict:** War and instability cause people to flee, leading to sparse populations.
 - **Social Amenities:** Access to good schools, hospitals, and security attracts people.

4. Population Concepts: Over, Under, and Optimum

- **Overpopulation:** Occurs when the population of an area exceeds its available resources, leading to a low standard of living, environmental degradation, and strain on infrastructure.
- **Underpopulation:** Occurs when there are too few people to fully utilize the available resources, potentially leading to slow economic growth and labor shortages.
- **Optimum Population:** A theoretical ideal where the population size is in perfect balance with the available resources, allowing for the highest possible standard of living.

5. Population Pyramid (Age-Sex Diagram) A graphical illustration that shows the distribution of various age groups and gender in a population.

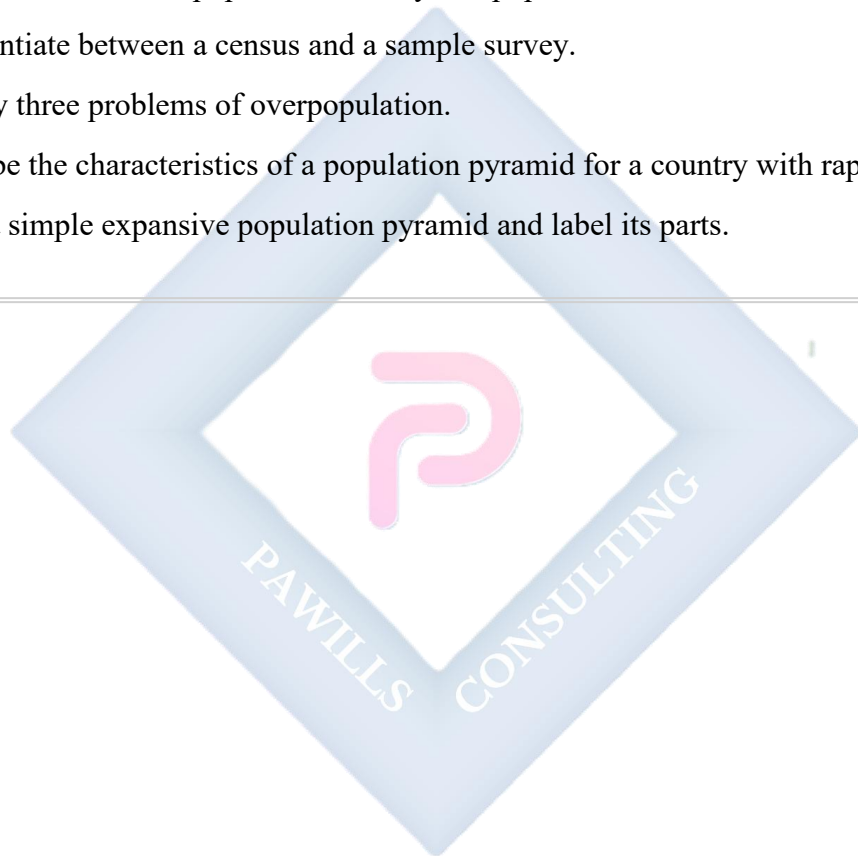
- **Components:**
 - **Vertical Axis:** Represents age cohorts (e.g., 0-4, 5-9).
 - **Horizontal Axis:** Represents percentage of the population. Males are on the left, females on the right.
- **Types and Interpretation:**
 - **Expansive (Triangular):** Wide base, narrowing top. Indicates a high birth rate, high death rate, and a youthful population. Typical of **developing countries** (e.g., Nigeria, Niger). Implication: Need for more schools, jobs, and high dependency ratio.
 - **Constrictive (Beehive/Urn-shaped):** Narrowing base. Indicates a declining birth rate and an aging population. Typical of **developed countries** (e.g., Japan, Germany). Implication: Strain on pension systems, need for healthcare for the elderly.
 - **Stationary (Column-shaped):** Roughly equal proportions in most age groups. Indicates low birth and death rates and a stable population.

EVALUATION

1. (WAEC 2016 Q2) Define census and list problems in conducting it (e.g., high cost, illiteracy, religious beliefs, inaccurate data).
2. (WAEC 2017 Q4) Explain overpopulation, underpopulation, and optimum population.
3. (NECO 2018 Q3) Draw and label a population pyramid and describe the two main types.
4. (WAEC 2020 Q5) State five factors affecting population distribution in West Africa.

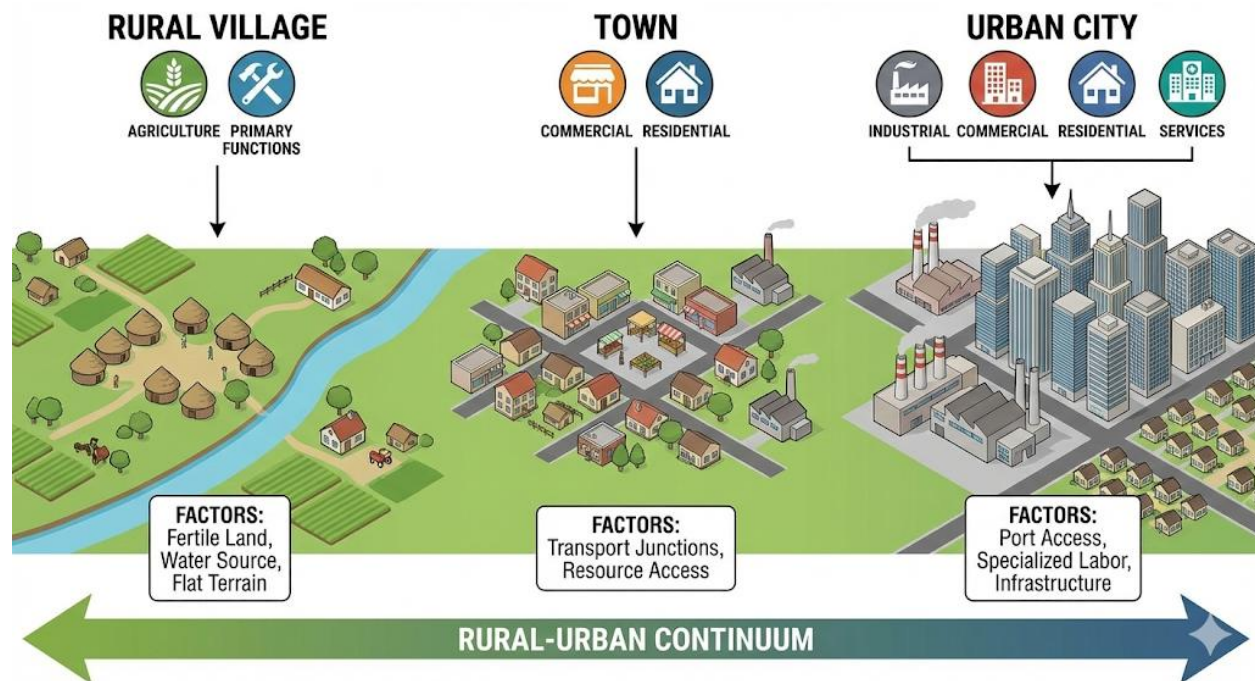
ASSIGNMENT

1. Differentiate between population density and population distribution.
 2. Differentiate between a census and a sample survey.
 3. Identify three problems of overpopulation.
 4. Describe the characteristics of a population pyramid for a country with rapid growth.
 5. Draw a simple expansive population pyramid and label its parts.
-



WEEK 4 - SETTLEMENTS

Introduction to Settlements A settlement is any place where people live and carry out social and economic activities. It is the physical expression of human organization on the landscape.



1. Meaning and Origin of Settlements Settlements originated when nomadic hunter-gatherers began practicing agriculture during the Neolithic Revolution. This led to a sedentary lifestyle, requiring permanent homes near farmlands and water sources. These early clusters evolved into the complex settlements we see today.

2. Types of Settlements

- **Temporary Settlements:** Places of abode not intended for long-term use (e.g., refugee camps, nomadic herders' camps, logging camps).
- **Permanent Settlements:** Established for long-term habitation with durable infrastructure (e.g., villages, towns, cities).

3. Classification of Settlements: Rural vs. Urban

- **Rural Settlements:**

- **Characteristics:** Small size, low population density, economy based on primary activities (farming, fishing, mining).
- **Examples:** Hamlets (tiny clusters), Villages.

- **Urban Settlements:**

- **Characteristics:** Large size, high population density, economy based on secondary (manufacturing) and tertiary (services) activities, complex social and administrative structures.
- **Examples:** Towns, Cities, Conurbations (e.g., Lagos Metropolitan Area), Megalopolis (e.g., the merging of Boston, New York, and Washington D.C. in the USA).

4. Settlement Patterns The spatial arrangement of buildings and houses.

- **Nucleated/Clustered:** Buildings are grouped closely together, often around a central feature like a crossroads or market. Promotes social interaction and is common in fertile agricultural areas.
- **Dispersed/Scattered:** Buildings are spread out over a wide area. Common in mountainous or extensive farming regions where land is less fertile.
- **Linear:** Buildings are arranged in a line, typically along a road, river, valley, or coastline. Facilitated by transport and access to resources.

5. Settlement Hierarchy Settlements can be ranked based on their size, population, and the number and range of services they offer.

- **Hierarchy (smallest to largest):** Isolated Dwelling → Hamlet → Village → Town → City → Conurbation/Megalopolis.
- **Sphere of Influence:** The area served by a settlement. Larger settlements have a wider sphere of influence and provide higher-order goods and services (e.g., universities, specialized hospitals) that people are willing to travel further for.

6. Functions of Settlements Most settlements are **multi-functional**, but they often have a primary function.

- **Residential:** Primarily for living (e.g., satellite towns).
- **Commercial:** Centres of trade and finance (e.g., Onitsha, Lagos).
- **Industrial:** Centred on manufacturing (e.g., industrial estates in Port Harcourt).
- **Administrative:** Host government headquarters (e.g., Abuja, State Capitals).
- **Cultural/Religious:** Of historical or spiritual significance (e.g., Jerusalem, Mecca, Ile-Ife).
- **Port/Transport:** Located at hubs of transportation (e.g., Apapa Port, Murtala Muhammed Airport).

EVALUATION

1. (WAEC June 2018) Distinguish between rural and urban settlements. Highlight four functions of urban settlements. Describe three rural settlement patterns.
2. (NECO June 2020) Define "settlement." Explain temporary and permanent settlements. State three factors influencing nucleated patterns.
3. (WAEC June 2015) Explain "settlement hierarchy." Describe the relationship between settlement size and function.

ASSIGNMENT

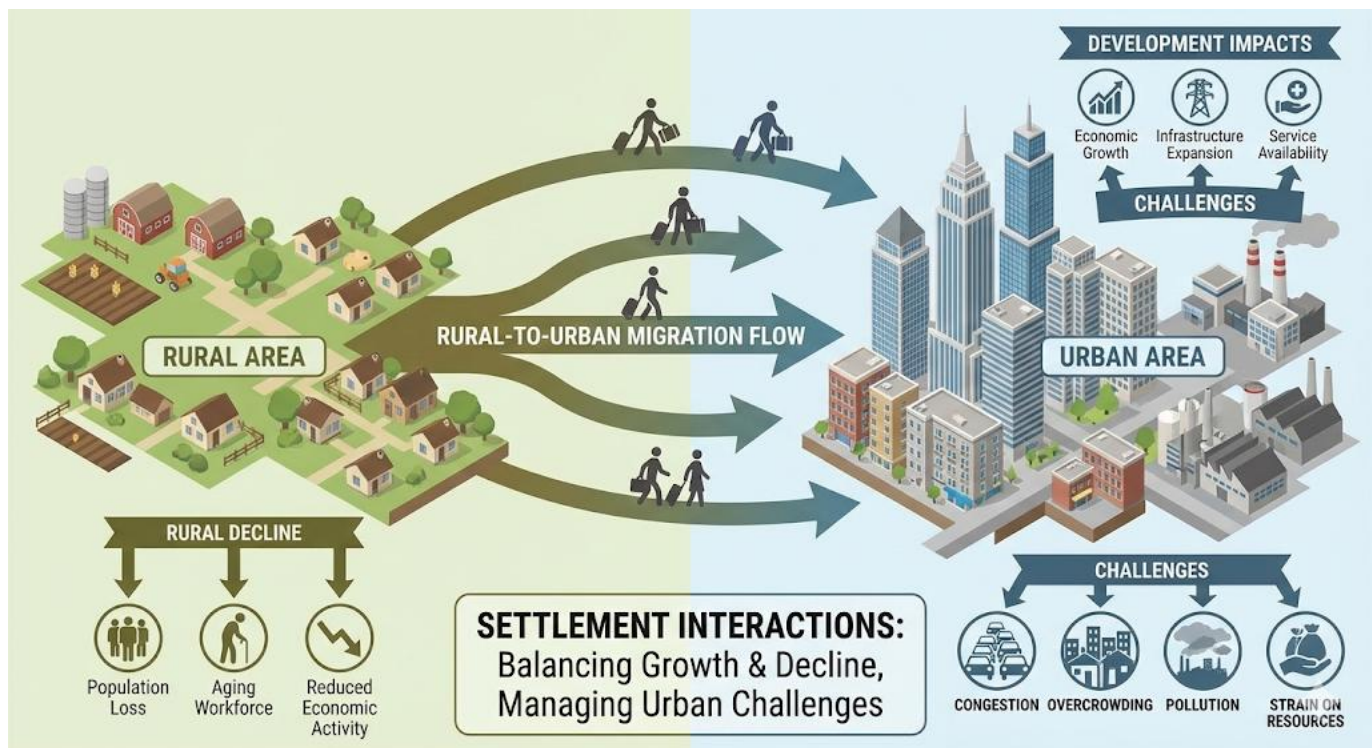
1. Define settlement and outline two reasons for their establishment (e.g., defence, trade, water supply).
2. Classify: Hamlet (Rural), Town (Urban), Village (Rural), Megalopolis (Urban).
3. Differentiate between linear and nucleated patterns with examples.
4. Identify four functions of an urban settlement in Nigeria (e.g., Lagos: commercial, industrial, port, administrative).
5. Draw labelled diagrams of nucleated and dispersed settlement patterns.

(Weeks 5 and 7 are assumed to be for revision, tests, or other school activities, so we proceed to Week 6.)



WEEK 6 - SETTLEMENT INTERACTION

Introduction: The Symbiotic Relationship Settlement interaction describes the complex, interdependent relationships and exchanges that occur between different settlements, most notably between rural and urban areas. This interaction is not a one-way street but a dynamic, symbiotic system where each zone relies on the other for survival and growth. The flows of people, goods, capital, and information create a functional regional economy and shape the



development of a nation.

1. Rural-Urban Migration: The Human Flow This is the demographic component of settlement interaction, involving the movement of people from rural to urban areas. It is one of the most significant human geography processes in developing nations.

- **Driving Forces: The Push-Pull Framework**
 - **Push Factors (Forces Expelling People from Rural Areas):**

- **Economic:** Lack of diverse employment, low agricultural income, land fragmentation, and poverty.
- **Social:** Limited access to quality higher education, specialized healthcare, and entertainment.
- **Environmental:** Land degradation, soil erosion, drought, and other climate-related challenges that threaten livelihoods.
- **Political/Infrastructural:** Perceived government neglect, lack of investment in rural infrastructure (electricity, good roads, internet).
- **Pull Factors (Attractions of Urban Areas):**
 - **Economic:** Perception of abundant jobs in diverse sectors (industry, services), higher wages, and greater opportunities for entrepreneurship.
 - **Social:** Access to prestigious universities, advanced hospitals, cultural institutions (museums, cinemas), and a more modern lifestyle.
 - **Psychological:** The allure of "city life" associated with progress, anonymity, and freedom from traditional constraints.
- **Multifaceted Effects:**
 - **On Rural Areas (Origin):**
 - **Negative:** Brain drain (loss of the most educated and youthful population), an aging demographic left behind (often called the "geriatrification" of the countryside), abandonment of farmland, and the breakdown of traditional community structures.
 - **Positive:** Inflow of **remittances** (money sent back by migrants) which can improve living standards for remaining families and stimulate local investment. Exposure to new ideas and technologies from urban areas.
 - **On Urban Areas (Destination):**
 - **Negative: Over-urbanization**—where urban population growth outpaces the development of infrastructure and jobs. This leads to:

- **Proliferation of Slums:** Unplanned settlements with poor sanitation, insecure tenure, and overcrowding (e.g., Makoko in Lagos).
- **Strained Services:** Overcrowded schools, overwhelmed healthcare facilities, and chronic water and power shortages.
- **Rising Unemployment & Underemployment:** A surplus of low-skilled labor leads to a large informal sector.
- **Increased Social Vices:** Higher rates of crime, prostitution, and social dislocation.
- **Positive:** Supply of a large, often low-wage labor force that fuels urban industries and services. Cultural diversification and a vibrant market for goods and services.

2. Movement of Goods and Services: The Economic Flow This is the economic bloodstream of the nation, connecting areas of production with areas of consumption and processing.

- **From Rural to Urban:**
 - **Agricultural Produce:** Foodstuffs (grains, vegetables, fruits, meat) to feed the urban population.
 - **Raw Materials:** Cash crops (e.g., cocoa, cotton), timber, minerals, and water resources that fuel urban industries.
 - **This flow ensures urban food security and supplies industries.**
- **From Urban to Rural:**
 - **Manufactured Goods:** Processed foods, clothing, electronics, machinery, fertilizers, and construction materials.
 - **Services:** Banking, insurance, specialized healthcare, higher education, and government administration.
 - **This flow modernizes rural life, improves agricultural productivity, and provides essential services.**

- **The Role of Market Centers:** **Periodic markets** in rural areas and **central markets** in urban areas act as critical nodes where these flows converge, facilitating exchange and setting regional prices.

3. Transportation and Communication Links: The Circulatory System These are the physical and digital networks that make interaction possible.

- **Transportation Networks:**
 - **Roads:** The most critical for rural-urban links, especially for perishable goods.
 - **Railways:** Efficient for bulk cargo over long distances (e.g., moving grains or minerals).
 - **Waterways:** Used in riverine and coastal areas for moving heavy goods cheaply.
 - **The quality and density of this network directly determine the strength and cost of interaction.**
- **Communication Technologies:**
 - **Mobile Phones & Internet:** Have revolutionized interaction by allowing:
 - **Price Information:** Farmers can check market prices in urban centers before selling.
 - **Financial Transactions:** Mobile money and banking facilitate remittances.
 - **Social Connection:** Migrants maintain strong ties with their rural families.
 - **The "Digital Divide"** between urban and rural areas in terms of internet speed and reliability remains a significant barrier to full integration.

4. Urbanization and Associated Problems: The Consequences of Concentration

Urbanization is the increasing percentage of a population living in urban areas. While a natural correlate of development, rapid and unplanned urbanization leads to severe challenges.

- **Key Urban Problems:**
 - **Environmental Degradation:**

- **Air Pollution:** From industries and dense vehicle traffic, leading to respiratory problems.
- **Water Pollution:** Inadequate sewage treatment and industrial waste contaminate water bodies.
- **Solid Waste Management:** Cities struggle to collect and dispose of the massive volumes of garbage generated, leading to littering and illegal dumpsites.
- **Infrastructural Deficit:** Chronic shortage of affordable housing, leading to slums; inadequate road networks causing perpetual traffic congestion; and unreliable water and power supply.
- **Socio-Economic Challenges:** High cost of living, growth of an urban underclass, and increased social inequality.
- **Managing Urbanization:**
 - **Integrated Development Planning:** Creating master plans that anticipate growth.
 - **Investment in Public Transport:** To reduce congestion and pollution.
 - **Promotion of Rural Development:** The most sustainable solution is to reduce the push factors by improving economic opportunities and quality of life in rural areas through electrification, road construction, and establishing agro-based industries.

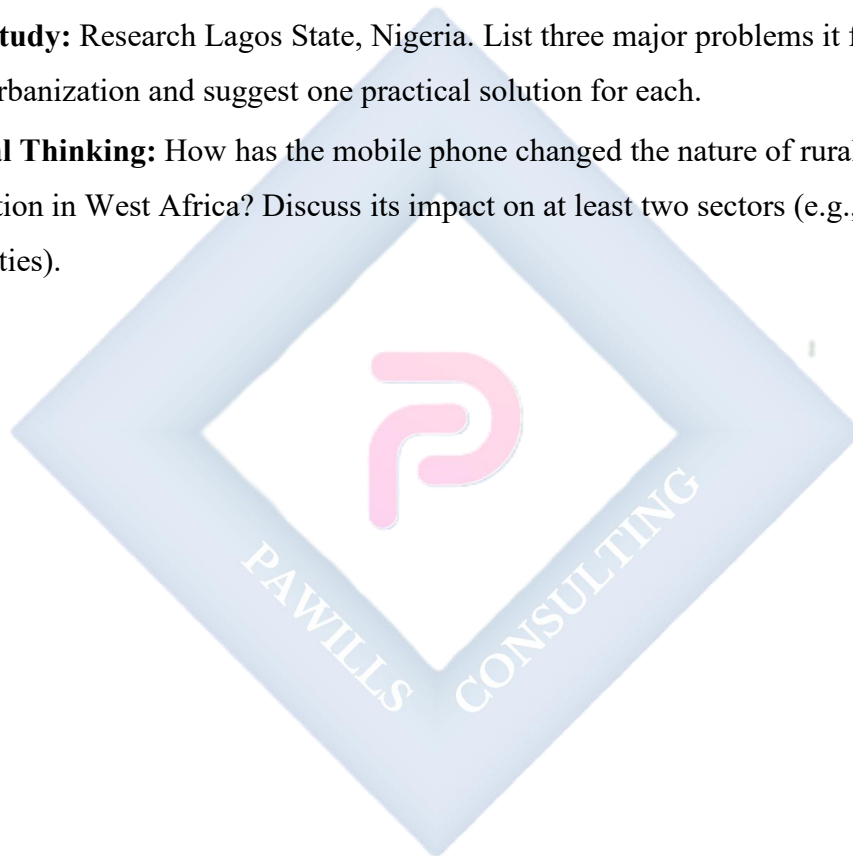
EVALUATION (WAEC/NECO Past Questions)

1. (WAEC June 2017): "Outline three effects of rural-urban migration on rural communities." *(Now, you can discuss both positive/remittances and negative/brain drain).*
2. (NECO 2018): "Explain any four problems associated with rapid urbanization in developing countries." *(You can now structure this by environmental, infrastructural, and social problems).*
3. (WAEC June 2014): "Describe three ways through which rural settlements interact with urban settlements." *(You can elaborate on the movement of goods, people, and information).*

4. (NECO 2020): "State four major causes of rural-urban migration in Nigeria." (*Use the detailed push-pull factors*).

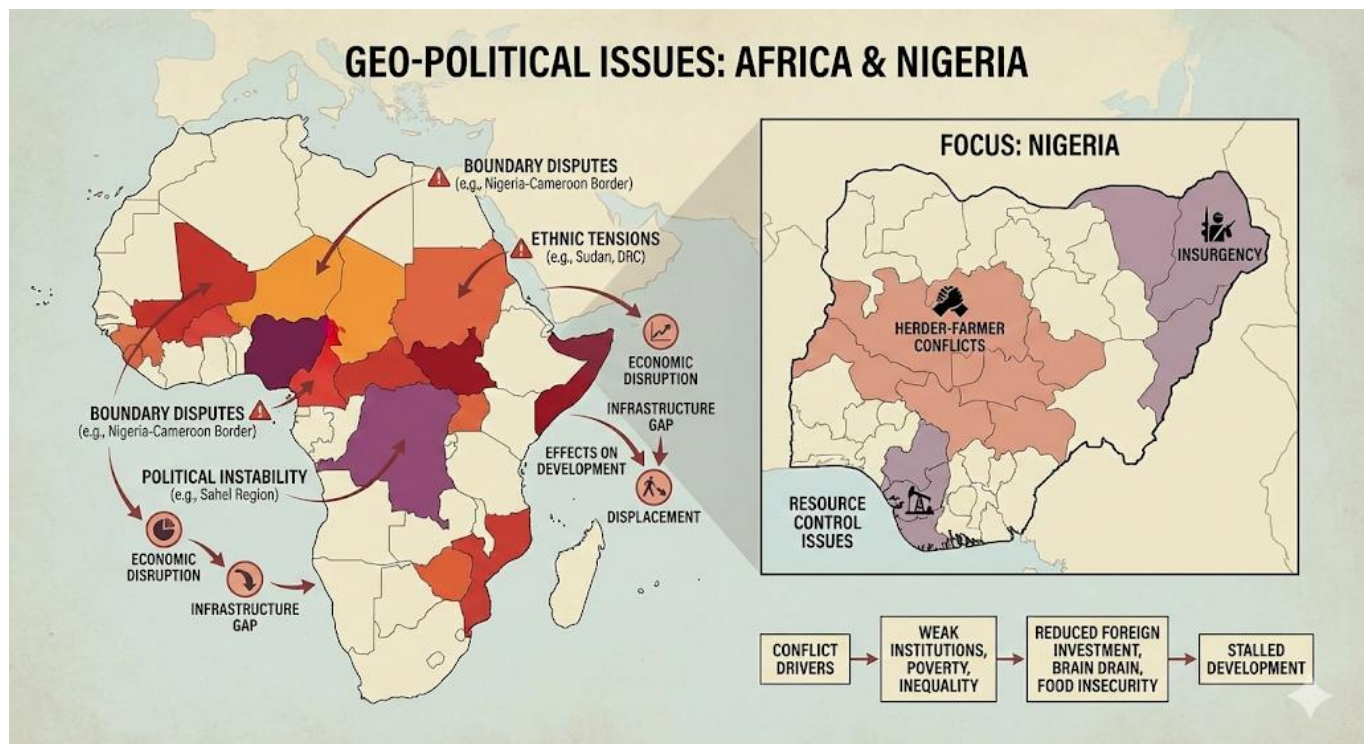
ASSIGNMENT

1. **Analyze:** "The positive effects of rural-urban migration on urban areas are outweighed by the negative effects." Do you agree with this statement? Give reasons for your answer.
2. **Diagram:** Draw a well-labelled flow diagram to show the interdependence between a rural farming village and a major city.
3. **Case Study:** Research Lagos State, Nigeria. List three major problems it faces due to rapid urbanization and suggest one practical solution for each.
4. **Critical Thinking:** How has the mobile phone changed the nature of rural-urban interaction in West Africa? Discuss its impact on at least two sectors (e.g., commerce, family ties).



WEEK 8 - GEO-POLITICAL ISSUES

Introduction: The Geography of Power Geo-politics examines how geographical factors—location, resources, size, and population—influence political power, international relations, and domestic governance. In Nigeria's context, it involves understanding how the country's immense diversity is managed through political structures to maintain unity while acknowledging regional differences.



1. Meaning of Geo-Politics and Geo-Political Zones

- **Geo-Politics:** The study of the effects of geography (human and physical) on international politics and international relations. It analyzes how factors like territory, population, resource distribution, and strategic location shape a country's power and foreign policy.
- **Geo-Political Zones:** In Nigeria, these are regional groupings of states primarily based on geographical location and ethno-cultural affinities. They serve as a framework for administrative convenience, political representation, and resource allocation, aiming to ensure equity among Nigeria's diverse groups.

2. Historical Origin of Geo-Political Zoning in Nigeria The current six-zone structure is not in the Nigerian constitution but is a political creation with deep historical roots:

- **Pre-Independence Regions:** Initially, Nigeria was divided into three regions: Northern, Western, and Eastern (1946-1963).
- **Mid-Western Region:** A fourth region (the Mid-West) was created in 1963.
- **State Creation Era:** From 1967, these regions were progressively broken down into states to weaken regional power blocs and bring government closer to the people.
- **The Six-Zone Framework:** The current six geo-political zones were formally proposed and institutionalized during the constitutional conferences of the 1990s, particularly under General Sani Abacha's regime. The primary goal was to create a more balanced structure for power-sharing and national integration.

3. Nigeria's Six Geo-Political Zones: Detailed Analysis

Geo-Political Zone	States	Key Features & Economic Base	Major Ethnic Groups	Unique Characteristics
North Central (Middle Belt)	Benue, Kogi, Kwara, Nasarawa, Niger, Plateau, FCT	Agriculture (Nigeria's "food basket"), hydropower (Kainji, Shiroro dams), solid minerals (tin, columbite).	Very diverse: Nupe, Tiv, Berom, Igala, etc.	Ethnic & Religious Diversity; serves as a bridge between North and South; hosts the Federal Capital Territory (Abuja).
North East	Adamawa, Bauchi, Borno, Gombe, Taraba, Yobe	Agriculture (livestock, cotton), trading .	Hausa, Kanuri, Fulani	Insurgency Challenge; the epicentre of the Boko Haram conflict; has the largest landmass of the zones.
North	Jigawa,	Commerce (Kano is	Predominantly	Highest Population

Geo-Political Zone	States	Key Features & Economic Base	Major Ethnic Groups	Unique Characteristics
West	Kaduna, Kano, Katsina, Kebbi, Sokoto, Zamfara	a historic Sahel trade hub), agriculture (sorghum, millet), textiles .	Hausa-Fulani	Density in the North; strong Islamic heritage; faces challenges of desertification.
South East	Abia, Anambra, Ebonyi, Enugu, Imo	Commerce & Industry (Nnewi - auto parts; Aba - manufacturing), oil & gas .	Predominantly Igbo	Entrepreneurial Spirit ; high population density; significant diaspora network; agitation for self-determination.
South South (Niger Delta)	Akwa Ibom, Bayelsa, Cross River, Delta, Edo, Rivers	Oil & Gas Production (heart of Nigeria's petroleum industry), fishing .	Very diverse: Ijaw, Ibibio, Urhobo, Itsekiri, etc.	Resource Control Agitation ; faces severe environmental degradation from oil spills; hosts major seaports.
South West	Ekiti, Lagos, Ogun, Ondo, Osun, Oyo	Commerce & Finance (Lagos is the economic capital), agriculture (cocoa, rubber), media & tech .	Predominantly Yoruba	Strong Educational & Infrastructure Base ; most urbanized zone; vibrant media landscape.

4. Purpose, Benefits, and Criticisms of the Zoning System

- **Purpose and Benefits:**

- **Power Rotation (Zoning):** Ensures the presidency and other key political offices rotate among the zones to prevent dominance by one region.
- **Federal Character Principle:** Mandates fair representation of all zones in federal government appointments, promotions, and project allocation.
- **Conflict Management:** Aims to reduce ethnic tension and the agitation for marginalization by creating a sense of inclusion.
- **Development Planning:** Allows for regional development strategies that address zone-specific challenges.
- **Criticisms and Negative Implications:**
 - **Entrenchment of Division:** Can reinforce regional and ethnic identities at the expense of national unity.
 - **Merit vs. Representation:** The federal character principle is sometimes criticized for prioritizing geographical representation over merit and competence.
 - **Internal Marginalization:** The focus on zonal balance can mask the marginalization of minority groups *within* each zone.
 - **Political Tool:** Can be manipulated by the political elite to bargain for power rather than genuinely address development needs.

5. Boundary Conflicts and National Integration

- **Causes of Interstate Boundary Conflicts:**
 - **Unclear/Un-demarcated Boundaries:** Inherited imprecise colonial boundaries.
 - **Competition for Resources:** Especially land for agriculture and, crucially, areas with mineral resources (e.g., oil-rich border areas between Delta and Imo, or Cross River and Akwa Ibom).
 - **Political and Administrative Rivalry.**
 - **Ethnic Kinship:** Communities of the same ethnic group split by state boundaries may have conflicting loyalties.
- **Notable Examples:**
 - **Kogi vs. Enugu/Anambra:** Over the Odeke/Odolu oil-rich area.

- **Cross River vs. Akwa Ibom:** Long-standing dispute over 76 oil wells and territory.
- **Lagos vs. Ogun:** Largely over administrative control of rapidly urbanizing border communities.
- **Implications of Boundary Conflicts:**
 - **Loss of lives and property** during communal clashes.
 - **Displacement** of populations, creating internal refugees.
 - **Stifled development** as governments avoid investing in disputed territories.
 - **Breach of law and order.**

6. Challenges to National Integration in Nigeria National integration is the process of creating a sense of national identity and loyalty among diverse groups.

- **Ethno-Religious Loyalties:** Strong identification with one's ethnic group or religion often supersedes national identity.
- **Resource Control Agitations:** Perceived unfairness in the distribution of national wealth, particularly from the Niger Delta region.
- **Secessionist Movements:** Such as the Indigenous People of Biafra (IPOB) in the South East and various groups in the Niger Delta.
- **Political Marginalization:** The perception by some zones that they are systematically excluded from central power and resources.
- **Uneven Development:** Visible disparities in infrastructure and social amenities between zones fuel feelings of neglect.
- **Strategies for Promoting National Integration:**
 - **The National Youth Service Corps (NYSC):** Post-national service program that posts graduates to states other than their state of origin.
 - **Federal Character Principle:** As enshrined in the constitution.

- **Equitable Resource Allocation:** Through statutory bodies like the Niger Delta Development Commission (NDDC) and the Federal Allocation Account.
 - **National Orientation:** Promoting national symbols, a unified history curriculum, and fostering patriotic values.
 - **Strong Institutions:** Building impartial institutions that treat all citizens fairly, regardless of origin.
-

EVALUATION (WAEC/NECO Past Questions)

1. (WAEC, 2019): List the six geo-political zones and name two states in each.
2. (NECO, 2020): Explain three reasons for the creation of geo-political zones in Nigeria. *(Discuss history, diversity management, and power sharing).*
3. (WAEC, 2016): Highlight four effects of boundary conflicts on national development. *(Mention social, economic, and political effects).*
4. (NECO, 2018): Discuss three challenges of national integration in Nigeria. *(e.g., ethno-religionism, resource control, uneven development).*

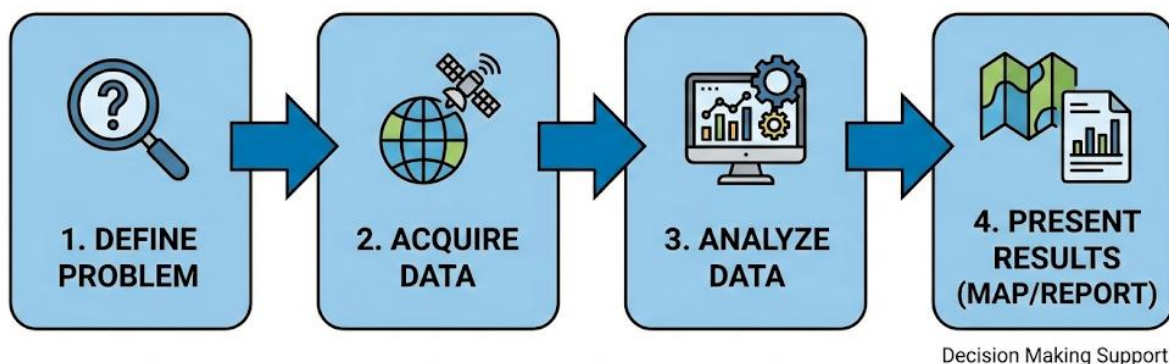
ASSIGNMENT

1. Draw a well-labelled map of Nigeria showing the six geo-political zones. Use different colours for each zone and indicate the location of the Federal Capital Territory (FCT).
2. Choose ONE geo-political zone and write a brief profile (200 words) describing its main economic activities, cultural features, and one major challenge it faces.
3. "The geo-political zoning system does more to divide Nigeria than to unite it." Do you agree or disagree? Present a well-reasoned argument for your position.
4. Suggest five concrete measures, beyond the NYSC, that the government could implement to strengthen national integration among Nigerian youth.

WEEK 9 - GEOGRAPHIC INFORMATION SYSTEMS (GIS)

Introduction: The Digital Revolution in Geography Geographic Information Systems (GIS) represent a fundamental transformation in how we collect, analyze, and visualize spatial data. It is a technological field that integrates geography with computer science, allowing us to understand patterns, relationships, and trends in the world around us in ways that were previously impossible. GIS has become an essential tool in virtually every field that deals with location-based information.

GIS WORKFLOW

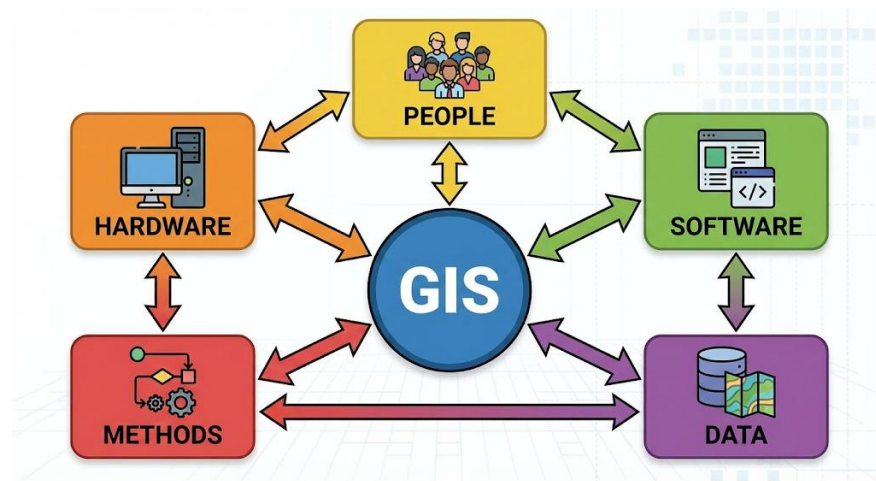


1. Meaning of GIS and Spatial Data

- **Geographic Information System (GIS):** A computer-based system designed to capture, store, manipulate, analyze, manage, and present all types of geographical data. It allows users to create interactive queries, analyze spatial information, edit data, and present the results of operations on maps.
- **Spatial Data:** Information that has a geographical or locational component. It answers two fundamental questions: "Where is it?" and "What is there?". Spatial data can be:
 - **Absolute:** Precise coordinates (latitude/longitude).
 - **Relative:** Described in relation to other features (e.g., "next to the river," "north of the highway").

2. Core Components of a GIS

A functional GIS requires five key components working together:



- **Hardware:** The physical equipment—computers, servers, GPS receivers, drones, scanners, and printers used to operate the GIS.
- **Software:** The applications and tools that provide the functionality. Examples include:
 - **Commercial:** ArcGIS (by Esri), MapInfo.
 - **Open-Source:** QGIS, GRASS GIS.
 - **Web-based:** Google Earth Engine, ArcGIS Online.
- **Data:** The most critical component. GIS data can be expensive to collect but is the foundation of any analysis. It includes both **spatial data** (the "where") and **attribute data** (the "what"—descriptive information in tabular form).
- **People:** The GIS specialists, technicians, analysts, and decision-makers who manage the system, design projects, and interpret the results. Without skilled people, GIS is just software and data.
- **Methods:** The standard operating procedures, workflows, and analysis techniques that ensure consistent and accurate results.

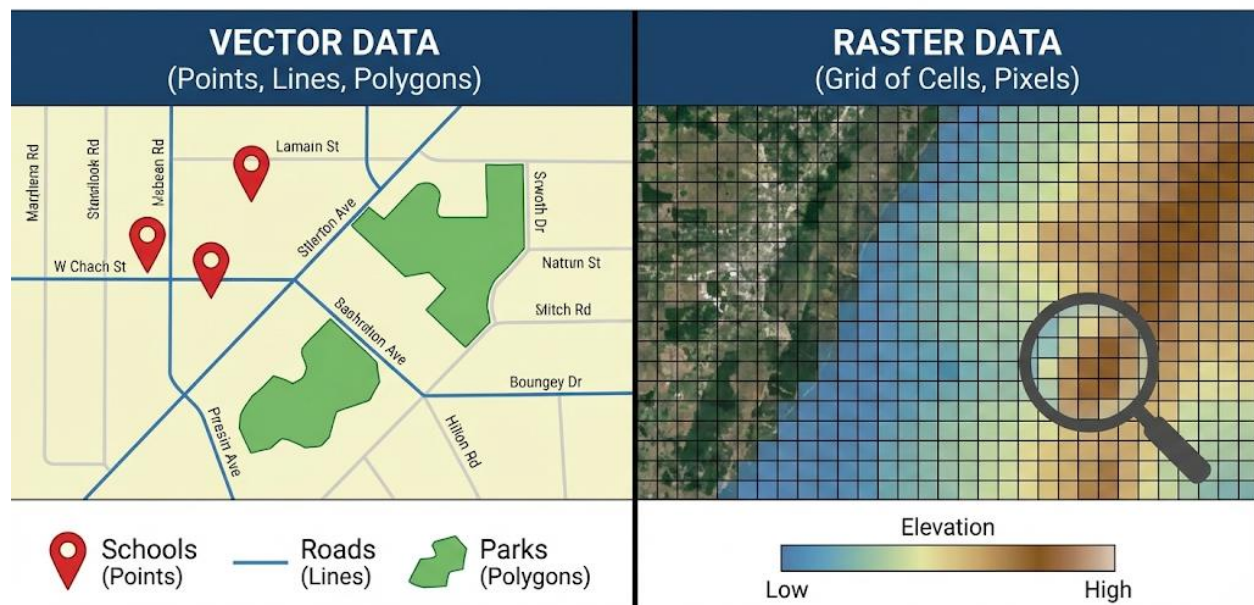
3. Types of GIS Data: Vector and Raster

GIS data is fundamentally stored in two primary formats, each suited for different types of information and analysis.

Feature	Vector Data	Raster Data
Structure	Composed of points, lines, and polygons	Composed of a grid of cells

Feature	Vector Data	Raster Data
	(areas).	(pixels), each with a value.
Best For	Representing discrete, well-defined features.	Representing continuous phenomena.
Examples	Points: Schools, boreholes, streetlights. Lines: Roads, rivers, pipelines. Polygons: Land parcels, lakes, administrative boundaries.	Satellite imagery, aerial photos, Digital Elevation Models (DEMs), temperature maps, rainfall maps.
Advantages	Precise representation, efficient storage, clear boundaries.	Effective for representing surfaces, good for mathematical analysis.
Disadvantages	Poor at representing continuous data, can be complex.	Can be large file sizes, "blocky" appearance when zoomed in.

Vector vs. Raster Visual Comparison



4. Sources of GIS Data Data can be collected from a variety of sources:

- **Primary Data Sources (Direct Collection):**

- **Global Positioning System (GPS):** Used for precise field data collection (e.g., mapping tree locations, accident sites).
- **Remote Sensing:** Capturing data from satellites (e.g., Landsat, Sentinel) or drones. Provides up-to-date, often real-time information.
- **Field Surveys:** Traditional surveying techniques and direct measurements.
- **Secondary Data Sources (Existing Data):**
 - **Topographic Maps:** Scanned and digitized.
 - **Census Data:** Population statistics linked to administrative boundaries.
 - **Government Agencies:** e.g., National Population Commission, Geological Survey agencies.
 - **Open Data Portals:** Many governments and organizations provide free spatial data.

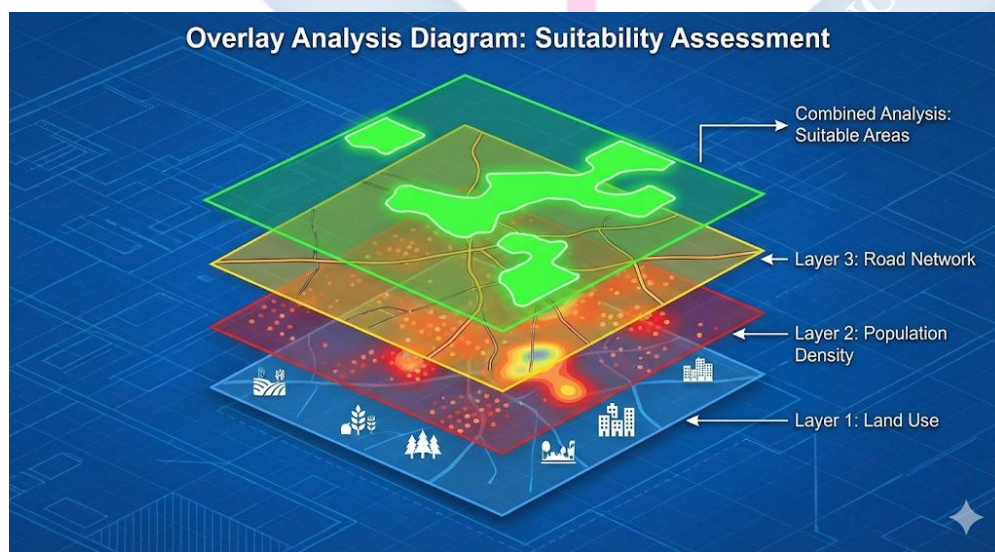
5. The Power of GIS: Overlay Analysis One of the most powerful functions of GIS is the ability to layer multiple datasets on top of one another to find relationships. For example, to find a suitable location for a new school, a planner could overlay:

1. **Land Use Layer** (to find vacant land).
2. **Population Density Layer** (to find areas with many children).
3. **Road Network Layer** (for accessibility).
4. **Flood Zone Layer** (to avoid risky areas). By combining these layers, GIS can identify the optimal site that meets all the criteria.



6. Applications of GIS in Geography and Beyond

- **Urban and Regional Planning:** Land-use planning, zoning, site selection for facilities, managing urban sprawl.
- **Environmental Management:** Tracking deforestation, monitoring pollution, managing natural resources, conservation planning.
- **Disaster Management and Response:** Flood risk mapping, wildfire management, coordinating emergency services during a crisis.
- **Agriculture:** Precision farming, soil analysis, yield prediction, and managing irrigation.
- **Public Health:** Mapping disease outbreaks (epidemiology), analyzing access to healthcare facilities.
- **Business and Marketing:** Analyzing demographic data for store location planning, logistics and delivery route optimization.
- **Transportation:** Planning new road/rail routes, managing traffic flow, and public transit planning.



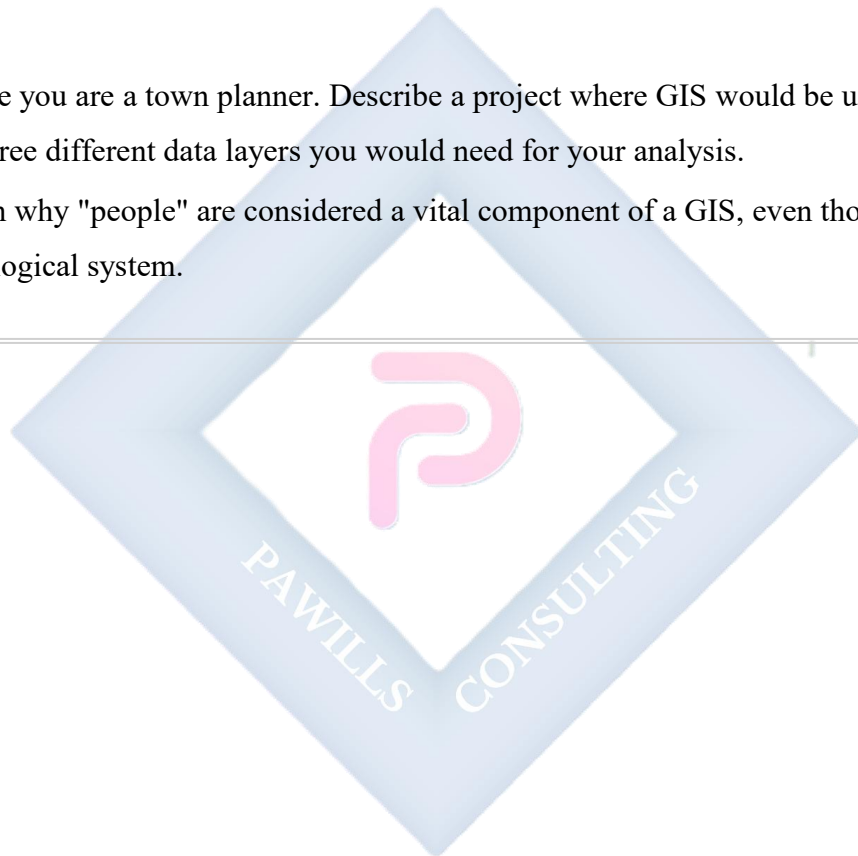
EVALUATION (WAEC/NECO Past Questions)

1. (WAEC June 2016) a. What is a Geographic Information System (GIS)? b. List four components of GIS. c. Mention two uses of GIS in environmental monitoring.

2. (NECO 2019) a. Define spatial data. b. Differentiate between raster and vector data in GIS.
3. (WAEC 2018) a. Mention four sources of data used in GIS. b. Highlight three applications of GIS in urban planning.

ASSIGNMENT

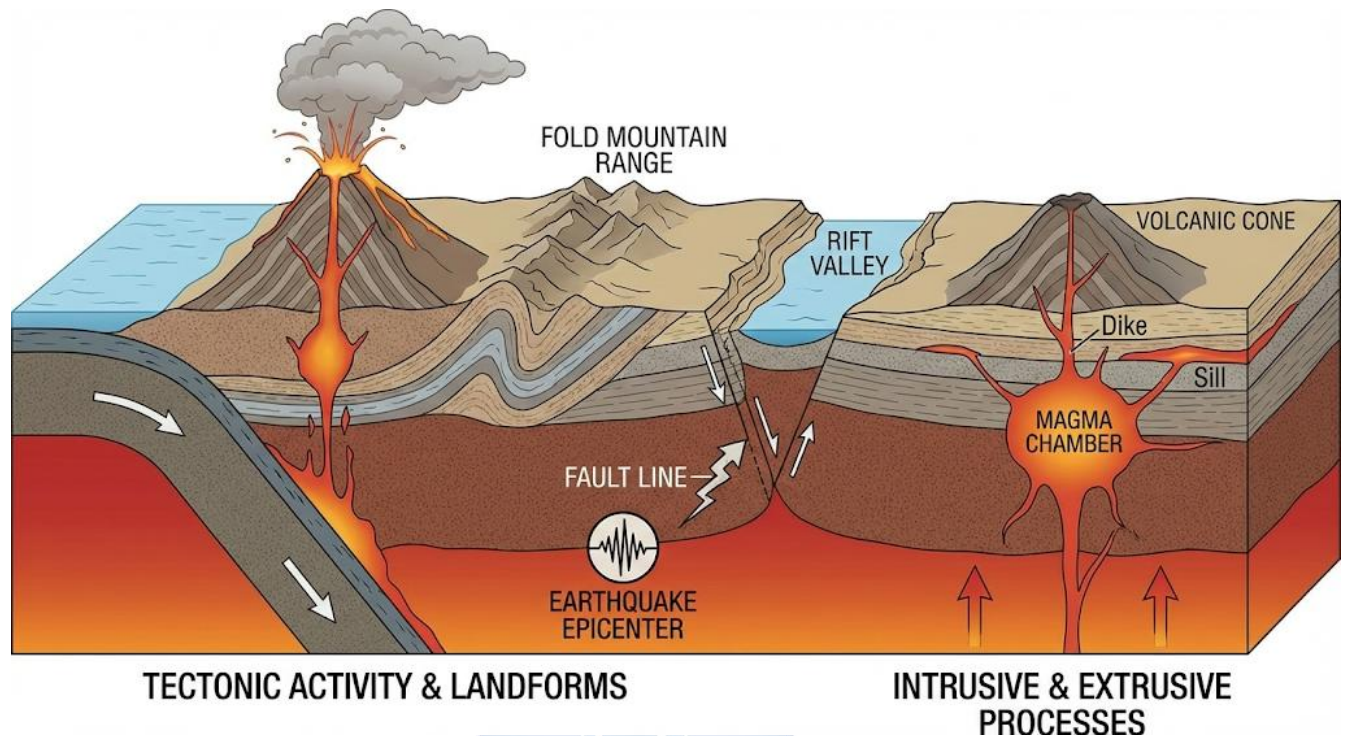
1. Define GIS and explain the role of its five key components.
2. Create a table to differentiate between vector and vector data, providing two examples of each.
3. Imagine you are a town planner. Describe a project where GIS would be useful. List at least three different data layers you would need for your analysis.
4. Explain why "people" are considered a vital component of a GIS, even though it is a technological system.



WEEK 10 - EARTH'S INTERNAL PROCESSES

Introduction: The Dynamic Earth

The Earth is a dynamic and ever-changing planet. While external processes like weathering and erosion wear down the surface, internal processes—driven by immense heat from the Earth's core—continuously build it up, creating mountains, ocean basins, and continents. These powerful forces are responsible for the planet's most dramatic landscapes and hazardous events.



1. The Earth's Internal Structure: The Engine Within Understanding internal processes requires knowledge of the Earth's layered structure:

- **Crust:** The thin, solid, outermost layer (5-70 km thick). It is fragmented into tectonic plates.
- **Mantle:** A thick layer of hot, semi-solid rock (2,900 km thick). Convection currents within the mantle are the primary driving force of plate tectonics.
- **Outer Core:** A layer of liquid iron and nickel. Its movement generates the Earth's magnetic field.
- **Inner Core:** A solid ball of iron and nickel, under extreme heat and pressure.

2. Types of Internal Processes

- **Diastrophism (Tectonic Movements):** Large-scale deformation of the Earth's crust.
 - **Folding:** The bending of rock layers under slow, compressional forces. This creates fold mountains (e.g., the Alps, Himalayas).
 - **Faulting:** The fracturing and displacement of rock layers. This creates landforms like rift valleys (e.g., East African Rift) and block mountains (horsts) like the Ruwenzori Mountains.
- **Volcanism:** The movement of molten rock (magma) from the Earth's interior to or towards the surface.
 - **Intrusive Volcanism:** Magma cools and solidifies below the surface, forming features like **batholiths** (large underground rock masses), **dykes** (vertical intrusions), and **sills** (horizontal intrusions).
 - **Extrusive Volcanism:** Magma reaches the surface as lava, forming volcanoes, lava plateaus (e.g., Deccan Plateau in India), and volcanic cones.
- **Earthquakes:** Sudden, violent shaking of the ground caused by the rapid release of energy along fault lines. The point of origin underground is the **focus**; the point directly above it on the surface is the **epicenter**.

3. The Unifying Theory: Plate Tectonics Plate Tectonics is the overarching theory that explains the large-scale movements of the Earth's lithosphere (the crust and upper mantle). The lithosphere is broken into several rigid **tectonic plates** that float on the semi-fluid asthenosphere below.

4. Types of Plate Boundaries and Associated Landforms The interaction of plates at their boundaries is where most internal processes and landform creation occur.

Boundary Type	Movement	Processes & Landforms	Real-World Examples
Divergent	Plates move apart	Rifting, sea-floor spreading, creation of mid-ocean ridges and rift valleys . Volcanism is	Mid-Atlantic Ridge, East African Rift Valley.

Boundary Type	Movement	Processes & Landforms	Real-World Examples
		common.	
Convergent	Plates move together	Subduction (one plate dives under another), intense folding and faulting, creating fold mountains, volcanic arcs, and deep-ocean trenches.	Oceanic-Continental: Andes Mountains. Oceanic-Oceanic: Japanese Islands. Continental-Continental: Himalayas.
Transform	Plates slide past each other	Shearing stress causes major strike-slip faults and powerful earthquakes. No volcanism.	San Andreas Fault (USA).

5. Causes of Earth's Internal Processes The ultimate driver is **heat from radioactive decay** in the Earth's core. This heat generates **convection currents** in the mantle—slow, rolling movements of semi-molten rock that drag the overlying tectonic plates.

6. Consequences and Importance of Internal Processes

- **Negative Consequences:**
 - Earthquakes and volcanic eruptions cause loss of life and destruction of property.
 - Tsunamis can be generated by undersea earthquakes.
- **Positive Consequences & Importance:**
 - **Formation of Continents and Mountains:** Shapes the very geography of our planet.
 - **Mineral Formation:** Creates concentrated deposits of valuable minerals and metals.
 - **Fertile Soils:** Volcanic ash weathers to form highly fertile soils ideal for agriculture.
 - **Geothermal Energy:** Heat from the Earth's interior can be harnessed for power.

- **Creation of New Land:** Volcanic activity creates new islands and landmasses (e.g., Surtsey in Iceland, the Hawaiian Islands).
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EVALUATION (WAEC/NECO Past Questions)

1. (WAEC 2020) With the aid of diagrams, describe the processes of folding and faulting, and explain how they influence the landscape.
2. (NECO 2019) Describe two effects of volcanic activities and give two examples each of intrusive and extrusive volcanic landforms.
3. (WAEC 2017) Differentiate between earthquakes and volcanoes. State three effects of each.
4. (NECO 2018) What is diastrophism? Mention and describe two forms of diastrophic movements.
5. (WAEC 2016) Explain three causes and three effects of earthquakes.

ASSIGNMENT

1. With the aid of diagrams, differentiate between divergent and convergent plate boundaries.
2. List any four major tectonic plates and the continents they affect.
3. Explain three benefits and three disadvantages of volcanic eruptions to humans.
4. (WAEC 2015) Discuss the role of internal processes in the formation of major landforms on the Earth's surface.
5. (NECO 2021) Explain the difference between intrusive and extrusive volcanic activities and give two examples of landforms formed by each.
6. (WAEC 2018) With examples, describe three ways in which internal processes affect human activities.